

2019-20

Assessment of Groundwater Quality around Industrial Clusters

Study Area: Ankleshwar, Surat (Sachin),
Vapi, Vadodara (Nandesari) and
Ahmedabad (Vatva) Industrial Cluster



Gujarat Environment Management Institute (GEMI)

An Autonomous Institute of Government of Gujarat

"ISO 9001:2015, 14001:2015 & ISO 45001:2018 Certified Institute"

"We Provide Environmental Solutions"

Report on

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(Nandesari) and Ahmedabad (Vatva) Industrial Clusters**

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About this Document

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Foreword

Industrialization in the State is on rapid pace posing direct threat to natural resources, one of such important resource is groundwater which today need attention both from its quality and quantity point of view. In the view of conserving and protecting an important resource of the state and to get an updated status about the quality of groundwater with a special focus around the most polluted industrial clusters of the State, Gujarat Environment Management Institute (GEMI) has taken up the task of '*Regional Groundwater Quality Monitoring of Gujarat*' from Forests and Environment Department, Government of Gujarat since 2013 and as a part of the same the Institute for the year 2019-20 has carried out an extensive monitoring around five industrial clusters namely; Ankleshwar, Nandesari, Sachin, Vapi and Vatva GIDC clusters for both pre and post monsoon seasons.

The Institute has carried out sampling and monitoring within 5 km buffer of five industrial clusters and collected around 20 groundwater samples from sources such as open wells, hand pumps, and bore wells of each clusters which were analysed at GEMI's Laboratory for physico-chemical parameters like pH, Color, Total Dissolved Solids (TDS), Total Hardness (TH), Calcium Hardness (Ca^{++}H), Magnesium Hardness (Mg^{++}H), Total Alkalinity, Electrical Conductivity (EC), Sulphate, Fluoride, Nitrate, Ammonical Nitrogen and a few heavy metal parameters. The analytical results so obtained were compared with the Indian Standards for Drinking Water (IS 10500:2012) to evaluate the groundwater quality and its suitability for various uses especially, for drinking water potability. The industrial cluster wise detailed discussions about the groundwater quality of respective industrial cluster has been incorporated in this report. However, in broad it was found that the overall groundwater quality for majority of the samples collected from industrial cluster in respect to all physico-chemical and heavy metals parameters except chemical oxygen demand is found to be fairly good and fit for drinking, domestic and irrigation purposes. Further, few monitored sources from industrial clusters mainly Ankleshwar, Nandesari, and Sachin are found to be contaminated where a population dependent ranging from 40 – 400 on such sources for various purposes mainly other than drinking requires to be looked after and studied further for restoration and remediation of the same.

**A K Jha, IFS
Director, GEMI**

DEDICATED TEAM

Overall Guidance and Supervision

Mr. R. D. Kamboj, IFS (Retd.)

Former Director

(1st November, 2018 to 16th July, 2019)

Shri. A. K. Jha, IFS

Director

Dr. Nitasha Khatri

Sr. Scientific Officer & Laboratory Head

Project Head

Mr. Paresh Chavda

Deputy Environmental Engineer & Unit Head

Project Manager

Ms. Smruti Patel

Assistant Environmental Engineer

Contribution by

Ms. Prachi Rana

Project Assistant

Field Team (Data Collection)

Mr. Janak Prajapati, Project Assistant

Mr Mayur Patel, Project Assistant

Mr. Dinesh Makwana, Project Assistant

Ms. Prachi Rana, Project Assistant

Laboratory Analysis

GEMI's Laboratory,

Gandhinagar

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Chapter: 1 Introduction

1.0 Introduction

The term groundwater is usually referred for the subsurface water that occurs beneath the water table in soils and geologic formation that are fully saturated. Ground water plays a vital role in the development of arid and semi-arid zones. It is believed to be comparatively much clean and free from pollution than surface water. ^[1]

However, groundwater contamination has become one of the most serious problems in the world in the last decades. In addition, groundwater contamination is generally irreversible i.e. once it is contaminated; it is difficult to restore the original water quality of the aquifer. ^[2] Groundwater quality depends on the quality of recharged water, atmospheric precipitation, inland surface water, and on sub-surface geochemical processes. Temporal changes in the origin and constitution of the recharged water, hydrologic and human factors may cause periodic changes in groundwater quality. ^[3]

Due to rapid industrialization and urbanization, prolonged discharged of industrial effluents, domestic sewage and solid waste dump causes the groundwater to become polluted, create health problem as well as causing severe environmental pollution. Most of the industries discharge their effluent without proper treatment into nearby open pits or pass them through unlined channels resulting in the contamination of groundwater. ^[1]

Therefore, assessment of groundwater near industrial zone becomes necessary with the purpose of checking its suitability for various usages i.e. drinking, domestic, industrial and irrigation.

The Institute has carried out Regional Groundwater Quality Assessment study of Gujarat's various districts under the project '*Regional Groundwater Quality of Gujarat*' since 2013. Every year a study with an approach of assessing overall groundwater quality of randomly selected district is conducted but this time an approach to monitor the groundwater quality around the industrial GIDC cluster was made for five randomly selected GIDC areas namely; Ankleshwar GIDC, Surat (Sachin) GIDC, Vapi GIDC, Vadodara (Nandesari) GIDC and Ahmedabad (Vatva) GIDC. Samples from each nearby GIDC region have been collected and analyzed for pre-monsoon and post-monsoon seasons.

1.1 Objective of the study

The objective of study is to monitor groundwater quality around Ankleshwar GIDC, Surat (Sachin) GIDC, Vapi GIDC, Vadodara (Nandesari) GIDC and Ahmedabad (Vatva)

GIDC (within 5 km buffer zone) in order to identify contaminated sites and to derive overall groundwater quality of selected GIDC area.

1.2 Scope of the study

The broad scope of the study is described as follows:

- To carry out preliminary visit nearby GIDC region and inquire in person about the contamination of groundwater.
- To analyze the groundwater quality of the nearby area (within 5 km buffer zone) of selected five GIDCs in order to assess the quality of groundwater around industrial area
- To check whether the groundwater is contaminated by means of physico-chemical parameters.
- To carry out sampling and monitoring of the identified sampling locations for pre and post monsoon seasons in order to assess the quality and suitability of groundwater for drinking purpose.
- To collect required data, and interpret the data using appropriate methods.
- To derive the overall groundwater quality of the selected study areas.

1.3 Criteria for site selection

From the time when rapid industrialization is occurring in the State, there arises a need to assess the groundwater quality of nearby industrial area, as groundwater is the major source of water for both industries and population residing in nearby area. In the present study, five GIDCs cluster are selected which are also declared as the critically polluted GIDC area by Comprehensive Environmental Pollution Index (CEPI) and are listed as:

- 1) Ankleshwar GIDC
- 2) Vapi GIDC
- 3) Sachin GIDC, Surat
- 4) Nandesari GIDC, Vadodara
- 5) Vatva GIDC, Ahmedabad

Chapter: 2 Literature Review

Numerous research papers were studied in order to understand the cause and effects of industrial pollution on groundwater quality as well as study methodology. Some of these research works are described here in brief by covering research title, authors, journal's information and findings. The same are as follows:

Table 2.1 Literature Review

Sr. No:	Paper Title	Authors	Journal's Information	Findings
1.	A Groundwater Quality Analysis of GIDC Naroda	1) Kartikey Tiwari 2) Shivendra Jha 3) Utkarsh Srivastava	<i>International Journal of Engineering Science Invention Research & Development,</i> Vol. III, Issue IX, pp 527 - 532 (March 2017) e-ISSN: 2349-6185	• The residential population resides only in phase 1-2, ground water is used by tea stalls and slums for drinking purposes all around GIDC. So, it wouldn't be true to say that water quality as per IS code 10500 needs to be maintained only in first two phases. Rather, the quality of water provided all around the GIDC should fulfil all the criteria of potable water. • The results of pH, TDS, Turbidity and COD tests show that the samples of ground water have them at excess which might cause certain health issues like, nutritional deficiency, bitter or salty taste, incrustation, films, precipitate or corrosion on fixtures, harmful for peoples with compromised immunity, newborns or elderly and microbial diseases.
2.	Groundwater Quality Assessment of Industrial Zone of Faisalabad Using Geographic	1) Muhammad Salman Nasir 2) Abid Sarwar 3) Imran Shaukat	<i>International Journal of Science and Research (IJSR),</i> Vol. 3, Issue 10, pp 2016 - 2020 (October	• The spatial distribution map of Hardness, Alkalinity, cadmium, arsenic, nickel, lead and fluoride shows that these parameters are vary accordingly in study area. • The interpreted water quality with respect to alkalinity indicates that more than 90% of the study area groundwater lies in bad range for drinking purposes. • The spatial distribution map of hardness concentrations illustrates that the majority of the samples are within the

	Information System		2014) ISSN (Online): 2319-7064	permissible limit. There is an increasing trend of trace elements in study area so there is need some strategic solution to overcome this trend. • It is recommended that the groundwater of industrial zone of Faisalabad should not be used for drinking purpose. The contamination in groundwater may be due to the old and worn out sewerage system so it should be repaired. • The industry is also playing role in groundwater contamination so further studies should be done to evaluate the role of industry in contaminating groundwater. Drains should be lined and drinking water should use after treatment.
3.	Groundwater quality assessment using the Water Quality Index and GIS in Saveh-Nobaran aquifer, Iran	1) S. M. Sadat-Noori 2) K. Ebrahimi 3) A. M. Liaghat	<i>Environ Earth Sci,</i> DOI 10.1007/s12665-013-2770-8 (Published online: 08 September 2013)	• The GIS had a major role in the progress and is considered an effective tool, as with GIS software various maps showing spatial distribution of various water quality parameters were prepared and analyzed. • Spatial distribution maps were used to confine the locational distribution of water pollutants in a thorough producer and assisted in indicating groundwater contamination control and remedial measures in a comprehensive way. • The WQI spatial distribution map clearly shows best sites for drinking in the area. Water quality indices are used to provide valuable tools for decision makers to be able to understand the status of the water quality in a water source and to have the opportunity to make adequate decisions for better use in future.
4.	Analytical study of ground water contamination in Ankleshwar Taluka of Gujarat	1) Mukesh A Modi 2) Hettal M. Togle 3) Dr. N J Shrimali	<i>International Journal of Applied Engineering Research,</i> Vol. 14, No.1, pp 16 - 21 (2019 (Spl.)) ISSN 0973-4562	• The spatial distribution map of Total Dissolved Solids, Total Alkalinity, Total Hardness, Chloride and Fluoride shows that, these parameters were not within the permissible limit throughout the study area uniformly. • The study region and its nearby locations are going through a quantity of groundwater issues, such as water shortages, locally excessive pollution, and low recharge and over exploitation of the resource. • The important sources of pollution in the study area are negative unregulated landfills, contaminated river water and drain systems.

5.	A study on assessment of groundwater quality and its suitability for drinking in Vuyyuru, Krishna(dist.), Andhra Pradesh	1) B Vijaya Lalitha 2) K Sai Tejaswini	<i>International Journal of Engineering Development and Research</i> , Vol. 5, Issue 2, pp 1662 - 1668 (2017) ISSN 2321-9939	- Based on water quality index the suitability of groundwater in 10 sample stations of the study area is determined. The quality of ground water in KCP colony (SI.NO.5), Rotary eye hospital, Vuyyuru main center, Sowbhagya nagar (SI.NO.2,4,6) is Poor. The quality of water in KCP sugar industry (SI.NO.3), Eginipadu (SI.NO.9) is very poor. The quality of groundwater in Gandigunta, Rajesh nagar, Jagannadhapuram, Mantada (SI.NO.1,7,8,10), is found as good. - From the above results conclusions are, some areas in the study area are having good water quality standards. Some areas are having poor and very poor quality. By treating the water in that areas, the groundwater can be used for drinking purposes. And though the study area previously used for agriculture.
6.	Groundwater Pollution in and Around a Paper Wastewater-Irrigated Area, Northwest China	1) Jianhua Wu 2) Peiyue Li 3) Hui Qian 4) Jie Chen	<i>2013 Fourth International Conference on Digital Manufacturing & Automation</i> , Conference Publishing Services, pp 649 - 652 (30 March 2014)	- Groundwater in the study area has been contaminated, especially NO₃-N and COD pollution. Most physiochemical parameters usually have high variations in concentration. The samples with exceeding contents of NO₂-N have accounted for over 50% of all samples. Furthermore, the number of samples that have excess SO₄²⁻, F⁻, TDS, TH and NH₄-N has also exceeded 40%. - Groundwater pollution in the forest area is mainly caused by COD_{Mn}, NO₂-N, NO₃-N and F⁻, while groundwater in the alluvial plain may be attributed to excess TDS, TH, F⁻ and NO₂-N. Elevated concentrations of COD_{Mn}, NO₂-N, NO₃-N is due to anthropogenic intervention, while TDS, TH, pH and F⁻ may be influenced more by natural processes.
7.	Heavy Metal Assessment in Industrial Groundwater in and Around Vijayawada, Andhra Pradesh, India	1) T. Ravi Sankar 2) P. T. S. R. K. Prasada Rao	<i>Eur. Chem. Bull.</i> , Vol. 3, Issue 10, pp 1008 - 1013 (2014)	- The minimum and maximum trace metals concentrations in different areas for, Pb, Cu, Fe, Ag, Zn, Cd, Mn, Ni, Cd, and Co are 0.012-0.153, 0.00 and 1.85, 0.00 -1.347, 0.05- 0.20, 0.05 - 1.5, 0.2-0.446, 0.081 - 0.155, 0.096-0.646, .007-0.137, 0.093-0.157 mg/L respectively. - Trace metals like Cd, Cr, Fe, Mn and exceeded the maximum limits according to WHO and IS10500 2012 for drinking water in several sampled bore wells in the Vijayawada region.

				From the obtained results it is evident that, at present the metal ion concentration is not at the levels which could be harmful for humans. But still the study clearly points out that the concentrations of toxic metals are present in slight excess in some locations.
8.	Ground water quality and its health impact analysis in an industrial area	1) M. Govindarajan 2) T. Senthilnathan	<i>International Journal of Current Microbiology and Applied Sciences</i> , Vol. 3, Number 7, pp 1028 - 1034 (2014) ISSN 2319-7706	- Continuous disposal of industrial effluents on land, which has limited capacity to assimilate the pollution load, has led to groundwater pollution. - Ground water quality surrounding the industrial areas has deteriorated, and the application of polluted groundwater for potability has resulted in increased salt content of soils. - Water taken in the industrial area of Maraimalai Nagar town within 5 Km radius was found to be unpotable, whereas the water taken from the 10Km radius of the industrial town was found to be potable.
9.	Heavy Metal Contamination in Groundwater around Industrial Estate vs Residential Areas in Coimbatore, India	1) K. Mohankumar 2) V. Hariharan 3) N. Prasada Rao	<i>Journal of Clinical and Diagnostic Research</i> , Vol. 10(4), pp 5 - 7 (Apr 2016)	- Mercury, Arsenic and Cadmium were not detected in the areas surrounding Kurichi Industrial Cluster. Instead Arsenic and Cadmium were found in regions far from the industrial estate. - The ground water of the areas surrounding the industrial cluster does not contain significant amount of metals. Instead, Heavy metal contamination of ground water were observed in some residential areas of coimbatore. - The regulatory measures to contain and prevent ground water contamination by industries undertaken by Tamilnadu pollution control board may have led to absence of heavy metal contamination in Kurichi Industrial cluster, Coimbatore, India.
10.	Groundwater Pollution Around an Industrial Area in the Coastal Stretch of Maharashtra State, India	1) Pradeep K. Naik 2) Biranchi N. Dehury 3) Arun N. Tiwari	<i>Environmental Monitoring and Assessment</i> , 132, pp 207 - 233 (2007)	- Chemical analyses of 131 water samples show that the area in general is characterized by hard water possibly due to its hydraulic connection with the sea. - Maintenance of pH in the three effluent collecting sumps has not been adequate with pH value varying between 1.00 at Sump no. 3 to 5.66 at Sump no. 2. Even in the main pipeline which discharges effluents to the sea at Navapur, a pH value of 3.70 was observed.

				The Muramba Creek seems to be highly polluted. The pH, TDS, Na⁺ and Cl⁻ values start increasing seaward while SO₄⁻, Fe, Mn, Pb and Zn values start decreasing towards the sea.
11.	Effect of Polluted Surface Water On Groundwater: A Case Study Of Budha Nullah	1) Gursimran Singh 2) Dapinder Deep Singh 3) Prof. S. K. Sharma	<i>IOSR Journal of Mechanical and Civil Engineering,</i> Vol. 5, Issue 5, pp 01 – 08 (Mar. - Apr. 2013) e-ISSN 2278 - 1684	- The quality of the surface water at different stretches of Budha Nullah has been impaired as the stream progresses through the city. Untreated waste from the industries and human settlements are found to be the major cause for deterioration in the water quality in Ludhiana. The surface water analysis of Budha Nullah reveals high values of Total Dissolved solids up to 1642 mg/L, chlorides up to 400 mg/L, Chemical Oxygen Demand values up to 448mg/L, Biochemical Oxygen Demand varying between 52-195 mg/L, Most Potable Number varying from 240+ upto 2400+ per 100ml, heavy metal like Cr in the Budha Nullah has value 0.084 mg/L, Fe 0.913 mg/L, Mn 0.095 mg/L and Ni 0.222 mg/L. As per the water quality criteria given by Central Pollution Control Board (CPCB), the water quality of Budha Nullah comes under E class of water which is not suitable for drinking, outdoor bathing, propagation of wildlife and fisheries, irrigation and industrial cooling. - Most of the Heavy metals were within permissible limit except iron which was found to be more than permissible limit in samples of the Budha Nullah as well as groundwater. - The quality of ground water at many places was found to be unfit for drinking mainly due to high amount of total dissolved solids and most potable number being much higher than the prescribed limit under CPCB. - BOD values of groundwater was also high ranging from 2 to 12 due to which it falls under category E i.e. it is unfit for human consumption but can be used for irrigational purposes.

Chapter: 3 Study Methodology

3.1 Methodology Adopted

The industrial cluster selected as a study area has varied ground water extraction structures and also the groundwater scenario varies from cluster to cluster. Hence a proper methodology was carefully planned to carry out the groundwater quality assessment. Chart describing the methodology adopted for the study of assessment of groundwater quality around industrial cluster is mentioned below:

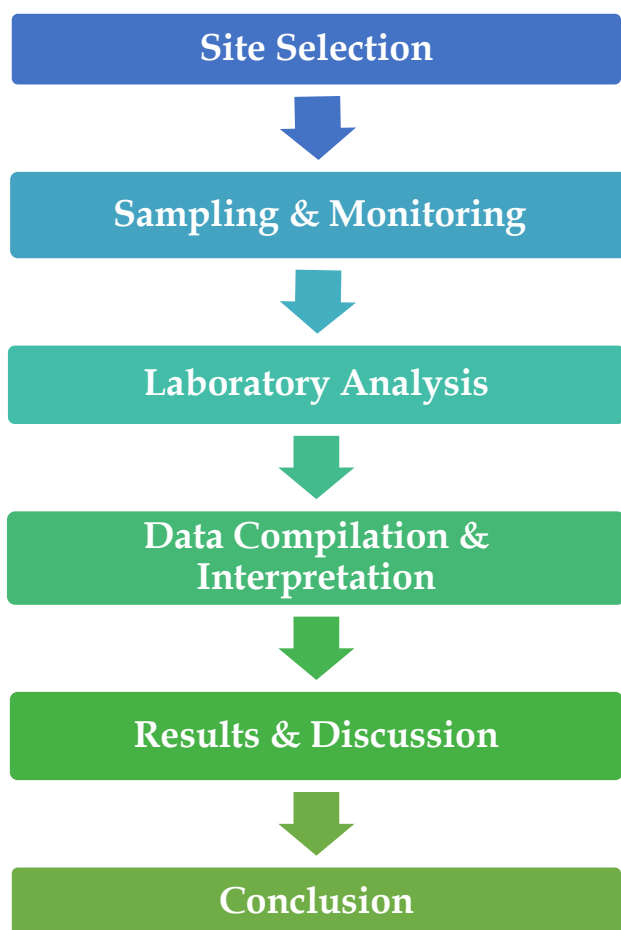


Figure 3.1 Flow Chart of Methodology adopted to conduct the Study

3.2 Site selection

Selection of sampling locations has been done on the basis of their respective geographical location, within 5 km of buffer zone area. Total 20 samples have been collected from each GIDC area and the same is denoted below.

In preliminary survey, identification of wells around the GIDC area within 5 km of buffer zone was carried out. Locations were identified around the five GIDCs which falls within 1, 3 and 5 km radius from the industrial clusters. Simultaneously, the basic information about the depth of the well, color of water, etc. were also collected during the sampling. The details regarding the samples and their respective information collected is described in respective chapters of the GIDC.

3.3 Sampling and Monitoring

Sampling being an important activity, it was carried out by trained and experienced field staff as per GEMI's Protocol for 'Sampling of Water and Wastewater' which is approved Government of Gujarat under provisions of Water (Prevention and Control of Pollution) Act, 1974. A total of 94 and 100 groundwater samples were collected during pre-monsoon and post-monsoon seasons respectively. The method used for collection of sample is Gab sampling method. Whereas, the steps followed for collection of samples is described as follows:

- Identify most suitable and safe spot for sample collection. Avoid any risk during sampling. Take all necessary safety measures.
- Properly rinse the sampling bucket/carboy for 2 to 3 times with the sample water before collecting the sample.
- All the samples were collected after running the water for 2 to 3 minutes from the valve in a polyethylene carboy, and further were distributed in other appropriate containers as per the specific requirements of parameters to be analysed.
- Care was taken that the carboy and other containers are properly rinsed 2 to 3 times with the sample water before collecting the same so as to avoid any kind of contaminant in the carboy affecting the quality of the sample.
- The sample requiring preservation were preserved at the site itself, using appropriate preservatives as prescribed in the Standard Methods for the examination of Water and Wastewater – American Public Health Association (APHA)

3.4 Transport and Delivery

The samples collected were transported to GEMI's laboratory through GEMI's vehicle. The samples when in transport are kept in ice cooled containers or boxes, whereas the vehicle is maintained in sufficient cooled condition by means of air conditioners and the samples were submitted in the GEMI's laboratory with the due procedure. ^[4]

3.5 Analysis of Groundwater

All the groundwater samples were analyzed for selected relevant physico-chemical parameters. The physical parameters include pH and Electrical Conductivity (EC). The

chemical parameters include Color, Turbidity, Fluoride (F⁻), Nitrate (NO₃-N), Sulphate (SO₄³⁻), Chloride (Cl⁻), Total Dissolved Solids (TDS), Total Suspended Solids (TSS), Total Hardness (TH), Calcium and Magnesium Hardness (Ca²⁺-H and Mg²⁺-H), Alkalinity, Chemical Oxygen Demand (COD) and Ammonical Nitrogen (NH₃-N). Some of these samples were also analyzed for Heavy Metals including Iron (Fe), Lead (Pb), Arsenic (As), Cadmium (Cd), Zinc(Zn), Total Chromium (Cr) and Hexavalent Chromium (Cr(VI)). All the parameters were analyzed in Gujarat Environment Management Institute's laboratory and the testing process in the laboratory was performed according to the procedures mentioned in the "Standard Methods for the examination of Water and Waste Water – American Public Health Association (APHA)" hand book. The parameters analyzed are compared with the Indian Standards Drinking Water – Specification since most of the groundwater of the study area is used for drinking purpose whereas, the analytical method uses for analysis is described in table 3.1.

Table 3.1 Parameters analyzed and their reference method

S. No.	Parameters	Units	Reference method
1	pH	-	APHA,23 rd Edition (section-4500-H ⁺ B)
2	Electrical Conductivity	μS/cm	APHA,23 rd Edition (section-2510B)
3	Total Dissolved Solids	mg/L	APHA,23 rd Edition (section-2540 C)
4	Chloride as Cl ⁻	mg/L	APHA,23 rd Edition(section-4500-Cl ⁻ B)
5	Total Hardness	mg/L	APHA,23 rd Edition (section-2340 C)
6	Ca ⁺⁺ Hardness	mg/L	APHA,23 rd Edition (section-3500-Ca ⁺⁺ B)
7	Mg ⁺⁺ Hardness	mg/L	APHA,23 rd Edition(section3500Mg ⁺⁺ B)
8	Alkalinity	mg/L	APHA,23 rd Edition (section-2320B)
9	Turbidity	NTU	APHA,23 rd Edition (Section -2130 B)
10	Fluoride	mg/L	APHA,23 rd Edition (section-4500-F D)
11	Sulphate	mg/L	APHA,23 rd Edition (section 4500-SO ₄ ²⁻ -E)
12	COD	Mg/L	APHA,23 rd Edition (section-5220 B)
13	Nitrate	mg/L	IS: 3025(Part 34)1989(reaffirmed 1988)
14	Lead	mg/L	APHA ICP 23 rd Edition (Section -3120 B)
15	Cadmium	mg/L	APHA ICP 23 rd Edition (Section -3120 B)
16	Iron	mg/L	APHA ICP 23 rd Edition (Section -3120 B)
17	Nickel	mg/L	APHA ICP 23 rd Edition (Section -3120 B)
18	Chromium	mg/L	APHA ICP 23 rd Edition (Section -3120 B)
19	Zinc	mg/L	APHA ICP 23 rd Edition (Section -3120 B)
20	Arsenic	mg/L	APHA ICP 23 rd Edition (section - 3120 B)
21	Colour	Hazen	APHA, 23 rd Edition (section - 2120 B)
22	Ammonical Nitrogen	mg/L	APHA,23 rd Edition (section -4500 NH ₃ C)
23	Hexavalent Chromium	mg/L	APHA ICP 23 rd Edition (Section -3120 B)



Chapter: 4 Ankleshwar GIDC

4.1 Overview of Ankleshwar Industrial Cluster

Ankleshwar is an industrial town located in the Bharuch district of Gujarat. The population of the town is approximately 150,000 most of which is directly or indirectly related to the chemical industry. The Ankleshwar Industrial Estate was set up by the Gujarat Industrial Development Corporation (GIDC) in the 1970s. The industrial estate, which is spread over an area of 1,600 hectares has close proximity to National Highway 8 (NH 8) and Delhi–Mumbai Railway Line. The industrial estate has more than 1,200 industries manufacturing diverse range of chemicals, pesticides, pharmaceuticals, bulk drugs, petroleum products, engineering, textiles, plastics, rubber, and packaging. Of these 1,200 units more than 600 are MSME units manufacturing various types of chemicals, like dyes, pigments, insecticides, specialty chemicals, petrochemicals, pharmaceuticals, and paints. [7]

Table 4.1: Ankleshwar district at a glance

S. No.	Particulars	Information
1.	Geographical information	21.6 °N 73 °E
2.	Geographical area (Bharuch) (sq. km)	6527
3.	Average annual rainfall (Bharuch)	600 mm
4.	Average temperature	40 °C (Summer), 32 °C (Monsoon), 23 °C (Winter)
5.	Population (Ankleshwar town)	1,50,000 (approx.)
6.	Literacy (Bharuch)	83.03 %
7.	Stage of Groundwater development	25.31% - Safe

(Source: Groundwater Brochure Bharuch District, March 2014. [6])

Products Manufactured

Chemical industries in Ankleshwar manufacture diverse range of products like dyes and dye intermediates, pigments, pesticides/insecticides, petrochemicals, agrochemicals, chlor-alkali, pharmaceuticals, paints, and so on. Some of the major categories of chemicals is shown in table:

Table 4.2: Chemical products manufactured in Ankleshwar GIDC

Category	Chemicals produced
Dyes and Pigments	Azo dyes, acid direct dyes, basic dyes, fats colour bases, oil-based dyes, whitening agents, organic pigment colours, pigment emulsion, food colours

Pesticides	Various pesticides and insecticides registered under the Insecticide Act, 1968
Petrochemicals	Synthetic fibres, fibre intermediates, polymer, performance plastics
Organic chemicals	Acetic acid, Acetone, Acetic anhydride, phenol, methanol, nitrobenzene, citric acid, chloromethane, aniline
Inorganic chemicals	Aluminium fluoride, calcium carbide, carbon black, potassium/sodium chlorate, titanium dioxide, and red phosphorous
Alkali	Soda ash, caustic soda, liquid chlorine

(Source: Cluster Profile Report – Ankleshwar Chemical Cluster. [7])

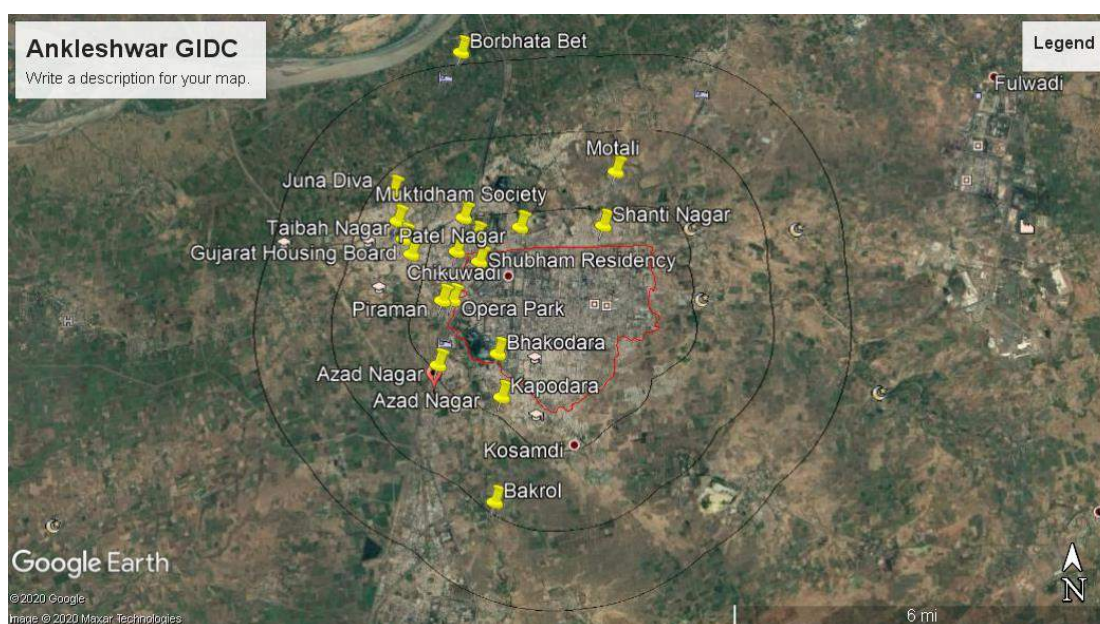
4.2 Sampling and monitoring

For the purpose of study sampling sites were selected within 5 km buffer region across the GIDC area. The details of the sampling sites selected is described in table 4.3 and figure 4.1.

Table 4.3 Sampling locations around Ankleshwar GIDC

Villages/locations selected for Sampling & Monitoring		
Within 1 km (10 samples)	Within 3 km (6 samples)	Within 5 km (3 samples)
Patel Nagar, Piraman, Chickoo Wadi, Muktidham Society, Shanti Nagar, Bhakodara, Kapodara, Shubham Residency, Opera Park (2), Koba Park	Motali, Juna Diva, Amboli, Taibah Nagar, Gujarat Housing Board, Azad Nagar	Borbhata Bet, Bakrol (2)

Figure 4.1 Sampling locations around Ankleshwar GIDC



Samples were collected from the identified location for both pre and post monsoon seasons. The information about sampling location, source type, color appearance at field, depth of source, population dependent on source, water usage and name of person contacted during the time of sampling is summarized in table 4.4.

Table 4.4 Details collected during sampling around Ankleshwar GIDC

S. ID.	Village/ Site	Location	Latitude Longitude	Source type	Color Appearance @ Field	Depth of well	Population dependent on source	Water Usage	Persons Contacted	Remarks
1.	Motali	Near household area	21.6497 73.0405	Bore well	Turbid	150-200	50-60	Domestic	Suraj bhai	Bore well is closed, hence post-monsoon sampling has not carried out
2.	Gujarat Housing Board	Water Tank (Residential Area)	21.62616 72.98814	Bore well	Clear	200-250	80	Drinking	Pankaj bhai	Water is used for drinking after filtration
3.	Amboli road	Bharuchi Naka	21.63592 72.98638	Bore well	Clear	250	320	Drinking	Kapil bhai	Water is used for drinking after filtration
4.	Juna Diva	Near Post Office	21.64753 72.97387	Bore well	Clear	180-200	60	Domestic	Amida ben	Source is surrounded by a lake and agricultural area
5.	Muktidham Society	Near household area	21.63877 73.00278	Bore well	Clear	60-70	10	Drinking	-	Water is used for drinking after filtration
6.	Patel Nagar	Near Masjid	21.6341 72.0165	Bore well	Clear	150-180	300-400	Domestic	Hiten Solanki	Sewage line is nearby the source.
7.	Piraman	Near School	21.61803 72.99842	Bore well	Clear	170-200	500-600	Drinking	Samiulha bhai	Use for drinking after mixing with other source of water
8.	Taibah Nagar	Near Antarnath Mahadev Temple	21.63117 72.98629	Bore well	Clear	200	200	Domestic	-	-
9.	Opera Park	Near Lupin Agrochemicals	21.61789 73.00006	Bore well	Turbid	180	320	Domestic	Pankaj bhai	Local people complains for salty water. Water is bit turbid,

										however quality is bit improved over the time.
10.	Opera Park (Palika bore well)	Near Masjid Garibnawaz	21.617913 72.999717	Bore well	Clear	200	200-400	Drinking	-	Water is used for drinking after filtration.
11.	Shanti Nagar	Near Hanumanji Temple	21.3842 73.0233	Bore well	Clear	170	100-150	Drinking, Domestic Industrial	Saroj bhai	-
12.	Bhakodara	Near Kedarnath Mahadev Temple	21.604879 73.011554	Bore well	Clear	200	450	Drinking	Baka bhai	Water is used for drinking after filtration.
13.	Kapodara	Near NewMahadev Temple	21.594013 73.007917	Bore well	Clear	200	300	Drinking	-	-
14.	Bakrol	Near Mahakali Temple	21.567204 73.003657	Bore well	Reddish	200	15	Domestic	-	GIDC is across the road.
15.	Bakrol	Near Ramnagar English School	21.566943 73.005194	Bore well	Reddish	180	150	Domestic	-	-
*	Bakrol	Near Ramnagar English School	21.566943 73.005836	Bore well	Reddish	180	150	Domestic	-	Source is different than previous, as previous one is closed, which is on the other side of the pond.
16.	Koba Park	Behind Umiyadham (Bhavan)	21.632113 73.005836	Bore well	Yellowish	200	40	-	Prabhat bhai	GIDC is 4 km away & 2 sewage line is nearby. At the time of sampling, water is not in use, however previously it was used for domestic purpose.
17.	Shubham Residency	Near Jalaram craft	21.626772 73.006832	Bore well	Brownish	170	100-200	Domestic	Janak bhai Sajupati	-

*	Shubham Residency	Near Jalaram craft	21.626772 73.006832	Bore well	Clear	140	100-200	Domestic	-	Previous source is closed due to contamination, hence sample taken from other source. Atlanta Back India Ltd. is adjacent to the source, however water seems clear in appearance but creates foam when used.
18.	Borbhata Bet	Village water tank	21.679221 73.002694	Bore well	Clear	180	-	Drinking Irrigation	Ravi bhai	-
19.	Chikuwadi	Near Sardar Patel Hospital	21.62891 73.00131	Bore well	Clear	180	60-80	Domestic	Imaran bhai	Foul smell
20.	Azad Nagar	Near Gausiya Jama Masjid	21.603469 72.996196	Bore well	Clear	200	100-150	Domestic	Firoz bhai	-

Note:

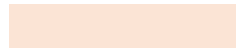
1. * different sample source has taken for post-monsoon season as previously monitored source was found sealed (not in use condition) due to contamination.
2. Color code given to sample ID



Color Code is given to samples collected within 1 km buffer



Color Code is given to samples collected within 3 km buffer



Color Code is given to samples collected within 5 km buffer



4.3 Result and Discussion

The physico-chemical parameters of groundwater samples with their acceptable and permissible limits as recommended by Indian Standards Drinking Water - Specification (Second Revision) (IS 10500: 2012) and analytical result range obtained for the present study for both pre and post monsoon seasons is shown in table 4.5:

Table 4.5 Result range of pre-monsoon and post-monsoon sampling

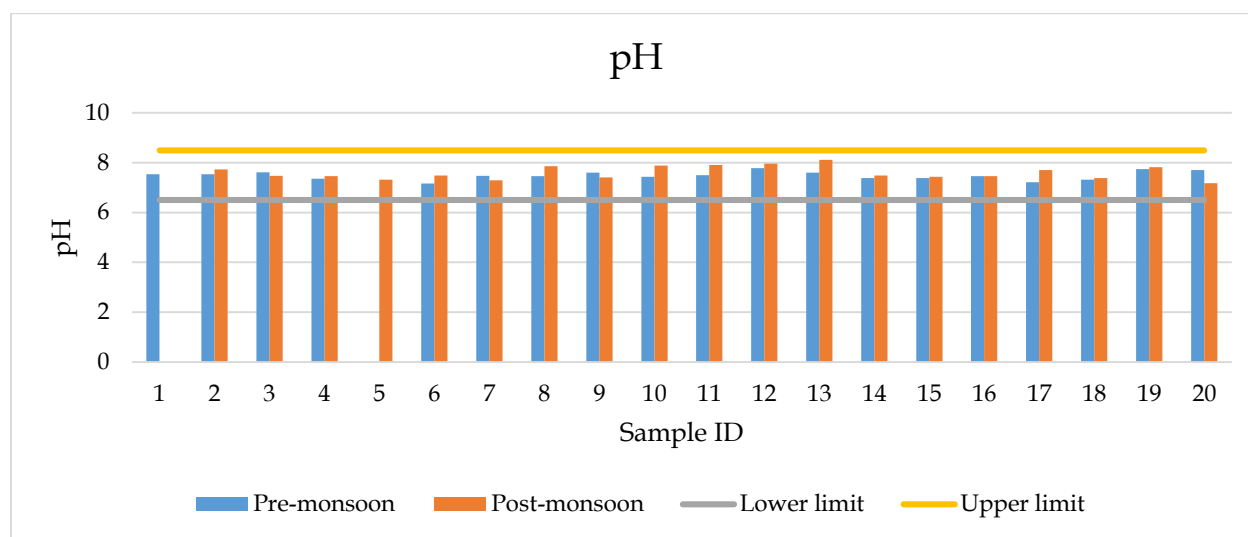
Sr. No.	Parameters	Measured Units	Requirement (Acceptable Limit)	Permissible limit in the absence of alternate source	Range of present study (Pre-monsoon)	Range of present study (Post-monsoon)
1	pH	-	6.5 – 8.5	No Relaxation	7.17 – 7.78	7.18 – 8.12
2	Colour	Hazen	5	15	5 - 1000	5 – 500
3	Electrical Conductivity	µS/cm	-	-	1634 - 6570	850 - 5580
4	TDS	mg/L	500	2000	868 – 3584	452 – 3082
5	TSS	mg/L	-	-	2 – 142	0 - 18
6	COD	mg/L	-	-	10 - 277	10 – 256
7	Chloride as Cl ⁻	mg/L	250	1000	222 - 1938	108 – 1235
8	Total Hardness	mg/L	200	600	200 – 1160	130 – 870
9	Ca ⁺⁺ Hardness	mg/L	75	200	90 – 540	30 – 310
10	Mg ⁺⁺ Hardness	mg/L	30	100	100 – 750	50 – 570
11	Alkalinity	mg/L	200	600	66 – 454	117 – 489
12	Turbidity	NTU	1	5	1 – 61.8	0.01 – 33.7
13	Fluoride	mg/L	1.0	1.5	0.31 – 1.78	0.40 – 1.24
14	Sulphate	mg/L	200	400	91 – 952	25 – 518
15	Nitrate	mg/L	45	No Relaxation	4 – 709	5 – 291
16	Ammonical Nitrogen	mg/L	-	-	0	0
17	Lead	mg/L	0.01	No Relaxation	0 – 0.023	0.002 – 0.018
18	Cadmium	mg/L	0.003	No Relaxation	0	0 – 0.002
19	Iron	mg/L	0.3	No Relaxation	0 – 1.90	0 – 0.57
20	Chromium	mg/L	0.05	No Relaxation	0 – 1.48	0 – 5.94
21	Zinc	mg/L	5	15	0	0
22	Arsenic	mg/L	0.01	0.05	0	0
23	Hexavalent Chromium	mg/L	-	-	-	-

Note: The limits are as per the **Indian Standards Drinking Water - Specification (Second Revision) (IS 10500: 2012)** [5]

Discussion

pH

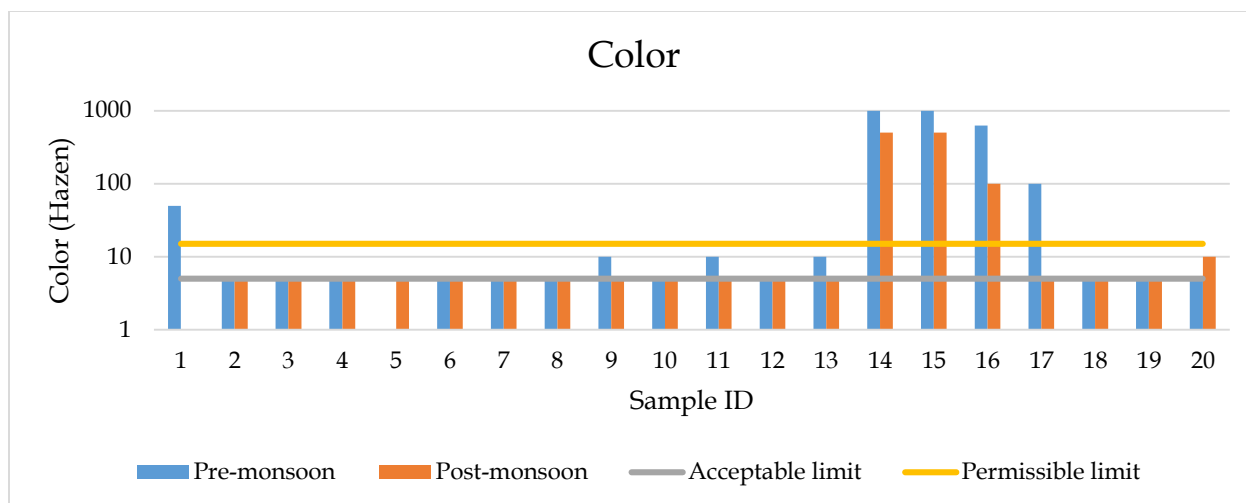
In common, water with a pH < 7 is considered acidic and with a pH > 7 is considered basic. The normal range for pH in surface water systems is 6.5 to 8.5 and for groundwater systems 6 to 8.5. The pH of pure water is 7 at 25° C. [23] In the present study, the pH ranged between 7.17 and 7.78 for pre-monsoon season whereas 7.18 and 8.12 for post-monsoon season. Each source analyzed was resulted within permissible limit of drinking water standards.



Graph 4.1 Seasonal variation of pH

Color (Hazen)

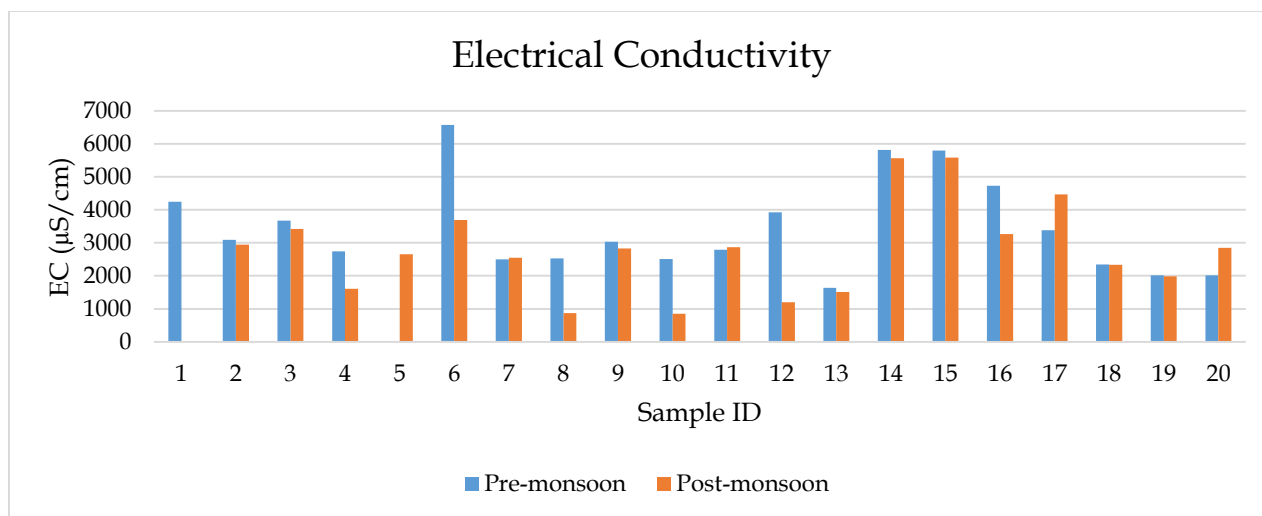
Suspended and dissolved particles in water influence color. Suspended material in water bodies may be a result of natural causes and/or human activity. Transparent water with a low accumulation of dissolved materials appears blue and indicates low productivity. [22] In the present study, color ranged between 5 Hazen and 1000 Hazen for pre-monsoon season whereas 5 Hazen and 500 Hazen for post-monsoon season. Sample IDs 1, 14, 15, 16, 17 and 14, 15, 16 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively, indicating intervention through pollution in groundwater.



Graph 4.2 Seasonal variation of Color

Electrical Conductivity ($\mu\text{S}/\text{cm}$)

Electrical conductivity is an indirect measurement of salinity, it is temperature dependent and good indicator of the total salinity, but it does not provide any information about the ionic composition within the water sample. ^[16] In the present study, EC ranged between 1634 $\mu\text{S}/\text{cm}$ and 6570 $\mu\text{S}/\text{cm}$ for pre-monsoon season, whereas 850 $\mu\text{S}/\text{cm}$ and 5580 $\mu\text{S}/\text{cm}$ for post-monsoon season. However, EC does not indicate pollution level in groundwater and also not a concerned parameter in drinking water standards.

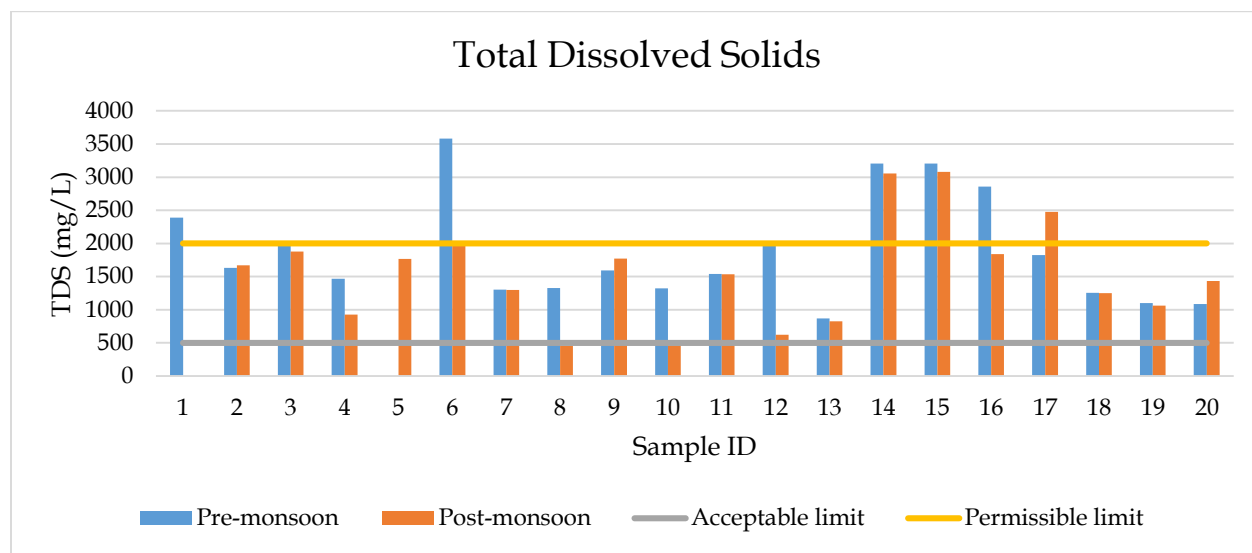


Graph 4.3 Seasonal variation of EC

Total Dissolved Solids (mg/L)

In general, the total dissolved solids (TDS) concentration is the sum of the cations (positively charged) and anions (negatively charged) ions in the water. Therefore, the

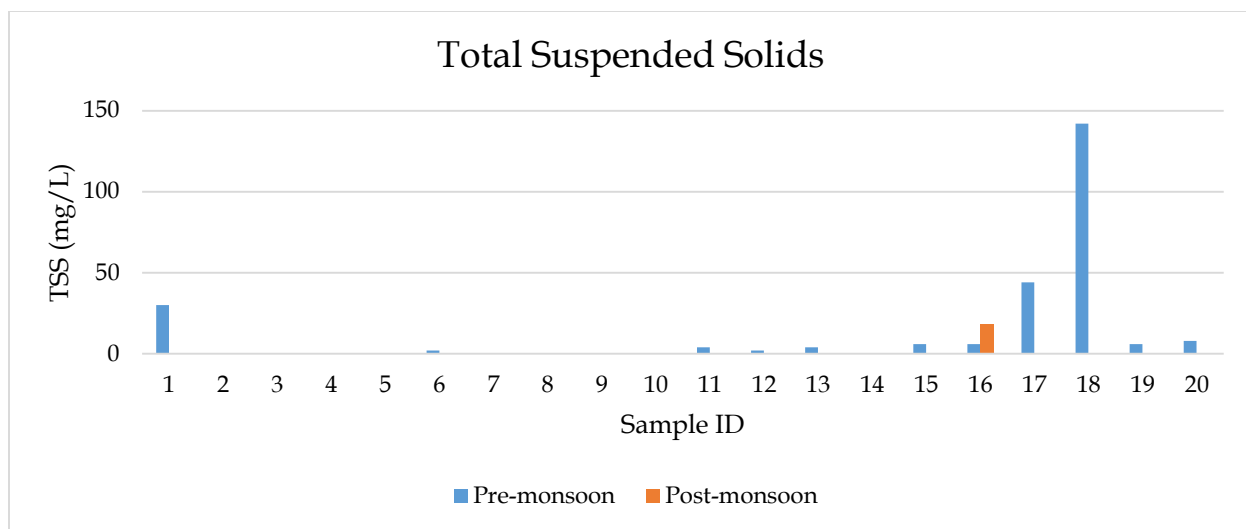
total dissolved solids test provides a qualitative measure of the amount of dissolved ions but does not tell us the nature or ion relationships. [24] In the present study, TDS ranged between 868 mg/L and 3584 mg/L for pre-monsoon season, whereas 452 mg/L and 3082 mg/L for post-monsoon season. Most of the samples were resulted within permissible limit of drinking water standards except for sample IDs 1, 6, 12, 14, 15, 16 and 6, 14, 15, 17; which were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 4.4 Seasonal variation of TDS

Total Suspended Solids (mg/L)

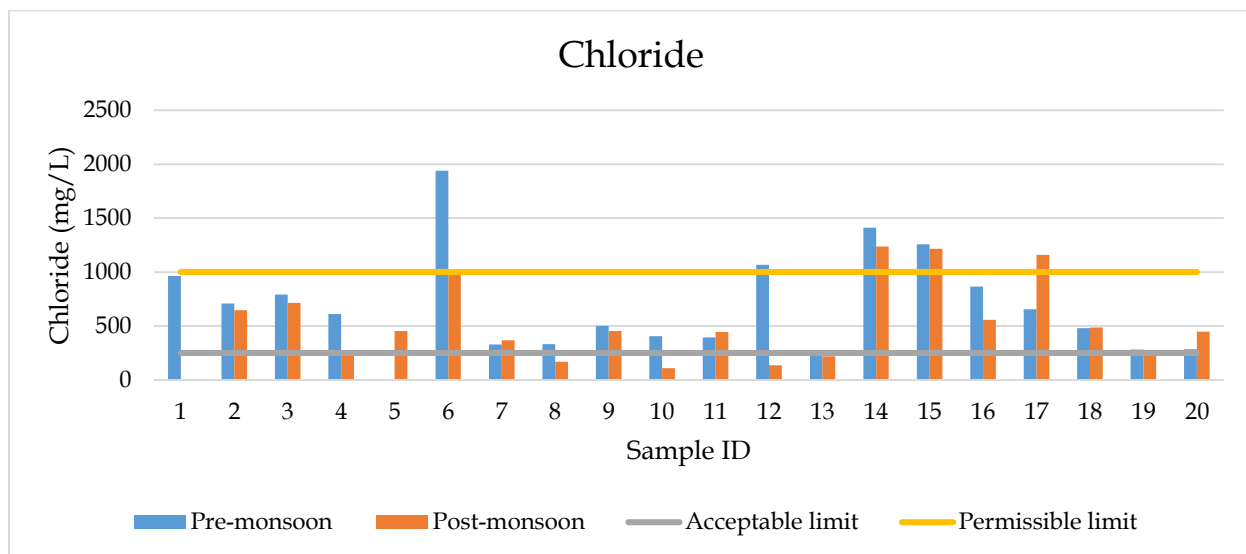
Total suspended solids (TSS) are particles that are larger than 2 microns found in the water column. Anything smaller than 2 microns (average filter size) is considered a dissolved solid. Most suspended solids are made up of inorganic materials, though bacteria and algae can also contribute to the total solids concentration. In the present study, TSS ranged between 0 and 142 mg/L for pre-monsoon season, whereas 0 and 18 mg/L for post-monsoon season. High variations were observed for this parameter, however due to presence of suspended solids, water needs to be filtered before using for drinking purpose.



Graph 4.5 Seasonal variation of TSS

Chloride (mg/L)

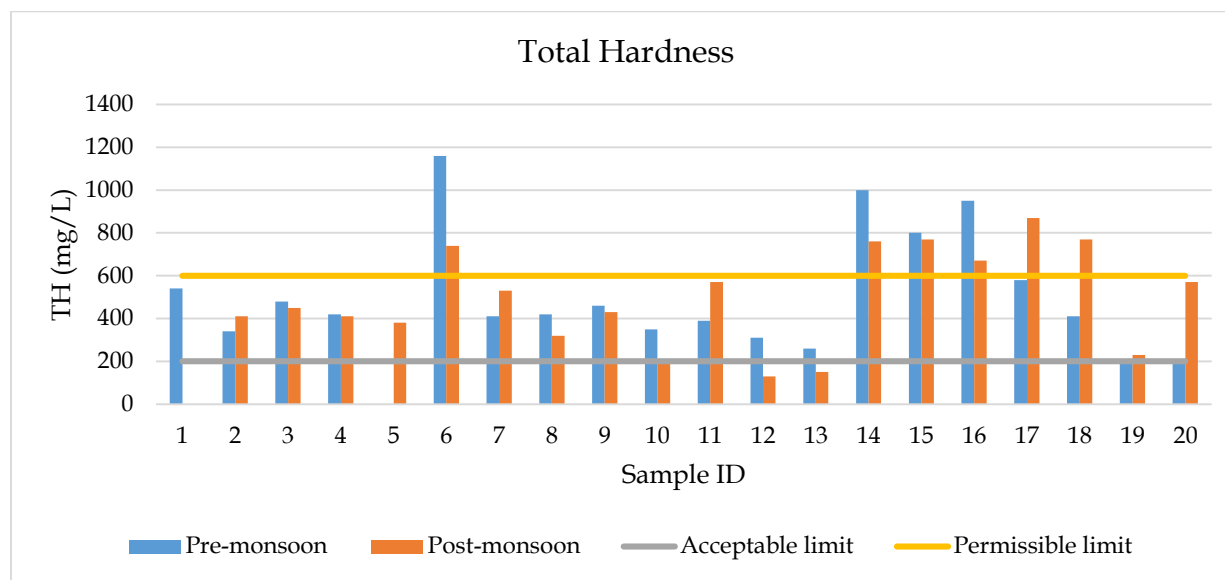
Chloride occurs in all natural water in widely varying concentration. The chloride content normally increases as the mineral content increases and it is also important in detecting the contamination of groundwater by sewage. ^[17] In the present study, Chloride ranged between 222 mg/L and 1938 mg/L for pre-monsoon season, whereas 108 mg/L and 1235 mg/L for post-monsoon season. Sample IDs 6, 12, 14, 15 and 6, 14, 15, 17 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively. Excess amount of Chloride results in increase of Total Dissolved Solids, which is a key parameter to be controlled if water is to be used for drinking purpose.



Graph 4.6 Seasonal variation of Chloride

Total Hardness (mg/L)

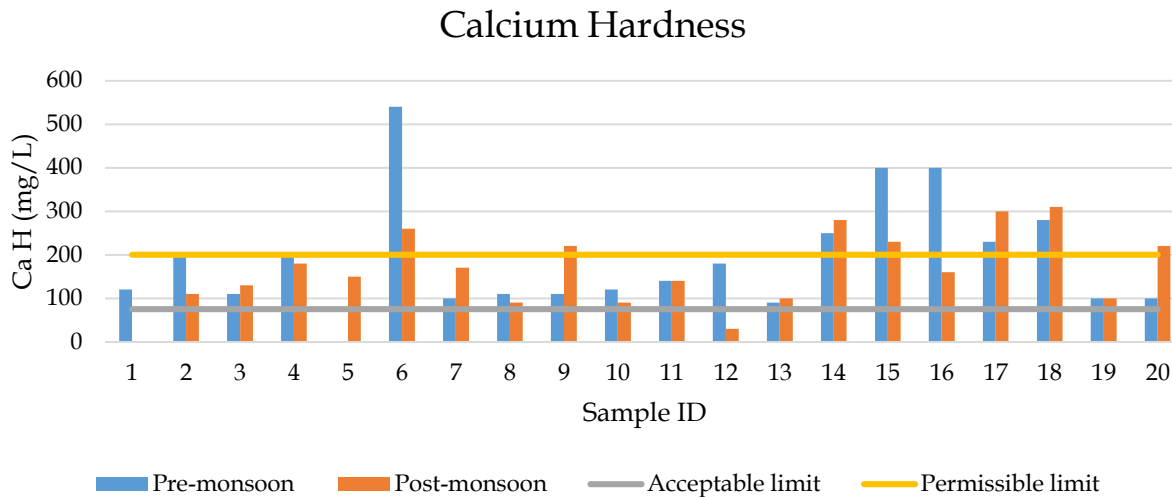
The optimum range of hardness in drinking water is from 80 mg/L to 100 mg/L. Water with hardness greater than 200 mg/L is considered hard and water with hardness greater than 500 mg/L is normally considered unacceptable for domestic purposes. [21] In the present study, TH ranged between 200 mg/L and 1160 mg/L for pre-monsoon season, whereas 130 mg/L and 870 mg/L for post-monsoon season. Sample IDs 6, 14, 15, 16 and 6, 14 to 18 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively. Scale buildup on plumbing fixtures & appliances, dry skin & hair, faded clothes, stained sinks, etc. are some adverse effects of hard water.



Graph 4.7 Seasonal variation of TH

Calcium Hardness (mg/L)

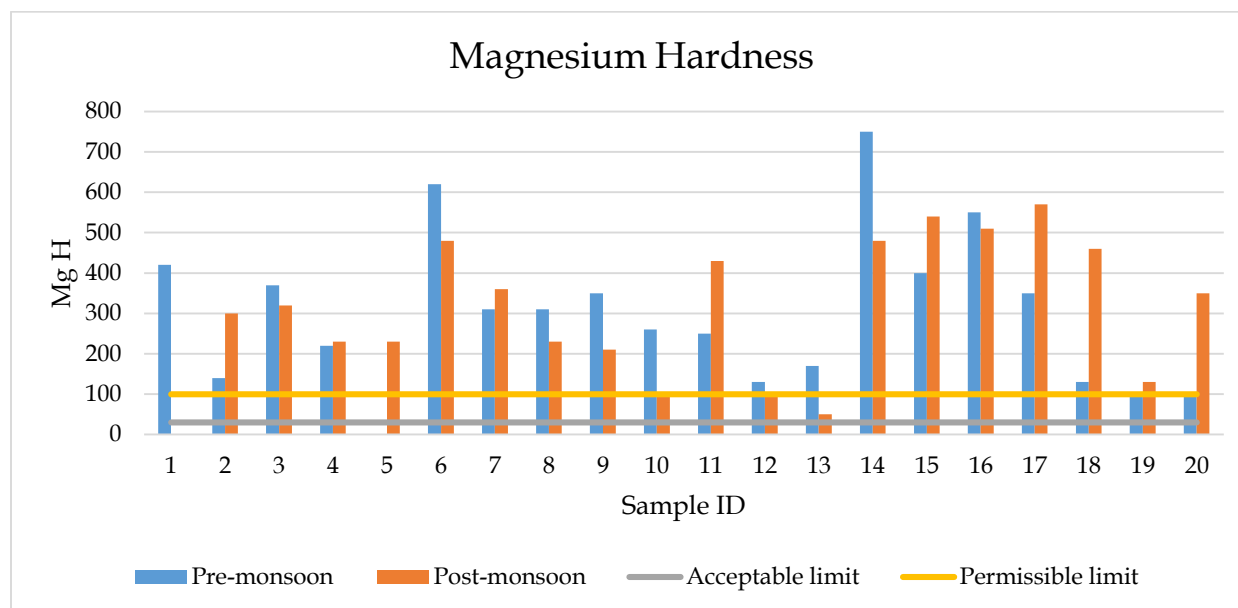
In the present study, Ca-H ranged between 90 mg/L and 540 mg/L for pre-monsoon season, whereas 30 mg/L and 310 mg/L for post-monsoon season. The lowest value found in Kapodara and Bhakodara for pre and post monsoon season, however the highest value found in Patel Nagar and Borbhata Bet for pre and post monsoon season respectively. Most of the samples were resulted below permissible limit of drinking water standards except for sample IDs 6, 14, 15, 16, 17, 18 and 6, 14, 15, 17, 18, 20; which were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 4.8 Seasonal variation of Ca H

Magnesium Hardness (mg/L)

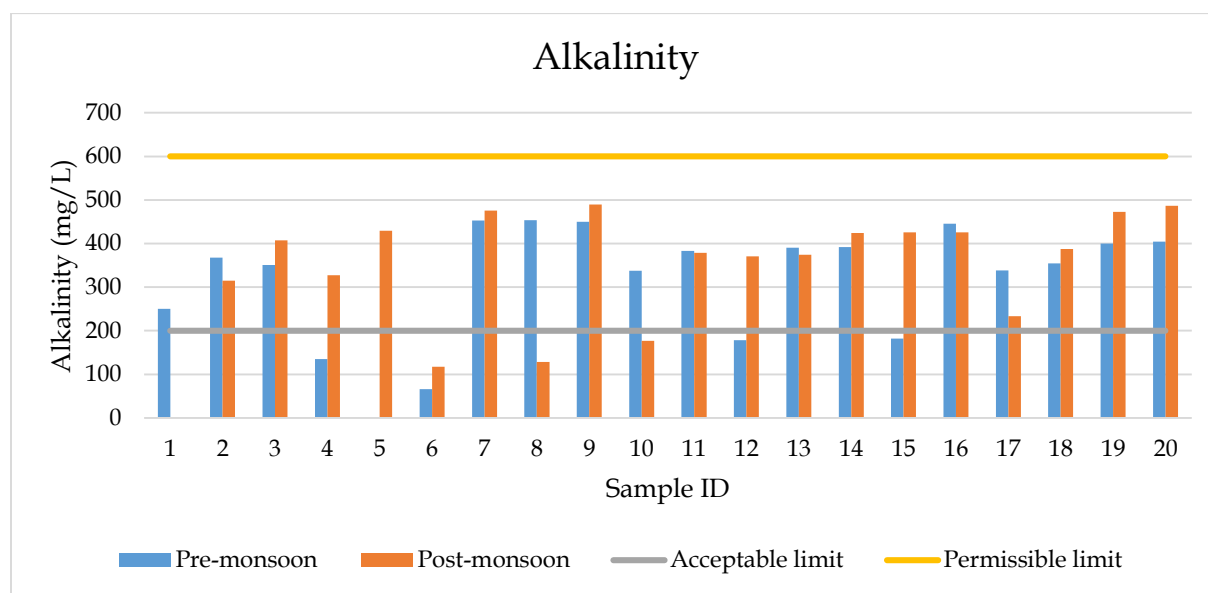
In the present study, Mg-H ranged between 100 mg/L and 750 mg/L for pre-monsoon season, whereas 50 mg/L and 570 mg/L for post-monsoon season. The lowest value found in both Chikuwadi & Azad Nagar and Kapodara for pre and post monsoon season, however the highest value found in Bakrol and Shubham Residency for pre and post monsoon season respectively. Almost all the samples were exceeding permissible limit of drinking water standards specifically sample IDs 1 to 4, 6 to 18 and 2 to 9, 11, 14 to 20; which were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 4.9 Seasonal variation of Mg H

Alkalinity (mg/L)

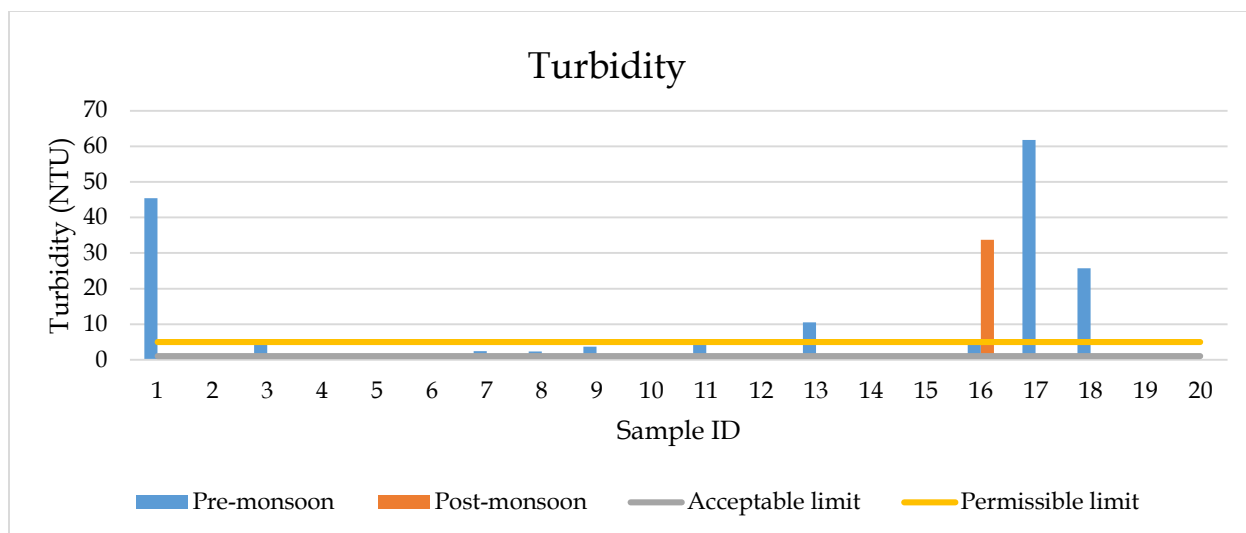
Alkalinity refers to the capacity of water to neutralize acid. In uncontaminated water, alkalinity is primarily a measure of dissolved bicarbonate and carbonate; concentrations of other acid - consuming solutes such as hydroxide, silicate, borate, and dissolved organic compounds are usually small compared to the amount of bicarbonate and carbonate. Alkalinity is reported as the equivalent amount of calcium carbonate in mg/L. [25] In the present study, Alkalinity ranged between 66 mg/L and 454 mg/L for pre-monsoon season, whereas 117 mg/L and 489 mg/L for post-monsoon season. All the samples analyzed were resulted within permissible limit and hence there is no trace of alkalinity pollution found in water.



Graph 4.10 Seasonal variation of Alkalinity

Turbidity (NTU)

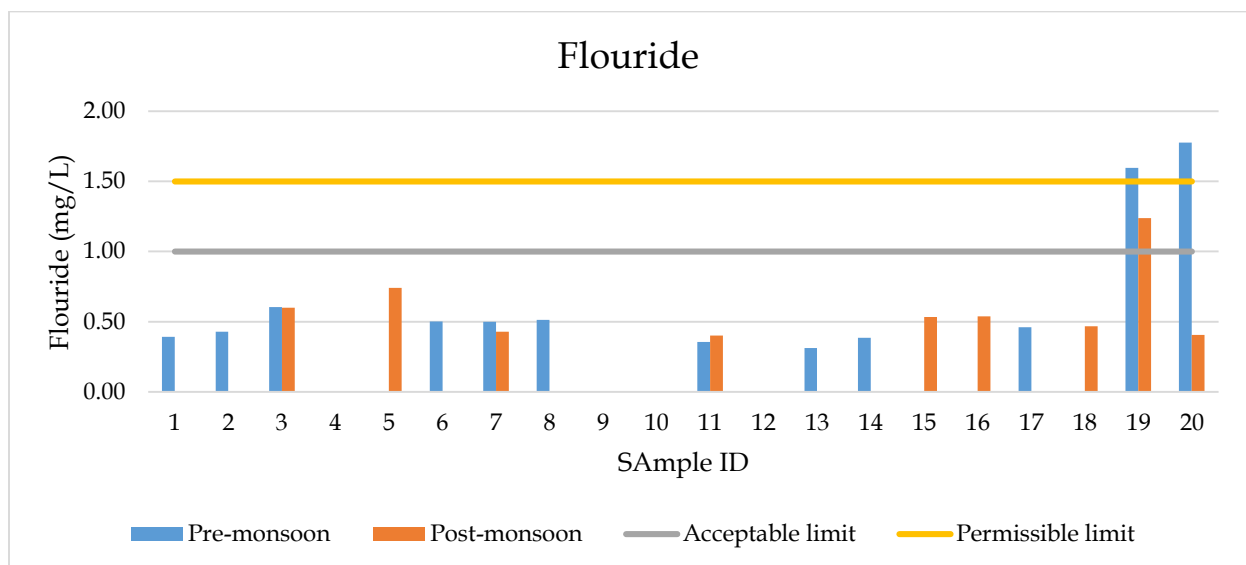
Turbidity ranged between 1 NTU and 61.8 NTU for pre-monsoon season, whereas 0.01 NTU and 33.7 NTU for post-monsoon season. Most of the samples analyzed showed turbidity level within permissible limit for drinking water standards except sample IDs 1, 3, 11, 13, 16, 17, 18 and 16; which showed extreme level of turbidity for pre-monsoon and post-monsoon respectively.



Graph 4.11 Seasonal variation of Turbidity

Fluoride (mg/L)

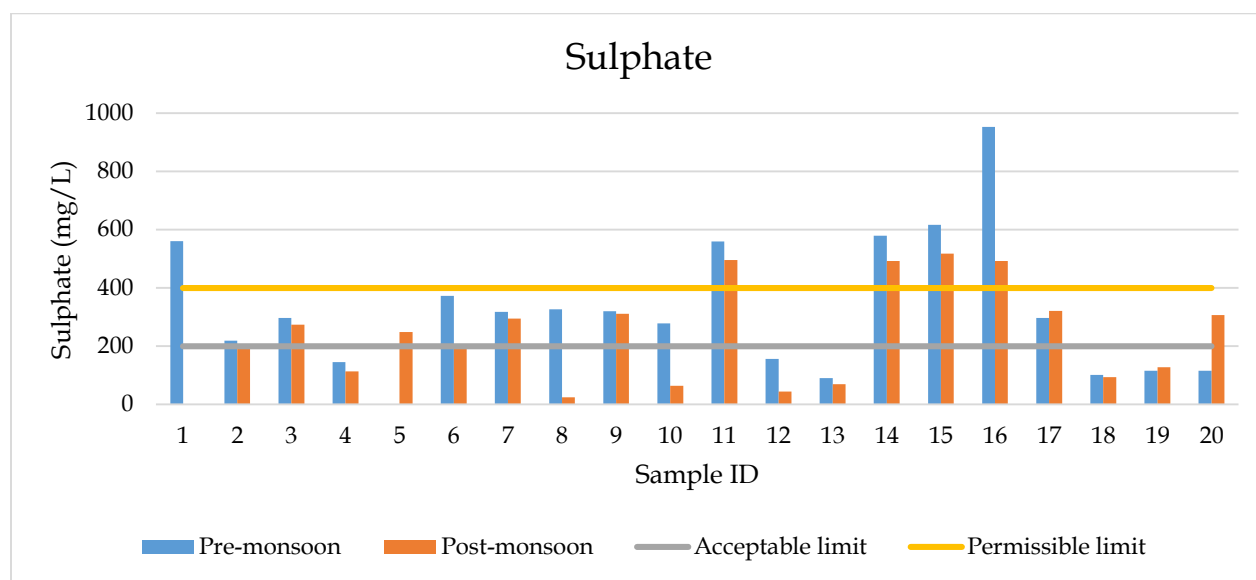
Fluorine is the lightest member of the halogen group and is one of the most reactive of all chemical elements. In groundwater, low or high concentrations of fluoride can occur, depending on the nature of the rocks and the occurrence of fluoride-bearing minerals. Fluoride is necessary in humans in minute quantities during tooth formation, to harden the tooth enamel. In the present study, Fluoride ranged between 0.31 mg/L and 1.78 mg/L for pre-monsoon season, whereas 0.40 mg/L and 1.24 mg/L for post-monsoon season. Sample IDs 19 and 20 were exceeding permissible limit for pre-monsoon season, however no sample resulted above permissible limit for post-monsoon season, which shows almost all the analyzed sources are free from Fluoride pollution.



Graph 4.12 Seasonal variation of Fluoride

Sulphate (mg/L)

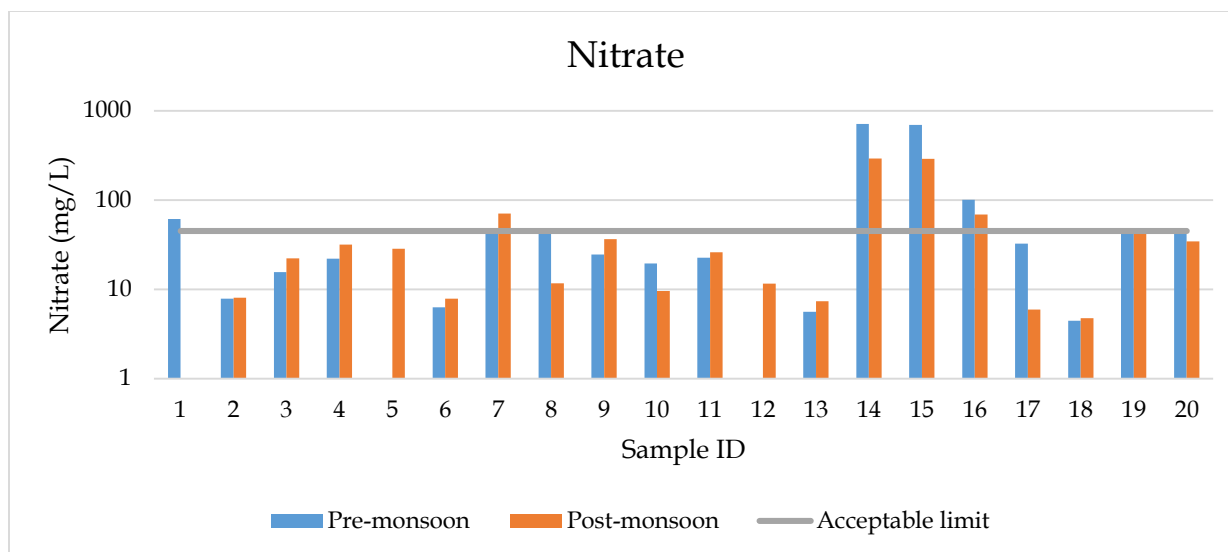
Sulphate concentration may be caused by the oxidation process of their ores, they are also present in industrial wastes. The sulphate ion is one of the major anions occurring in natural water. In the present study, Sulphate ranged between 90.7 mg/L and 952 mg/L for pre-monsoon season, whereas 25 mg/L and 518 mg/L for post-monsoon season. Concentration of Sulphate in most of the analyzed samples was found within the permissible limit except for few samples. Sample IDs 1, 11, 14, 15, 16 and 11, 14, 15, 16 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 4.13 Seasonal variation of Sulphate

Nitrate (mg/L)

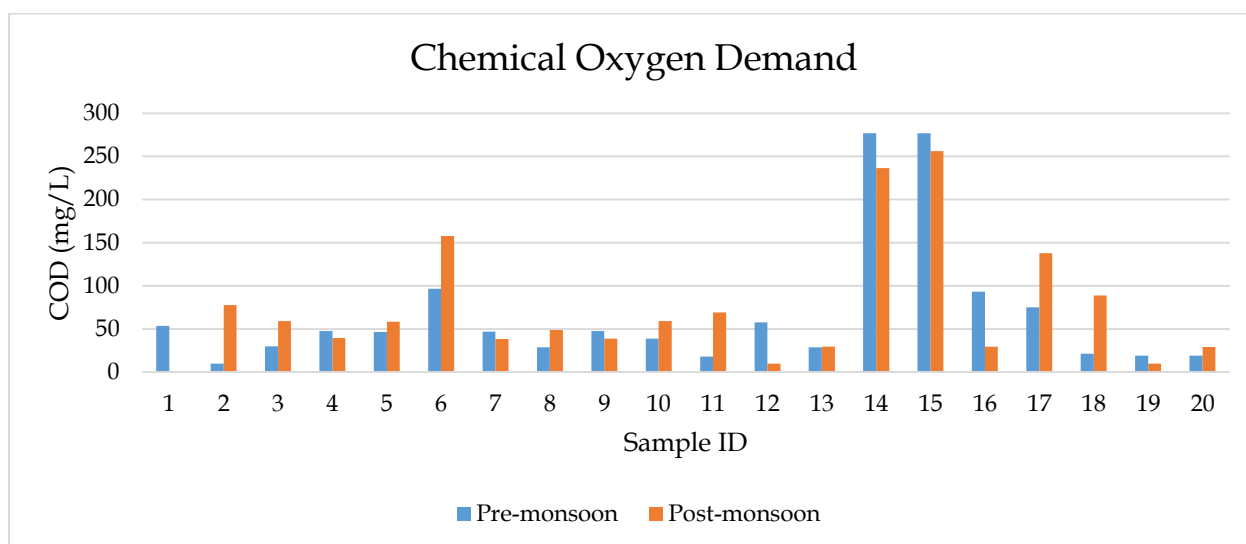
Relatively little of the nitrate found in natural waters is of mineral origin, most coming from organic and inorganic sources, the former including waste discharges and the latter comprising chiefly artificial fertilizers. However, bacterial oxidation and fixing of nitrogen by plants can both produce nitrate. Groundwater contamination can directly affect human health because excessive levels of nitrate in drinking water can produce negative health impacts on human well-being. In the present study, Nitrate ranged between 4.45 mg/L and 709 mg/L for pre-monsoon season, whereas 4.73 mg/L and 291 mg/L for post-monsoon season. Few samples were exceeding permissible limit of drinking water standards while remaining samples were found within the limits. Sample IDs 1, 7, 8, 14, 15, 16 and 7, 14, 15, 16, 19 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 4.14 Seasonal variation of Nitrate

Chemical Oxygen Demand (mg/L)

COD is defined as the amount of oxygen equivalents consumed in oxidizing the organic compounds of samples by strong oxidizing agents such as dichromate or permanganate. It is expressed in milligrams per liter (mg/L) that indicates the mass of oxygen consumed per liter of solution. The higher the chemical oxygen demand, the higher the amount of pollution in the water sample. ^[20] In the present study, COD ranged between 10 mg/L and 277 mg/L for pre-monsoon season, whereas 10 mg/L and 256 mg/L for post-monsoon season. Excessive amount of COD present indicates interference in the quality of groundwater and presence of chemical pollution.



Graph 4.15 Seasonal variation of COD

Ammonical Nitrogen

It is also commonly found in groundwater due to anthropogenic activities, primarily due to leaching from fertilizers, organic waste disposal or leaking sewage systems. ^[18] $\text{NH}_3\text{-N}$ in groundwater aquifer reacts through complex physical- chemical and biological processes to form other associated nitrogen bearing compounds such as NO_2 , NO_3 which makes the risk even more complicated. Their toxicity is attributed to the influence of pH, temperature, and dissolved oxygen concentration. ^[19] In the present study, Ammonical Nitrogen was analyzed for all the collected samples for both pre-monsoon and post-monsoon season. However, none of the sample exceeded concentration of 1 mg/L for both pre-monsoon and post-monsoon season.

Heavy Metals

Heavy Metals Zn, Cd, As, Pb, Cr and Fe were analyzed in a few samples collected during sampling for both pre and post monsoon seasons. The parameter wise inference drawn is covered in table 4.6 along with the details of sample IDs exceeding the permissible limit for respective seasons. Heavy metals in general were analyzed for 6 samples and 11 samples for pre and post monsoon season respectively.

Table 4.6 Details of Heavy Metals analyzed

Sample IDs analyzed for Heavy Metals	
Pre-monsoon	1, 9, 14, 15, 16, 17
Post-monsoon	4, 5, 6, 7, 9, 10, 12, 14, 15, 16, 17
Parameters	Inferences drawn
Zinc	Each sample analyzed showed concentration below 0.5 mg/L and hence no trace of zinc pollution is evident.
Cadmium	Cadmium with acceptable limit of 0.003 mg/L was not detected in any of samples collected for both pre and post monsoon seasons except for sample ID 12, which showed 0.002 mg/L for post monsoon sample.
Arsenic	Arsenic was not found in any of samples analyzed for both pre and post monsoon seasons and hence monitored source are free from arsenic pollution.
Chromium	Chromium contamination was recorded in three samples for sample ID 17 with 1.48 mg/L collected for pre-monsoon season and sample IDs 16 and 17 with concentration of 5.94 and 0.19 mg/L collected in post-monsoon season.
Lead	Two samples with sample ID 17 with 0.023 mg/L concentration of pre-monsoon and sample ID 12 with 0.018 mg/L concentration of post-monsoon exceeded the permissible limit of 0.01 mg/L.
Iron	Three samples with ID no. 1, 16 and 17 exceeded permissible limit of 0.3 mg/L for pre-monsoon season with concentration recorded as 1.6 mg/L, 0.36 mg/L and 1.90 mg/L. Whereas, only one sample

	with ID no. 16 exceeded permissible limit for post monsoon season with concentration of 0.57 mg/L.
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4.4 Conclusion

Groundwater sampling was carried out around the industrial cluster of Ankleshwar wherein; 20 samples were collected from 20 selected locations for both pre and post monsoon seasons. From the analytical results and monitoring carried out following conclusions are drawn:

- The overall groundwater quality around industrial cluster is found to be good for most of the samples collected expect a few with sample IDs and location as 6 (Bore well of Patel nagar near masjid), 14 (Bore well of Bakrol near Mahakali temple) 15, (Bore well of Bakrol near Ramnagar English school), 16 (Bore well of Koba park near umiyadham) and 17 (Bore well of shubham residency) whose quality is found to be inferior with respect to all physico-chemical parameters analyzed for the samples including heavy metal contamination in terms of Iron, Lead, Chromium contamination raising a serious concern on application and use of those particular sources. Chemical Oxygen Demand analyzed in the samples reflects presence of pollution in groundwater ranging from smaller to greater extent.
- In respect to information collected about sources during monitoring it is known that none of the objectionable sources are used for drinking purposes but however their application recorded is domestic purposes which poses substantial possibility of affecting the population ranging from 40 to 400 in number for each sources.
- Thus the overall groundwater quality of the monitored cluster is found to be fairly good expect for few samples.

Chapter: 5 Vapi GIDC

5.1 Overview of Vapi Industrial Cluster

Vapi is an industrial base for Chemicals, Textiles and Paper & Paper Pulp industry sectors located in Valsad. The district has also emerged as Horticulture hub of the State having significant food grain and crops production during the last decade. Over more than 300 Medium and Large Scale industries largely concentrated in Vapi thrust the economic growth of the district. One of the Asia's largest Common Effluent Treatment Plant (CETP) is located in Vapi owned by Vapi Waste & Effluent Management Company and promoted by Vapi Industrial Association provides healthy facilities to protect the environment. [9]

Table 5.1: Valsad district at a glance

S. No.	Particulars	Information
1.	Geographical information	20.07 to 21.05 N and 72.73 to 73.00 E
2.	Geographical area (sq. km)	3055
3.	Average annual rainfall	2223 mm
4.	Average temperature	36.6 °C (Summer), 17.8 °C (Winter)
5.	Population	17,03,068
6.	Literacy (Vapi)	89%
7.	Stage of Ground Water Development (Pardi tehsil)	53.12 % - Safe Category

(Source: Groundwater Brochure Valsad District [8])

Products Manufactured

Various industries in Vapi manufacture diverse range of products like dyes and pigments, pharmaceuticals, paints, pipes & fittings, automobiles parts, chemicals, inks, textiles, pulp & paper and so on. Some of the major categories of chemicals is shown in table:

Table 5.2: Chemical products manufactured in Vapi GIDC

Category	Material produced
Pipes & fittings	Polymer Pipes & Fittings, HDPE Pipes & Fittings, PP Pipes & Fittings, Polymer Profile, Polymer Welding & Solution Rods, MIPS Pipes & Fittings, PVDE Pipes & Fittings, Polymer Sheets & Profiles Roto Mould Marine Items, HDPE Sheets & Profiles, HIPS, ABS\PVDF Sheets, Roto Mould PP\HDPE\UDBE Marine Items & Profiles etc.
Dyes and Pigments	Pigment Intermediates
Pharmaceutical	Allopathic and other Pharmaceutical Preparation
Automobile	Forging of Automobile Parts, Connecting Rods, Connecting Rod Caps, STRG Arm and STRG Knuckle etc.

Chemicals	Organic Chemicals, chloroaniline, calcium phosphate, sulphuric acid, single super phosphate, lubricating chemicals
Inks	Printing inks, synthetic resins, prepared glues and adhesives, colour pigments
Textile	Weaving and processing of artificial and synthetic textile fibres, Cotton yarns
Pulp & Paper	Uncoated craft paper, newsprint and poster paper

(Source: Brief Industrial Profile of Valsad District [9])

5.2 Sampling and monitoring

For the purpose of study sampling sites were selected within 5 km buffer region across the GIDC area. The details of the sampling sites selected is described in table 5.3 and figure 5.1.

Table 5.3 Sampling locations around Vapi GIDC

Villages/locations selected for Sampling & Monitoring		
Within 1 km (12 samples)	Within 3 km (4 samples)	Within 5 km (4 samples)
Imran Nagar, Chanod (2), Godal Nagar, Sulpad, Candon Nagar, Desai Wadi, Karvad, Kanchan Nagar (2), Navi Nagari, Chharwada	Navdha, Shanthi Nagar, Valwada, Rajnigandha Appartment	Morai, Karambeli, Achchhari, Tukwada

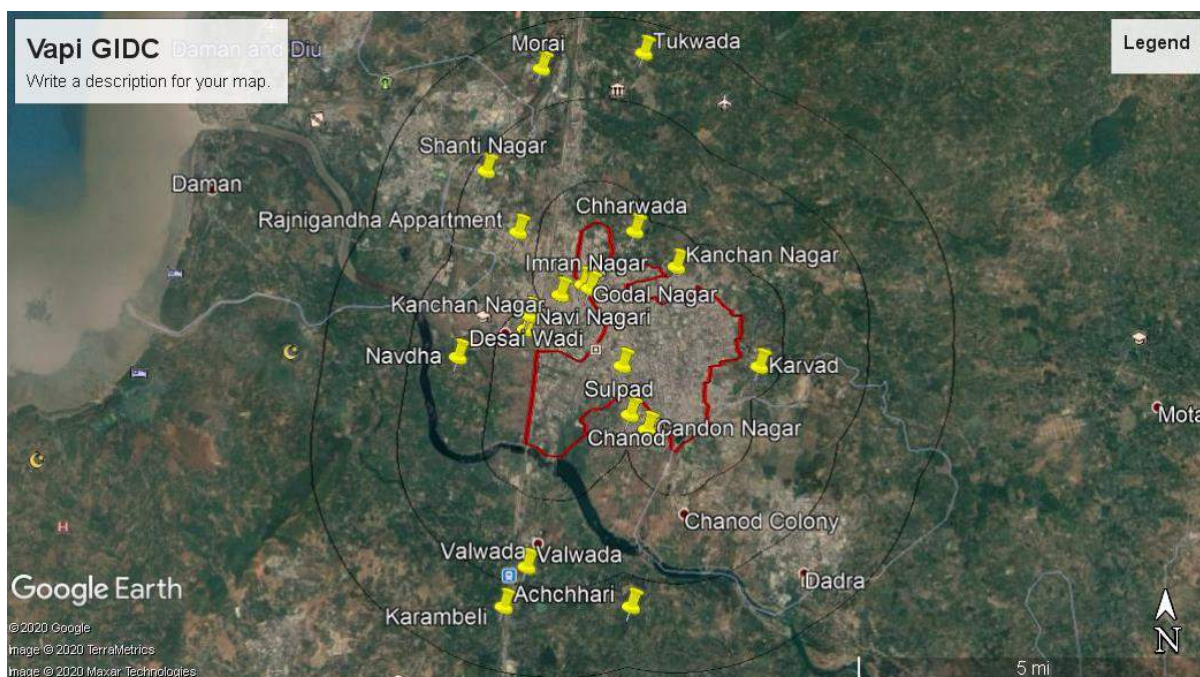


Figure 5.1 Sampling locations around Vapi GIDC

Samples were collected from the identified location for both pre and post monsoon seasons. The information about sampling location, source type, color appearance at field, depth of source, population dependent on source, water usage and name of person contacted during the time of sampling is summarized in table 5.4.

Table 5.4 Details collected during sampling around Vapi GIDC

S. ID	Village/ Site	Location	Latitude Longitude	Source type	Color Appearance @ Field	Depth of well	Population dependent on source	Water Usage	Persons Contacted	Remarks
1.	Desai Wadi	Near Geetanagar	20.36727 72.90453	Hand pump	Clear	100	40	Domestic	Nagesh bhai	-
2.	Imran Nagar	Near Gayatri temple	20.37305 72.91803	Bore well	Clear	150	20	Domestic	Kalu bhai	-
3.	Godal Nagar	Near Hedsi Manjil	20.37385 72.91653	Bore well	Clear	200	150	Domestic	Yusuf bhai	-
4.	Chanod	Shivaji Nagar	20.34639 72.9271	Bore well	Clear	300-350	300-400	Domestic	Dinesh bhai Vagher	-
5.	Chanod	Shreenathji Park, Nr. Pioneer bakery road	20.34866 72.93254	Bore well	Clear	150-170	160-170	Domestic	Ramesh bhai	Particles settle down & colour changes after some time
6.	Kanchan Nagar	Near vapi east, Geetanagar, Kaproli	20.3716 72.91137	Bore well	Clear	200	50	Domestic	-	-
*	Kanchan Nagar	Near Vapi east, Geetanagar, Kaproli	20.3716 72.91137	Bore well	Clear	200	50	Domestic	-	Previous source is closed, hence sample taken from nearby source
7.	Shanti Nagar	Near Post Office	20.398257 72.894297	Bore well	Clear	200	35	Domestic	-	-
*	Shanti Nagar	Near Post Office (Dungar Fadiya)	20.398257 72.894297	Bore well	Clear	120	35	Domestic	-	Previous source is closed, hence sample taken from nearby source

8.	Candon Nagar	Near Pay & Use	20.3433 72.93088	Hand pump	Clear	10	30-40	Domestic	Pinky ben	-
9.	Sulpad	Near Vegetable Market	20.3566 72.92529	Bore well	Clear	200	25-30	Domestic	Gansyam bhai	-
*	Sulpad	Near Vegetable Market	20.355358 72.924318	Bore well	Clear	50 - 80	25-30	Domestic	-	Previous source is closed, hence sample taken from nearby source
10.	Rajnigandha Appartment	Near Parvarish Children Hospital	20.38489 72.90158	Bore well	Clear	100	100-150	Drinking Domestic	-	Use in Drinking after filtration
11.	Valwada	Near Sahiyar Emporium	20.315016 72.904165	Open well	Clear	30	30-40	Drinking Domestic	Mahesh bhai	Water level rise 10 ft. after monsoon season
12.	Achchhari	Between shankar temple & post office, valwada village road	20.30682 72.927255	Open well	Clear	35	30-40	Drinking Domestic	Sunil bhai	Chlorine is added once in a month. It was added 2-4 days before the sampling date.
13.	Karambeli	Near Railway signal	20.30685 72.89907	Hand pump	Clear	80	10	Domestic	Pankaj bhai	Often water appears turbid.
14.	Navi Nagari	Near Dashamata temple	20.36434 72.90368	Bore well	Clear	100		Domestic	Ajay bhai	Open sewer line is nearby
15.	Navdha	Near Navdha Govt. School	20.3585 72.888308	Bore well	Clear	170		Domestic	Soma bhai	-
16.	Morai	Near Anganwadi	20.420795 72.90671	Bore well	Clear	80	300	Drinking Domestic	Rupesh bhai	-
17.	Tukwada	Near Avadh Utopia	20.424108 72.930405	Hand pump	Clear	80	15	Drinking Domestic	Bhupesh bhai	-
18.	Kanchan Nagar	Near old govt. school	20.377589 72.937671	Bore well	Clear	100-150	70-80	Domestic	-	Open sewer line is nearby

19.	Chharwada	Near Pruthvi Appartment	20.385309 72.928258	Hand pump	Clear	50	-	Domestic	Subhash bhai Patel	-
20.	Karvad	Near Bhavanimata temple	20.356399 72.956165	Hand pump	Yellowish	120	15	Domestic	-	-
*	Karvad	Near Bhavanimata temple	20.356439 72.951421	Hand pump	Clear	130	15	Domestic	-	Previous source is closed, hence sample taken from new source

Note:

1. * different sample source has taken for post-monsoon season as previously monitored source was found sealed (not in use condition) due to contamination.
2. Color code given to sample ID



Color Code is given to samples collected within 1 km buffer



Color Code is given to samples collected within 3 km buffer



Color Code is given to samples collected within 5 km buffer



5.3 Result and Discussion

The physico-chemical parameters of groundwater samples with their acceptable and permissible limits as recommended by Indian Standards Drinking Water - Specification (Second Revision) (IS 10500: 2012) are shown in below table:

Table 5.5 Result range of pre-monsoon and post-monsoon sampling

Sr. No.	Parameters	Measured Units	Requirement (Acceptable Limit)	Permissible limit in the absence of alternate source	Range of present study (Pre-monsoon)	Range of present study (Post-monsoon)
1	pH	-	6.5 – 8.5	No Relaxation	7.03 – 7.58	6.65 – 7.67
2	Colour	Hazen	5	15	5 - 50	5
3	Electrical Conductivity	µS/cm	-	-	528 – 3530	253 – 2020
4	TDS	mg/L	500	2000	274 – 2074	132 – 1168
5	TSS	mg/L	-	-	2 – 66	2 – 10
6	COD	mg/L	-	-	2 – 167	4 - 20
7	Chloride as Cl ⁻	mg/L	250	1000	24 – 866	38 – 427
8	Total Hardness	mg/L	200	600	150 – 800	110 – 630
9	Ca ⁺⁺ Hardness	mg/L	75	200	100 – 660	50 – 400
10	Mg ⁺⁺ Hardness	mg/L	30	100	30 – 420	50 – 230
11	Alkalinity	mg/L	200	600	177 – 451	86 – 619
12	Turbidity	NTU	1	5	0.1 – 52.4	0.1 – 15.7
13	Fluoride	mg/L	1.0	1.5	0.40 – 0.86	0.44 – 1.64
14	Sulphate	mg/L	200	400	5 – 322	5 – 147
15	Nitrate	mg/L	45	No Relaxation	1 – 82	2 – 53
16	Ammonical Nitrogen	mg/L	-	-	0	0
17	Lead	mg/L	0.01	No Relaxation	0	0.003 – 0.009

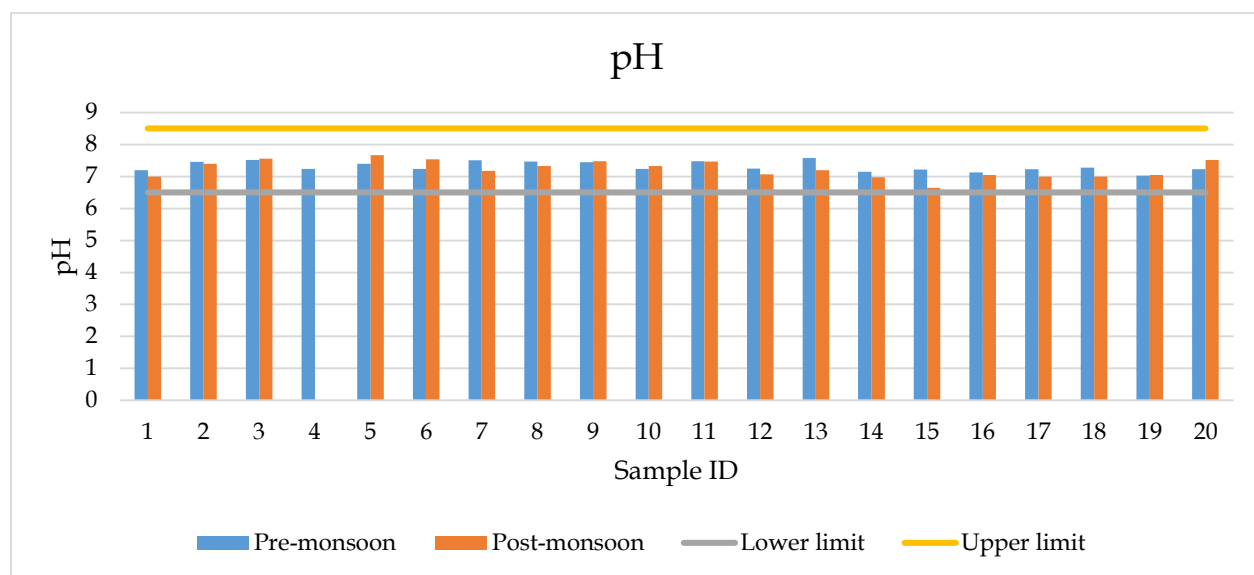
18	Cadmium	mg/L	0.003	No Relaxation	0	0
19	Iron	mg/L	0.3	No Relaxation	0 – 0.0001	0 – 0.253
20	Chromium	mg/L	0.05	No Relaxation	0 – 0.007	0 – 0.006
21	Zinc	mg/L	5	15	0	0
22	Arsenic	mg/L	0.01	0.05	0	0

Note: The limits are as per the **Indian Standards Drinking Water - Specification (Second Revision) (IS 10500: 2012)** ^[5]

Discussion

pH

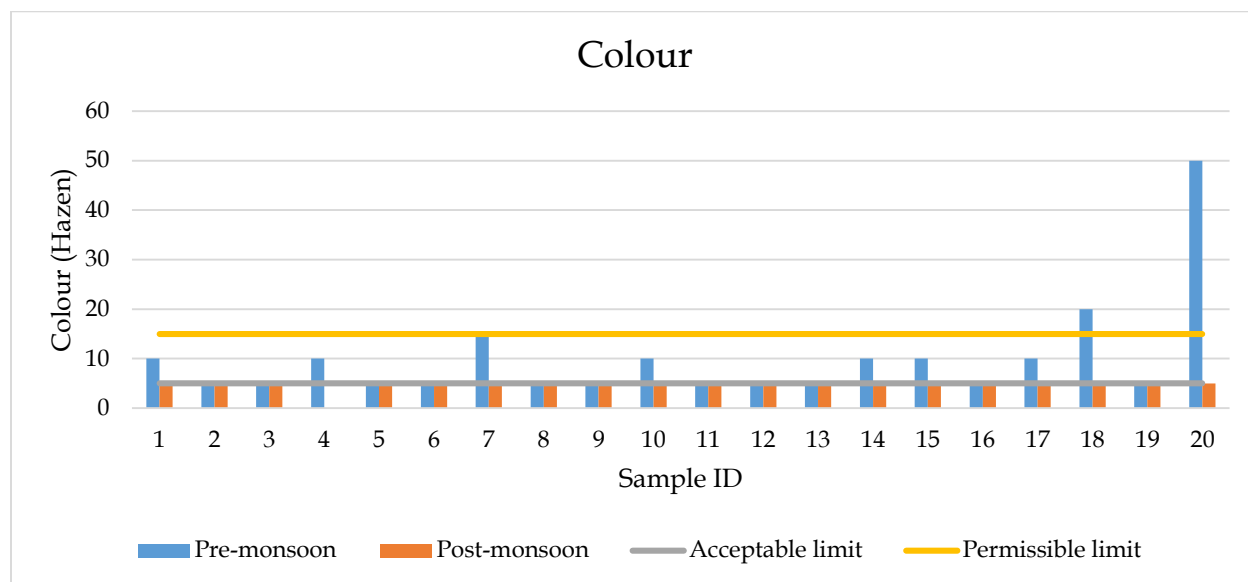
In common, water with a pH < 7 is considered acidic and with a pH > 7 is considered basic. The normal range for pH in surface water systems is 6.5 to 8.5 and for groundwater systems 6 to 8.5. The pH of pure water is 7 at 25° C. ^[23] In the present study, the pH ranged between 7.03 and 7.58 for pre-monsoon season whereas 6.65 and 7.67 for post-monsoon season. All the analyzed samples resulted within permissible limit of drinking water standards.



Graph 5.1 Seasonal variation of pH

Color (Hazen)

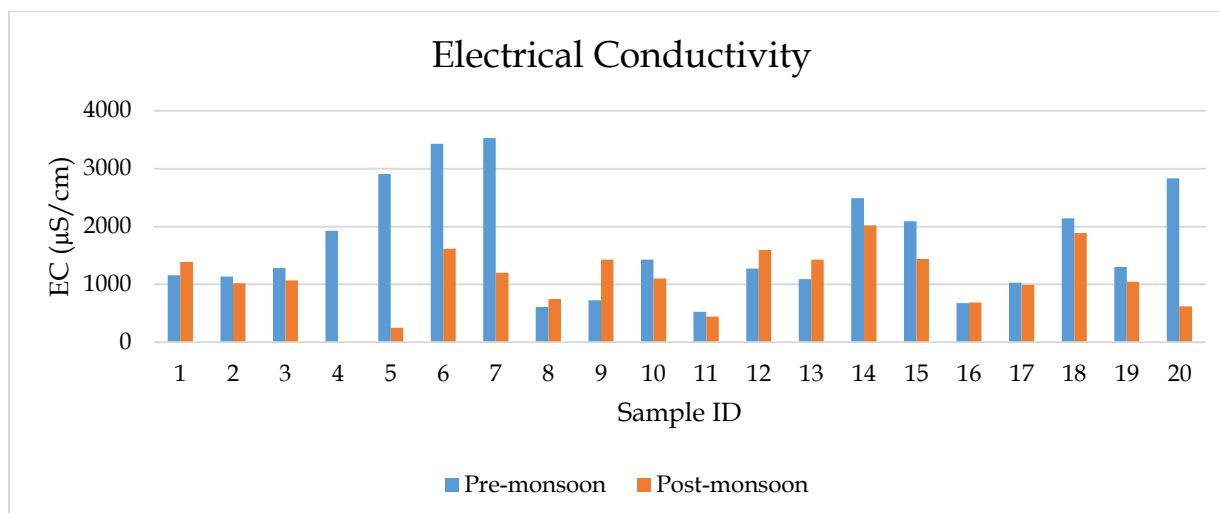
Suspended and dissolved particles in water influence color. Suspended material in water bodies may be a result of natural causes and/or human activity. Transparent water with a low accumulation of dissolved materials appears blue and indicates low productivity. [22] In the present study, color ranged between 5 Hazen and 50 Hazen for pre-monsoon season whereas every sample showed 5 Hazen for post-monsoon season. Most of the analyzed samples were found within permissible limit of drinking water standards except for 2 samples. Sample IDs 18 and 20 were exceeding permissible limit for pre-monsoon, however no sample exceeded permissible limit for post-monsoon season.



Graph 5.2 Seasonal variation of Colour

Electrical Conductivity ($\mu\text{S}/\text{cm}$)

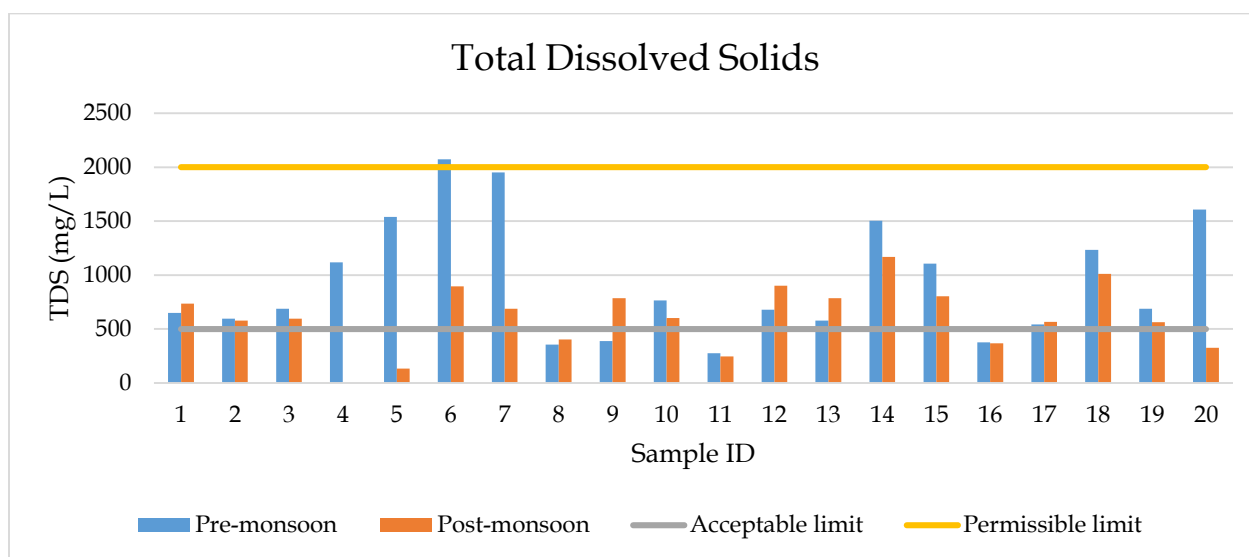
Electrical conductivity is an indirect measurement of salinity, and it is temperature dependent and good indicator of the total salinity, but it does not provide any information about the ionic composition within the water sample. [16] In the present study, EC ranged between 528 $\mu\text{S}/\text{cm}$ and 3530 $\mu\text{S}/\text{cm}$ for pre-monsoon season, whereas 253 $\mu\text{S}/\text{cm}$ and 2020 $\mu\text{S}/\text{cm}$ for post-monsoon season. High variations were observed in concentration of EC; however it does not indicate pollution of water.



Graph 5.3 Seasonal variation of EC

Total Dissolved Solids (mg/L)

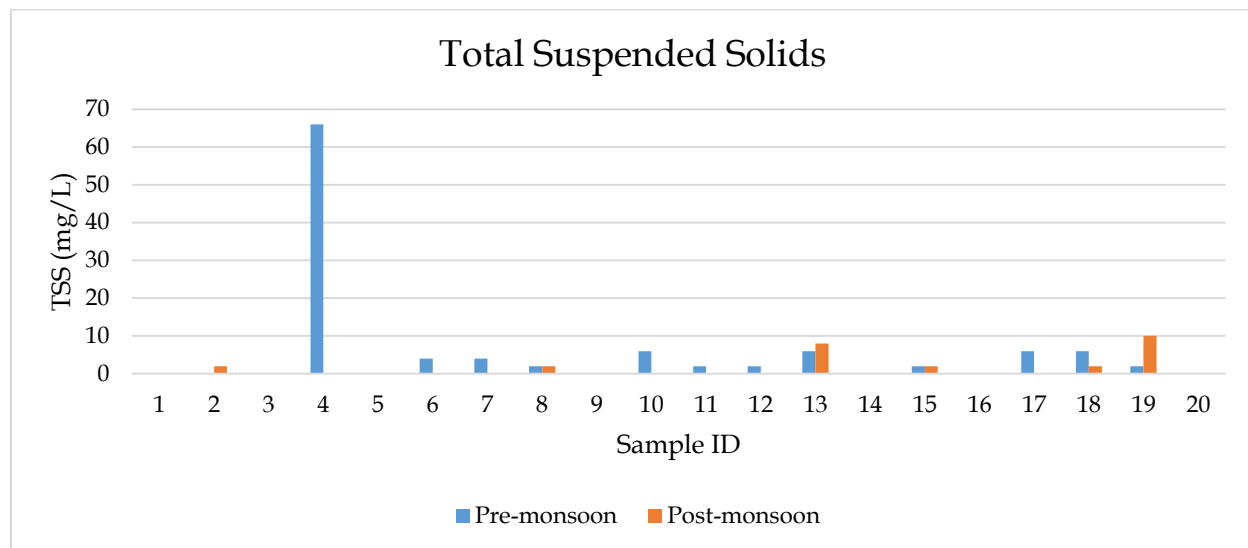
In general, the total dissolved solids concentration is the sum of the cations (positively charged) and anions (negatively charged) ions in the water. Therefore, the total dissolved solids test provides a qualitative measure of the amount of dissolved ions but does not tell us the nature or ion relationships. [24] In the present study, TDS ranged between 274 mg/L and 2074 mg/L for pre-monsoon season, whereas 132 mg/L and 1168 mg/L for post-monsoon season. Each sample analyzed was found within permissible limit except for sample ID 6; which was exceeding the limit for pre-monsoon, however no sample was exceeding permissible limit for post-monsoon season.



Graph 5.4 Seasonal variation of TDS

Total Suspended Solids (mg/L)

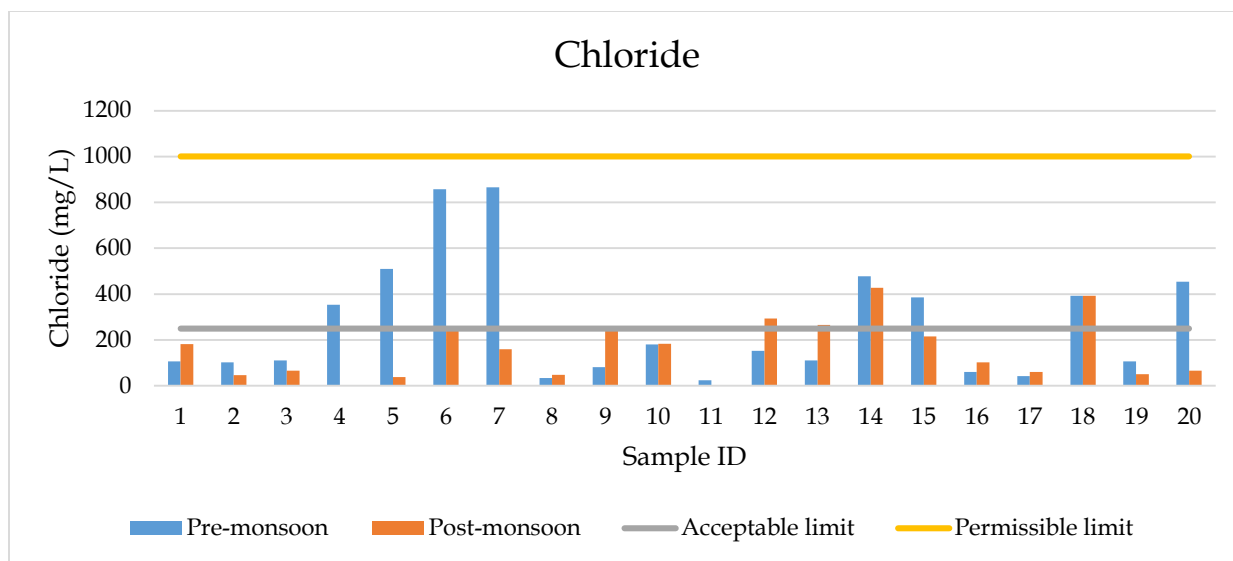
Total suspended solids (TSS) are particles that are larger than 2 microns found in the water column. Anything smaller than 2 microns (average filter size) is considered a dissolved solid. Most suspended solids are made up of inorganic materials, though bacteria and algae can also contribute to the total solids concentration. In the present study, TSS ranged between 0 and 66 mg/L for pre-monsoon season, whereas 0 and 10 mg/L for post-monsoon season. All the analyzed samples showed TSS concentration below 10 mg/L except for the sample ID 4 collected during pre-monsoon season.



Graph 5.5 Seasonal variation of TSS

Chloride (mg/L)

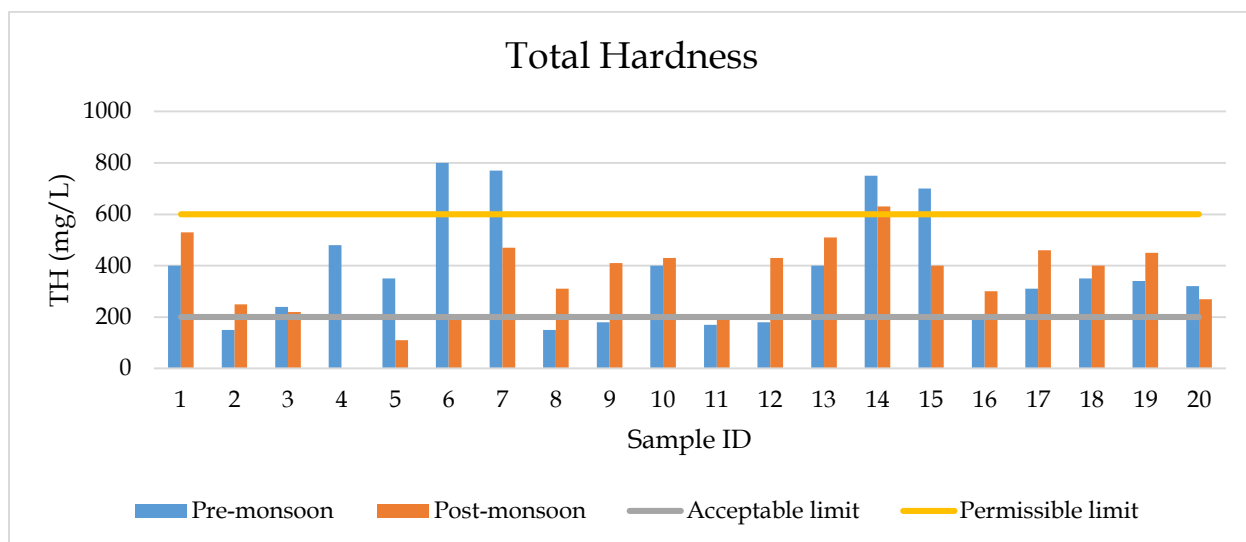
Chloride occurs in all natural water in widely varying concentration. The chloride content normally increases as the mineral content increases and it is also important in detecting the contamination of groundwater by sewage. ^[17] In the present study, Chloride ranged between 24 mg/L and 866 mg/L for pre-monsoon season, whereas 38 mg/L and 427 mg/L for post-monsoon season. All the analyzed samples showed Chloride concentration within permissible limit of drinking water standards.



Graph 5.6 Seasonal variation of Chloride

Total Hardness (mg/L)

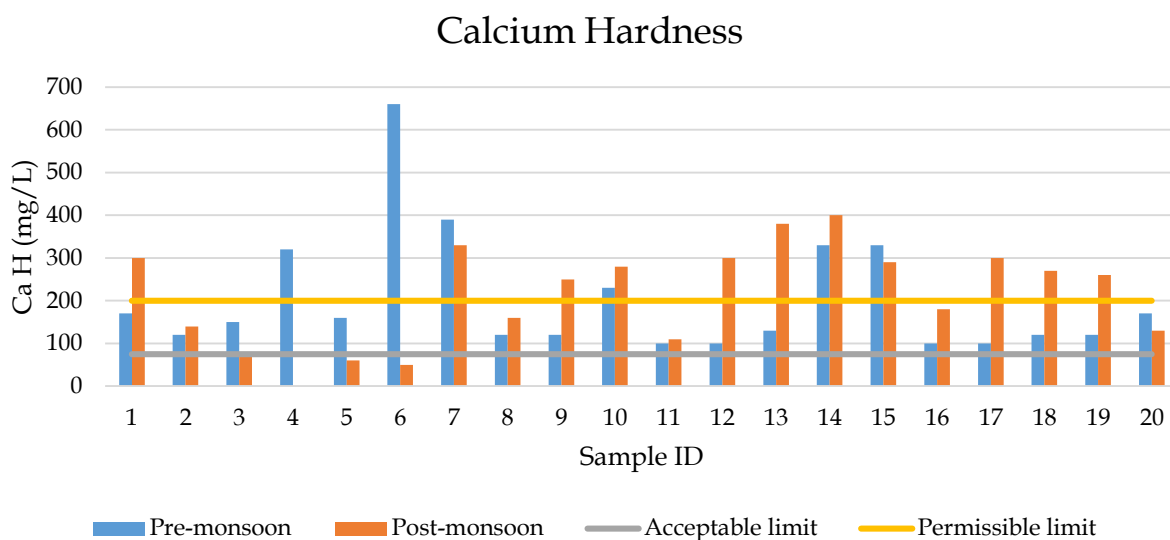
The optimum range of hardness in drinking water is from 80 mg/L to 100 mg/L. Water with hardness greater than 200 mg/L is considered poor in most regions of the province and water with hardness greater than 500 mg/L is normally considered unacceptable for domestic purposes. ^[21] In the present study, TH ranged between 150 mg/L and 800 mg/L for pre-monsoon season, whereas 110 mg/L and 630 mg/L for post-monsoon season. Sample IDs 6, 7, 14, 15 and 14 were exceeding permissible limit for pre-monsoon and post-monsoon seasons indicating the presence of moderate hardness in groundwater.



Graph 5.7 Seasonal variation of TH

Calcium Hardness (mg/L)

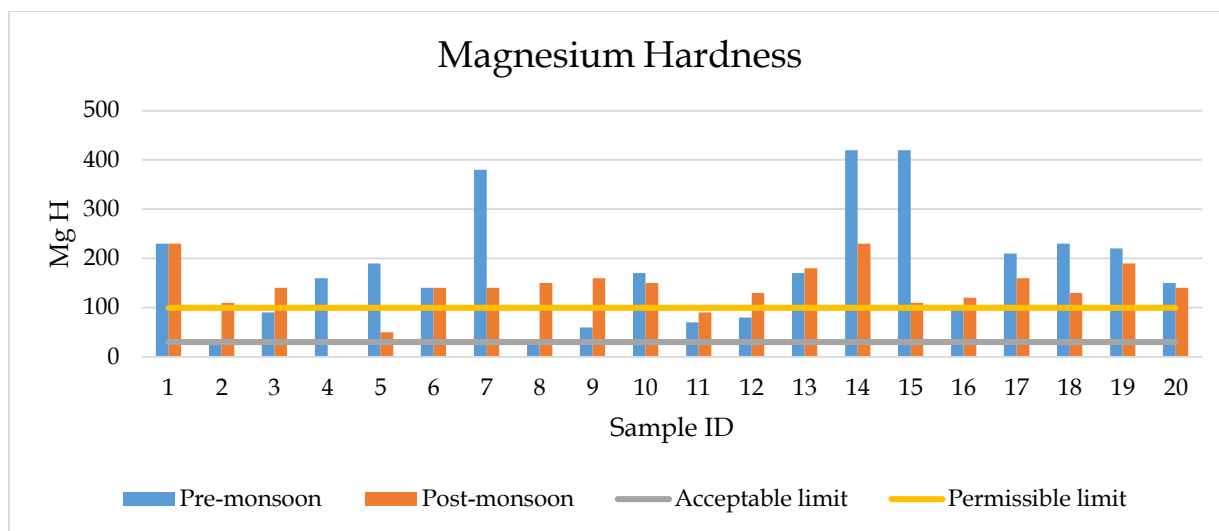
In the present study, Ca-H ranged between 100 mg/L and 660 mg/L for pre-monsoon season, whereas 50 mg/L and 400 mg/L for post-monsoon season. The lowest value found in few samples for pre monsoon season and in Kanchan Nagar for post monsoon season, however the highest value found in Kanchan Nagar and Navi Nagari for pre and post monsoon season respectively. Sample IDs 4, 6, 7, 10, 14, 15 and 1, 7, 9, 10, 12 to 15, 17, 18, 19 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 5.8 Seasonal variation of Ca H

Magnesium Hardness (mg/L)

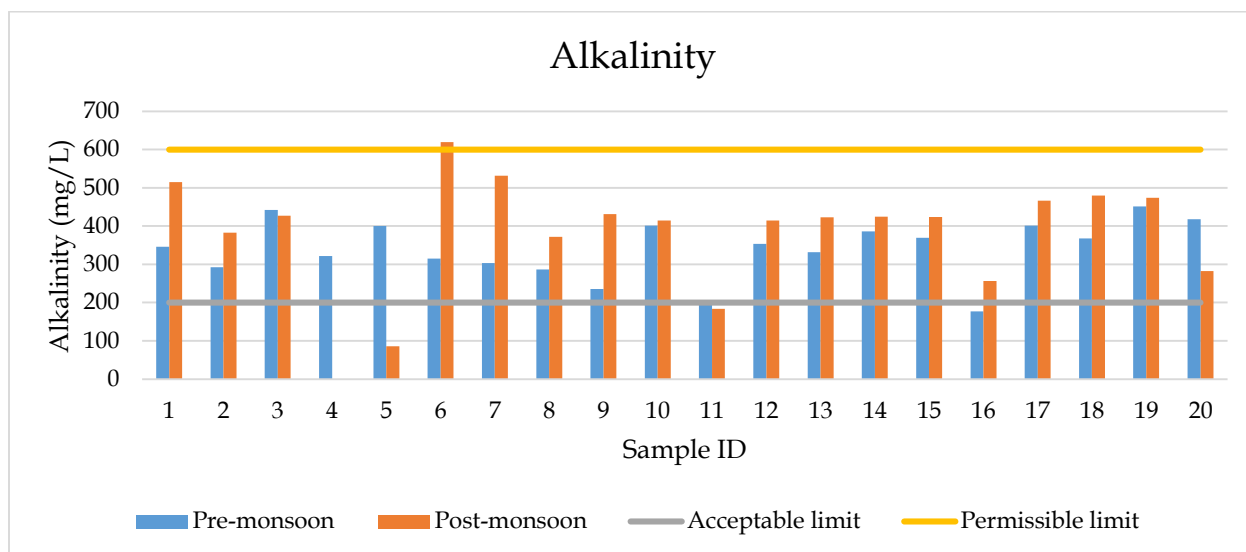
In the present study, Mg-H ranged between 30 mg/L and 420 mg/L for pre-monsoon season, whereas 50 mg/L and 230 mg/L for post-monsoon season. The lowest value found in both Imran Nagar & Candon Nagar and Chanod for pre and post monsoon season, however the highest value found in both Navi Nagari & Navdha and in both Desai Wadi & Navi Nagari for pre and post monsoon season respectively. Sample IDs 1, 4, 5, 6, 7, 10, 13, 14, 15, 17 to 20 and 1, 2, 3, 6 to 10, 12 to 20 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 5.9 Seasonal variation of Mg H

Alkalinity (mg/L)

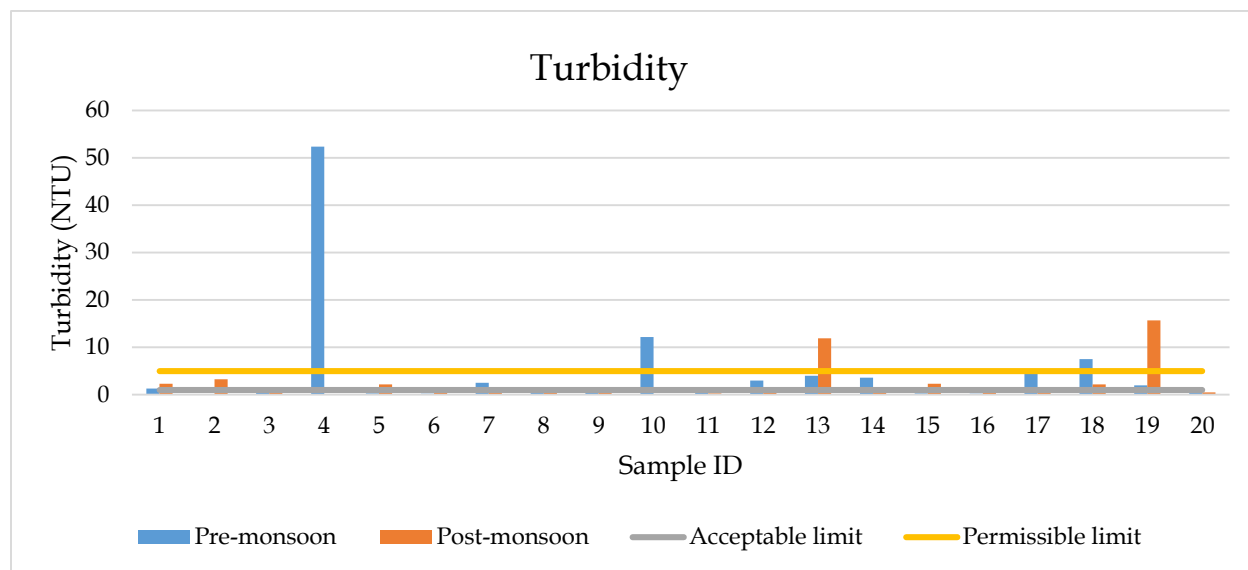
Alkalinity refers to the capacity of water to neutralize acid. In uncontaminated water, alkalinity is primarily a measure of dissolved bicarbonate and carbonate; concentrations of other acid-consuming solutes such as hydroxide, silicate, borate, and dissolved organic compounds are usually small compared to the amount of bicarbonate and carbonate. Alkalinity is reported as the equivalent amount of calcium carbonate in mg/L. [25] In the present study, Alkalinity ranged between 177 mg/L and 451 mg/L for pre-monsoon season, whereas 86 mg/L and 619 mg/L for post-monsoon season. Moderate variations are observed in alkalinity concentration in groundwater of the monitored industrial cluster.



Graph 5.10 Seasonal variation of Alkalinity

Turbidity (NTU)

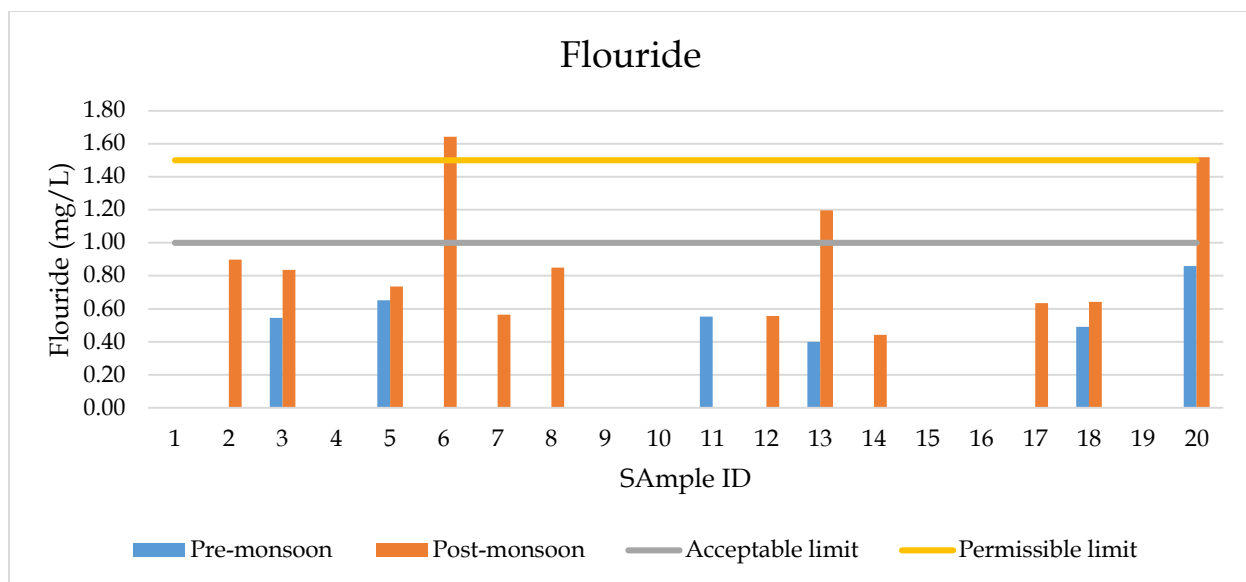
In the present study, Turbidity ranged between 0.1 NTU and 52.4 NTU for pre-monsoon season, whereas 0.1 NTU and 15.7 NTU for post-monsoon season. The lowest value found in Imran Nagar and Rajnigandha Apartment for pre and post monsoon season, however the highest value found in Chanod and Chharwada for pre and post monsoon season respectively. Sample IDs 4, 10, 18 and 13, 19 were exceeding permissible limit for pre-monsoon and post-monsoon indicating inferior groundwater quality.



Graph 5.11 Seasonal variation of Turbidity

Fluoride (mg/L)

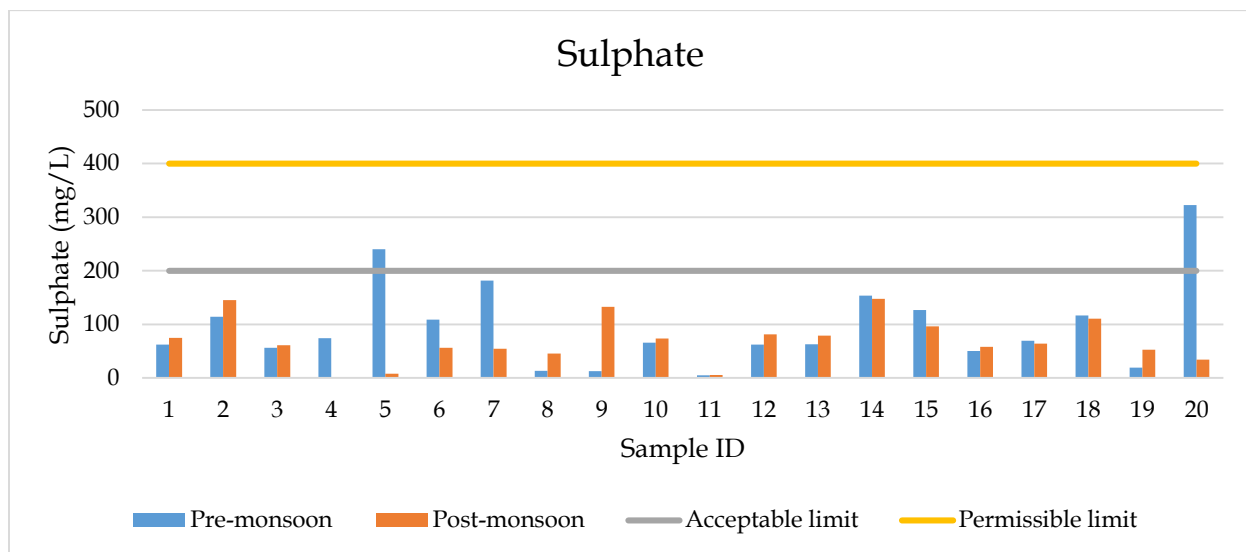
Fluorine is the lightest member of the halogen group and is one of the most reactive of all chemical elements. In groundwater, low or high concentrations of fluoride can occur, depending on the nature of the rocks and the occurrence of fluoride-bearing minerals. Fluoride is necessary in humans in minute quantities during tooth formation, to harden the tooth enamel. In the present study, Fluoride ranged between 0.40 mg/L and 0.86 mg/L for pre-monsoon season, whereas 0.44 mg/L and 1.64 mg/L for post-monsoon season. The fluoride concentration observed for the monitored sources indicates the quality of groundwater is meeting drinking water standards almost for all samples except for sample IDs 6 and 20 which were exceeding permissible limit for post-monsoon season.



Graph 5.12 Seasonal variation of Flouride

Sulphate (mg/L)

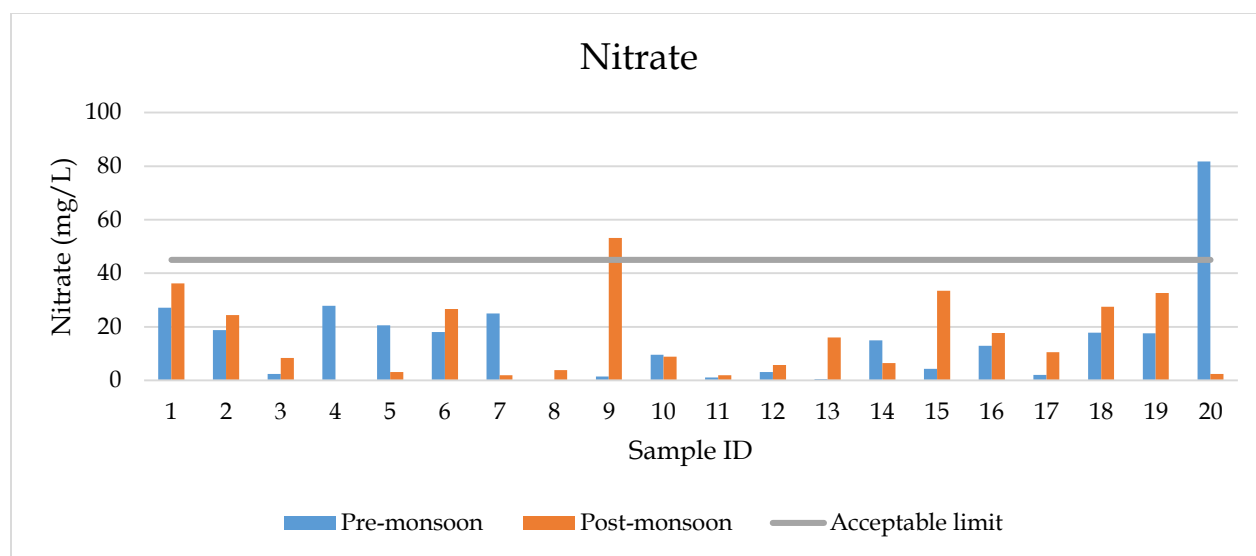
Sulphate concentration may be caused by the oxidation process of their ores, they are also present in industrial wastes. The sulphate ion is one of the major anions occurring in natural water. In the present study, Sulphate ranged between 5 mg/L and 322 mg/L for pre-monsoon season, whereas 5 mg/L and 147 mg/L for post-monsoon season. The sulphate concentration for all samples was found within permissible limit.



Graph 5.13 Seasonal variation of Sulphate

Nitrate (mg/L)

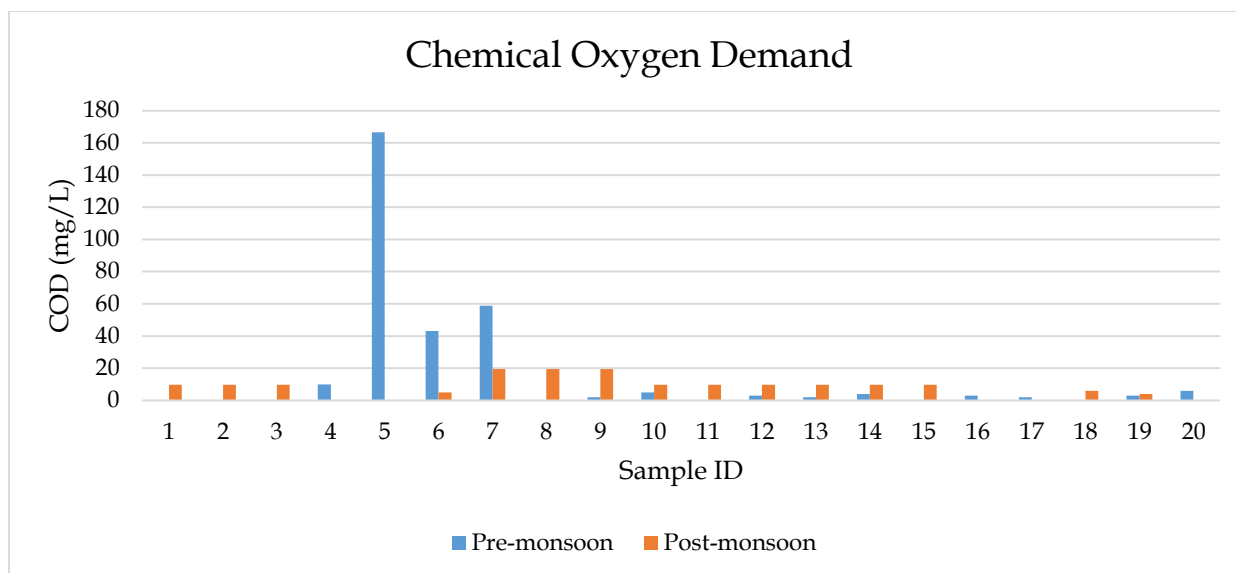
Relatively little of the nitrate found in natural waters is of mineral origin, most coming from organic and inorganic sources, the former including waste discharges and the latter comprising chiefly artificial fertilisers. However, bacterial oxidation and fixing of nitrogen by plants can both produce nitrate. Groundwater contamination can directly affect human health because excessive levels of nitrate in drinking water can produce negative health impacts on human well-being. In the present study, Nitrate ranged between 1 mg/L and 82 mg/L for pre-monsoon season, whereas 2 mg/L and 53 mg/L for post-monsoon season. In general, Nitrate concentration is found well within the acceptable limits for all the monitored sources except for sample IDs 20 and 9 which reported concentrations exceeding the limit.



Graph 5.14 Seasonal variation of Nitrate

Chemical Oxygen Demand (mg/L)

COD is defined as the amount of oxygen equivalents consumed in oxidizing the organic compounds of samples by strong oxidizing agents such as dichromate or permanganate. It is expressed in milligrams per liter (mg/L) that indicates the mass of oxygen consumed per liter of solution. The higher the chemical oxygen demand, the higher the amount of pollution in the water sample. ^[20] In the present study, COD ranged between 2 mg/L and 167 mg/L for pre-monsoon season, whereas 4 mg/L and 20 mg/L for post-monsoon season. Excessive COD reported in a few samples indicates the contamination of source by anthropogenic/industrial activities.



Graph 5.15 Seasonal variation of COD

Ammonical Nitrogen

It is also commonly found in groundwater due to anthropogenic activities, primarily due to leaching from fertilizers, organic waste disposal or leaking sewage systems. ^[18] $\text{NH}_3\text{-N}$ in groundwater aquifer reacts through complex physical- chemical and biological processes to form other associated nitrogen bearing compounds such as NO_2 , NO_3 which makes the risk even more complicated. Their toxicity is attributed to the influence of pH, temperature, and dissolved oxygen concentration. ^[19] In the present study, Ammonical Nitrogen was analyzed for all the collected samples for both pre-monsoon and post-monsoon season. However, none of the sample exceeded concentration of 1 mg/L for both pre-monsoon and post-monsoon season.

Heavy Metals

Heavy Metals Zn, Cd, As, Pb, Cr and Fe were analyzed in a few samples collected during sampling for both pre and post monsoon seasons. The parameter wise inference drawn is covered in table 5.6 along with the details of sample IDs exceeding the permissible limit for respective seasons. Heavy metals in general were analyzed for 1 samples and 4 samples for pre and post monsoon season respectively.

Table 5.6 Details of Heavy Metals analyzed

Sample IDs analyzed for Heavy Metals	
Pre-monsoon	20
Post-monsoon	5, 6, 7, 20
Parameters	Inferences drawn

Zinc	Each sample analyzed showed concentration below 0.5 mg/L and hence no trace of zinc pollution is evident.
Cadmium	Cadmium with acceptable limit of 0.003 mg/L was not detected in any of samples collected for both pre and post monsoon seasons.
Arsenic	Arsenic was not found in any of samples analyzed for both pre and post monsoon seasons and hence monitored source are free from arsenic pollution.
Chromium	Chromium with acceptable limit of 0.05 mg/L was not detected in any of samples collected for both pre and post monsoon seasons.
Lead	Concentration of Lead was recorded within permissible limit of 0.01 mg/L for both pre-monsoon and post-monsoon seasons.
Iron	The acceptable of Iron is 0.3 mg/L and however none of the analyzed sample have found concentration exceeding from 0.5 mg/L and hence no trace of iron pollution is evident.

5.4 Conclusion

Groundwater sampling was carried out around the industrial cluster of Vapi wherein; 20 samples were collected from 20 selected locations for both pre and post monsoon seasons. From the analytical results and monitoring carried out following conclusions are drawn:

- The groundwater quality around industrial cluster is found to be good in respect to all monitored physico-chemical and heavy metal parameters for almost all the samples collected during both pre and post monsoon seasons. Hence, no traces of major or minor contamination is observed during a period of study.
- Thus the overall groundwater quality of the monitored cluster is found to be fairly good and almost free from contamination.

Chapter: 6 Sachin GIDC

6.1 Overview of Surat (Sachin) Industrial Cluster

Industrial development in Surat district is attributed to the presence of a large number of diamond processing, textiles and chemical & petrochemical industries. It processes 10 out of 12 varieties of diamonds in the world contributing to Rs. 45,000 Crore, which is approximately 65 % of the total diamond exports from India. It is also known as synthetic capital of India hosts over 65,000 power looms and provides over seven lac jobs in the district. There are over 41,300 small and medium industries functioning in the district. Some of the main industries are textiles, chemicals, dying & printing, diamond processing, zari (silver) making, and; engineering and related activities (including manufacturing machines & equipments). Maximum number of (nearly 24,000 units) in small and medium enterprises is related to textile industry in the district followed by repairing & service industry with more than 11,000 units. ^[11]

Table 6.1: Surat district at a glance

S. No.	Particulars	Information
1.	Geographical information	21.00 to 21.23 °N and 72.38 to 74.23 °E
2.	Geographical area (sq. km)	4,326.97 sq. km
3.	Average annual rainfall	1307 mm
4.	Average temperature	40 °C (Summer), 32 °C (Monsoon), 23 °C (Winter)
5.	Population	6,079,231
6.	Literacy	89%
7.	Stage of Groundwater development	77.72% - Safe

(Source: District Ground Water Brochure Surat District ^[10])

Products Manufactured

Chemical industries in Surat manufacture diverse range of products like dyes and dye intermediates, pigments, pesticides/insecticides, petrochemicals, agrochemicals, chlor-alkali, pharmaceuticals, paints, and so on. Some of the major categories of chemicals is shown in table:

Table 6.2: Chemical products manufactured in Sachin GIDC

Category	Chemicals produced
Power & Steel	Naphtha, M. S.
Textile	Dyes and Pigments
Organic chemicals	NGL, LPG, Propine, Ethol Mercapton
Inorganic chemicals	Ammonia, Liquid Oxygen, Chlorine, Argon

(Source: Brief Industrial Profile of Surat District ^[11])

6.2 Sampling and monitoring

For the purpose of study sampling sites were selected within 5 km buffer region across the GIDC area. The details of the sampling sites selected is described in table 6.3 and figure 6.1.

Table 6.3 Sampling locations around Sachin GIDC

Villages/locations selected for Sampling & Monitoring		
Within 1 km (7 samples)	Within 3 km (7 samples)	Within 5 km (6 samples)
Sonari, Umbher, Pali Gam, Raj Nagar, Pardi Kande, Talangpor, Chandani Nagar	Unn (2), Gabheni, Bhestan, Budiya, Ganesh Nagar, Vanz	Udhna, Pandesara, Bamroli, Vaktana, Kharvasa, Althan

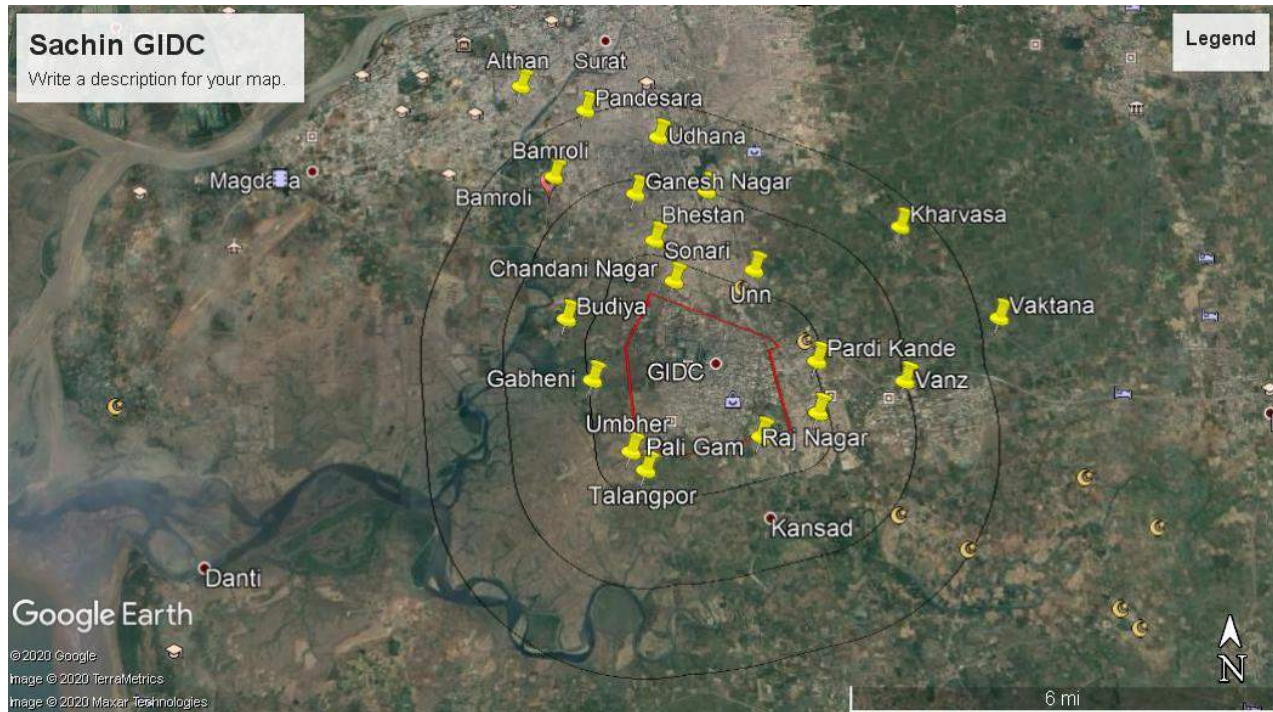


Figure 6.1 Sampling locations around Sachin GIDC

Samples were collected from the identified location for both pre and post monsoon seasons. The information about sampling location, source type, color appearance at field, depth of source, population dependent on source, water usage and name of person contacted during the time of sampling is summarized in table 6.4.

Table 6.4 Details collected during sampling around Sachin GIDC

S. ID	Village/ Site	Location	Latitude Longitude	Source type	Color Appearance @ Field	Depth of well	Population dependent on source	Water Usage	Persons Contacted	Remarks
1.	Bhestan	Near Canal Road	21.127276 72.861505	Bore well	Clear	170	70-80	Domestic & industrial	Sakti bhai	-
2.	Sonari	Near Hanumanji Temple	21.11658 72.846844	Hand pump	Clear	25	30-40	Domestic	Vijay bhai	-
3.	Unn	Near Bharat Petrol Pump	21.106665 72.850017	Hand pump	Clear	30	-		-	-
*	Unn	Near Bharat Petrol Pump	21.106665 72.850017	Hand pump	Clear	30	-	Domestic		Hand pump is not in working condition, hence sample taken from another source
4.	Gabheni	Near Zinga Farm	21.087915 72.821502	Bore well	Turbid	200	200	Domestic	Jitu bhai	Post-monsoon sampling was canceled for this location as bore well was closed and no other source available nearby
5.	Talangpor	Near kirana store	21.066864 72.839899	Hand pump	Clear	60	40-50	Domestic	-	-
*	Talangpor	Near Prafful Industries	21.09095 72.845877	Hand pump	Clear	30	40-50	Domestic	-	Previous source is closed, hence sample taken from other source
6.	Vaktana	Near Post Office	21.099241 72.924282	Bore well	Clear	150	60-80	Drinking & Domestic	-	-
7.	Kharvasa	Near Auto shop	21.121792 72.901522	Bore well	Turbid	170	100-120	Drinking & Domestic	-	-
8.	Althan	Near new citylite road	21.151257 72.80955	Bore well	Clear	160	30-40	Domestic	-	-

9.	Umbher	Near village lake	21.06743 72.84297	Open well	Clear	50	20	Domestic	Buddhabhai	-
10.	Pali Gam	Near Raghuvir Residency	21.07348 72.86986	Bore well	Clear	200	-	Irrigation	-	-
*	Pali Gam	Near Jay Ambe Residency	21.074022 72.869583	Bore well	Clear	150 - 200	-	Drinking & Domestic	-	Previous source is closed, sample taken from nearby source. Salty water from new source, hence used for drinking after filtration.
11.	Bamroli	Near Anand Clinic	21.138641 72.826755	Bore well	Clear	150	10	Domestic	Ketan bhai	Salty taste since last 10 years
12.	Ganesh Nagar	Near Sagar Cinema	21.15708 72.837206	Hand pump	Clear	120	30-40	Drinking & Domestic	Sanjay bhai	-
13.	Budiya	Near village lake	21.102482 72.822366	Open well	Clear	45	30-40	Drinking	Nilesh bhai	-
14.	Raj Nagar	Samarpan Society	21.079444 72.880668	Bore well	Clear	140	20-30	Domestic	Girish bhai	Salty taste. Sewage line is 2 km away and GIDC is 500 mtrs away from the source.
15.	Vanz	Near Ring Road	21.091622 72.903063	Bore well	Clear	150	10	Domestic	Pritesh bhai	-
*	Vanz	Near Ring Road	21.096622 72.903963	Open well	Clear	150	10	Irrigation	-	Previous source is closed, hence sample taken from other source
16.	Pardi Kande	Near Bapa Sitaram Company (Hima Processing)	21.088502 72.874627	Bore well	Clear	100	5	Domestic	Mehul bhai	Salty taste. Near to Bapa Sitaram Dying Company, also surrounded by agricultural land. Since last 5 years, company has closed drainage line, however earlier it was open.

17.	Pandesara	Near Gwalak, Maldhari Vasahat	21.161673 72.625477	Bore well	Clear	150	Animal husbandry	Domestic & animal bathing	Alpeshbhai	Salty taste
18.	Udhana	Near Rajeshwar Plaza	21.15596 72.843888	Bore well	Clear	100	30-40	Domestic	Jagdishbhai	-
19.	Chandani Nagar	Near Sachin-Magdalla Road	21.108357 72.846813	Hand pump	Yellowish	130	Shop cleaning	Domestic	Pavan Gupta	Slight change in colour
20.	Unn	Near Garib Navaz Masjid	21.11129 72.851267	Hand pump	Yellowish	150	Mazjid cleaning & cloth washing	Domestic	-	Change in color. Little turbid

Note:

- * different sample source has taken for post-monsoon season as previously monitored source was found sealed (not in use condition) due to contamination.
- Color code given to sample ID

	Color Code is given to samples collected within 1 km buffer
	Color Code is given to samples collected within 3 km buffer
	Color Code is given to samples collected within 5 km buffer



6.3 Result and Discussion

The physico-chemical parameters of groundwater samples with their acceptable and permissible limits as recommended by Indian Standards Drinking Water - Specification (Second Revision) (IS 10500: 2012) are shown in below table:

Table 6.5 Result range of pre-monsoon and post-monsoon sampling

Sr. No.	Parameters	Measured Units	Requirement (Acceptable Limit)	Permissible limit in the absence of alternate source	Range of present study (Pre-monsoon)	Range of present study (Post-monsoon)
1	pH	-	6.5 – 8.5	No Relaxation	7.09 – 8.09	6.87 – 8.31
2	Colour	Hazen	5	15	5 – 50	5 – 40
3	Electrical Conductivity	µS/cm	-	-	718 – 33700	879 – 25900
4	TDS	mg/L	500	2000	376 – 20926	474 – 15344
5	TSS	mg/L	-	-	2 – 780	0 – 4
6	COD	mg/L	-	-	2 – 307	10 – 657
7	Chloride as Cl ⁻	mg/L	250	1000	56 – 12617	70 – 9940
8	Total Hardness	mg/L	200	600	70 – 5750	260 – 9880
9	Ca ⁺⁺ Hardness	mg/L	75	200	50 – 3950	60 – 4560
10	Mg ⁺⁺ Hardness	mg/L	30	100	80 – 2800	20 – 5320
11	Alkalinity	mg/L	200	600	117 – 632	124 – 771
12	Turbidity	NTU	1	5	0.2 – 33.1	0.01 – 30.9
13	Fluoride	mg/L	1.0	1.5	0.43 – 1.51	0.49 – 2.82
14	Sulphate	mg/L	200	400	29 – 1737	21 – 780
15	Nitrate	mg/L	45	No Relaxation	1 – 86	2 – 82
16	Ammonical Nitrogen	mg/L	-	-	0	0
17	Lead	mg/L	0.01	No Relaxation	0 – 0.005	0.004 – 0.177
18	Cadmium	mg/L	0.003	No Relaxation	0	0

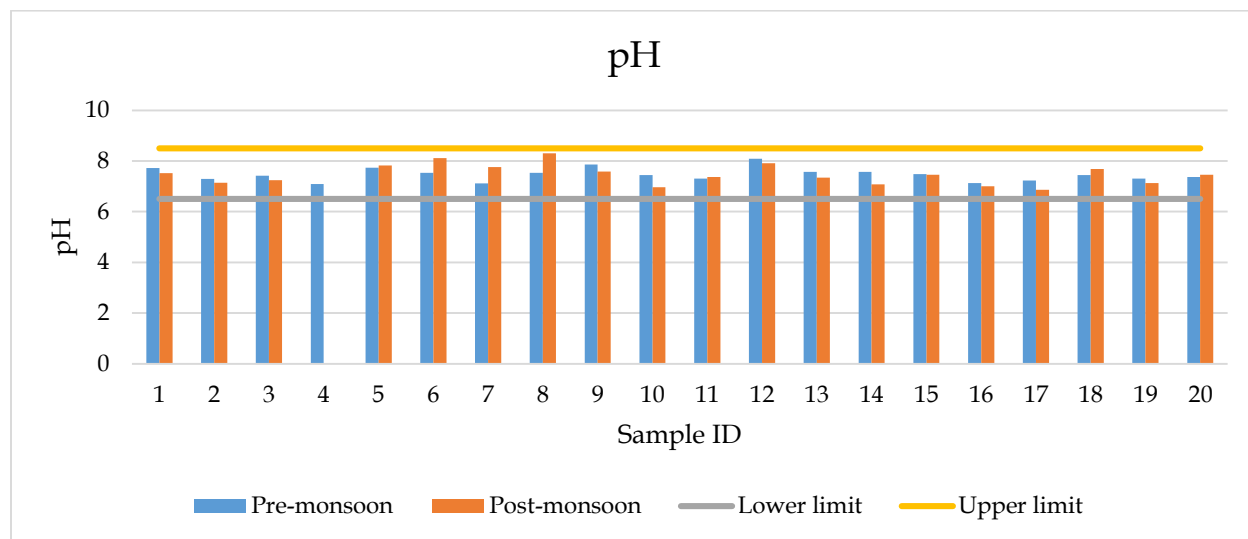
19	Iron	mg/L	0.3	No Relaxation	0.0001 – 0.008	0 - 2.120
20	Chromium	mg/L	0.05	No Relaxation	0.0001 – 0.016	0 – 0.023
21	Zinc	mg/L	5	15	0	0
22	Arsenic	mg/L	0.01	0.05	0	0 – 0.005
23	Hexavalent Chromium	mg/L	-	-	-	0

Note: The limits are as per the **Indian Standards Drinking Water - Specification (Second Revision) (IS 10500: 2012)** [5]

Discussion

pH

In common, water with a pH < 7 is considered acidic and with a pH > 7 is considered basic. The normal range for pH in surface water systems is 6.5 to 8.5 and for groundwater systems 6 to 8.5. The pH of pure water is 7 at 25° C. [23] In the present study, the pH ranged between 7.09 and 8.09 for pre-monsoon season whereas 6.87 and 8.31 for post-monsoon season. All the analyzed samples resulted within permissible limit of drinking water standards.

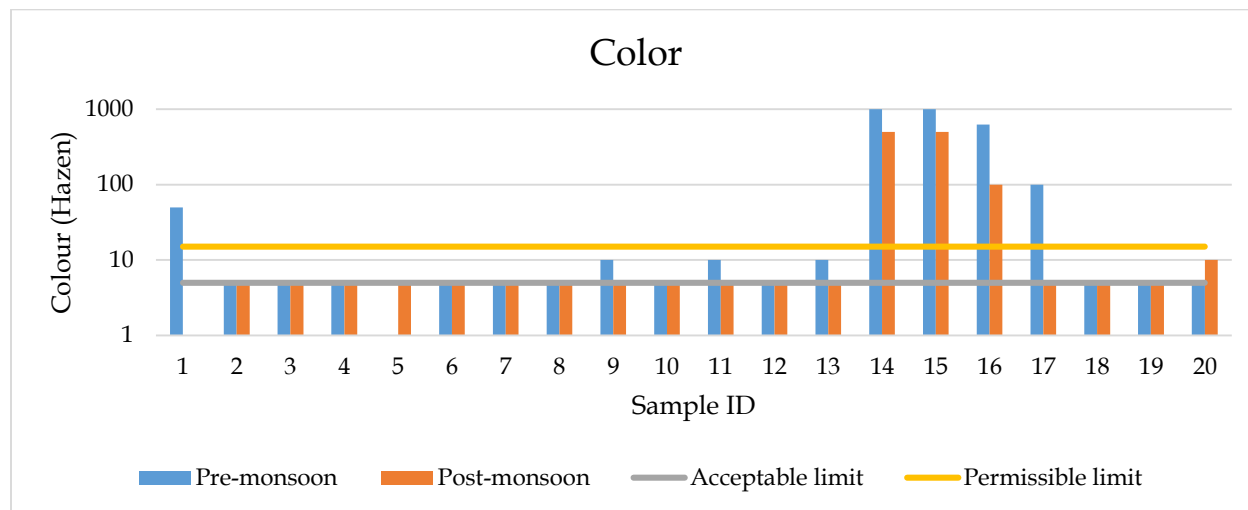


Graph 6.1 Seasonal variation of pH

Color (Hazen)

Suspended and dissolved particles in water influence color. Suspended material in water bodies may be a result of natural causes and/or human activity. Transparent water with a low accumulation of dissolved materials appears blue and indicates low productivity. [22] In the present study, color ranged between 5 Hazen and 50 Hazen for pre-monsoon season whereas 5 Hazen and 40 Hazen for post-monsoon season. Sample IDs 19, 20 and

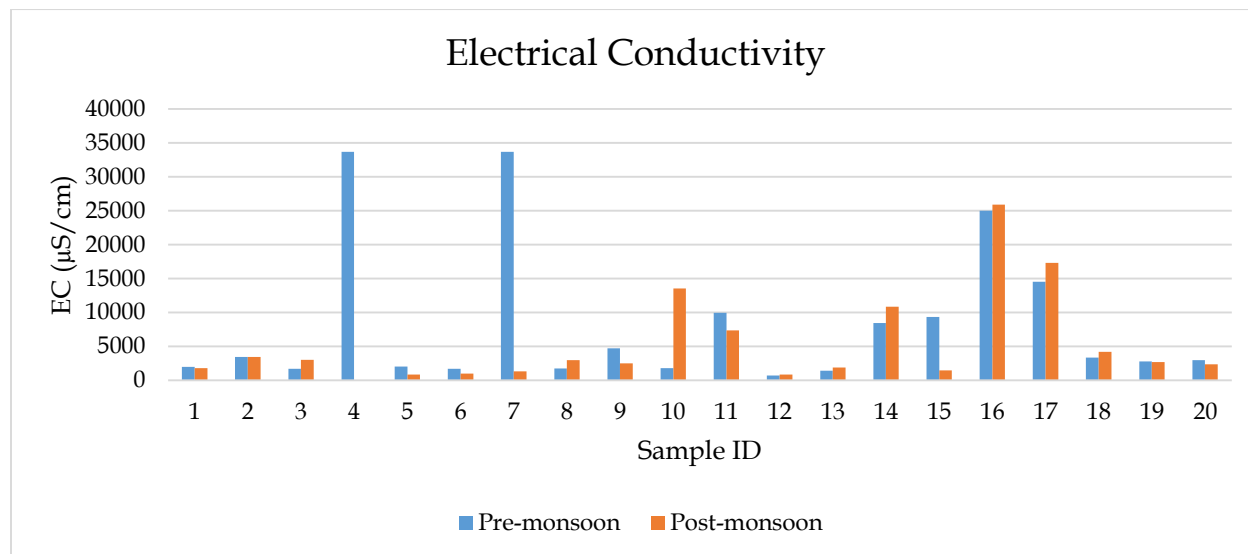
19 showed excess amount of color present in water for pre-monsoon and post-monsoon season respectively. Remaining samples resulted within permissible limit.



Graph 6.2 Seasonal variation of Color

Electrical Conductivity ($\mu\text{S}/\text{cm}$)

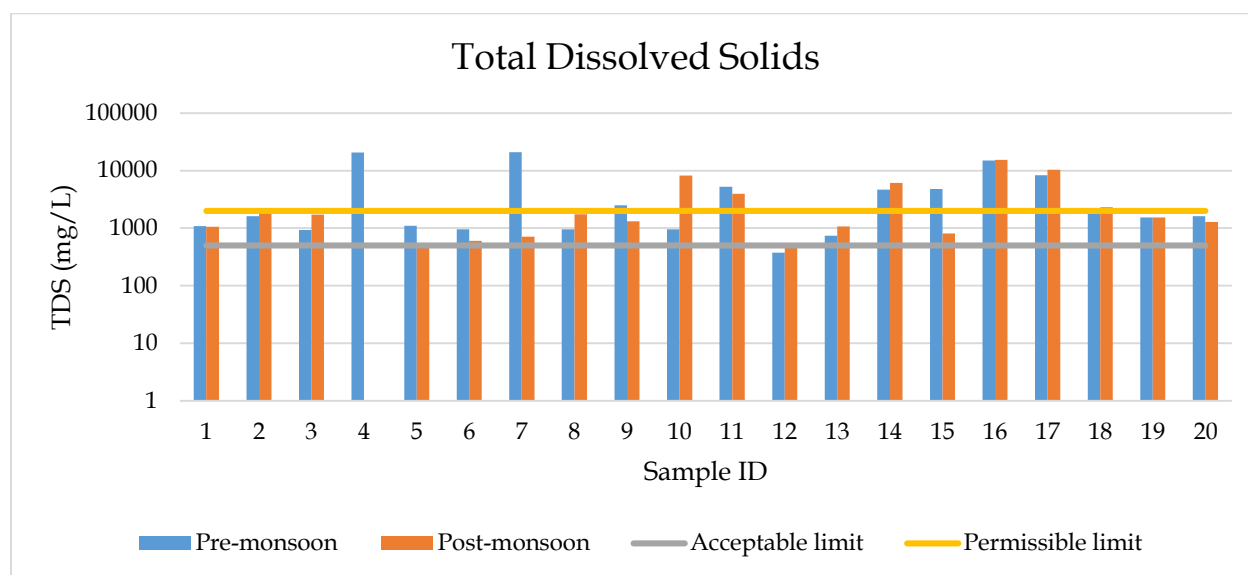
Electrical conductivity is an indirect measurement of salinity, and it is temperature dependent and good indicator of the total salinity, but it does not provide any information about the ionic composition within the water sample. ^[16] In the present study, EC ranged between 718 $\mu\text{S}/\text{cm}$ and 33700 $\mu\text{S}/\text{cm}$ for pre-monsoon season, whereas 879 $\mu\text{S}/\text{cm}$ and 25900 $\mu\text{S}/\text{cm}$ for post-monsoon season. High variations were observed for this parameter for both pre and post monsoon seasons.



Graph 6.3 Seasonal variation of EC

Total Dissolved Solids (mg/L)

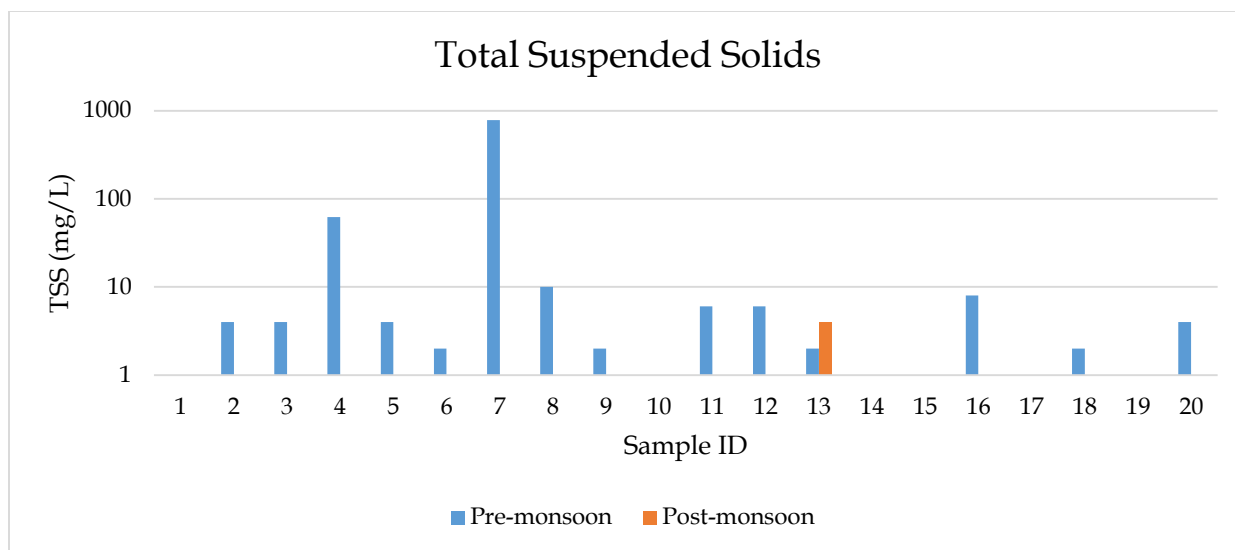
In general, the total dissolved solids concentration is the sum of the cations (positively charged) and anions (negatively charged) ions in the water. Therefore, the total dissolved solids test provides a qualitative measure of the amount of dissolved ions but does not tell us the nature or ion relationships. [24] In the present study, TDS ranged between 376 mg/L and 20926 mg/L for pre-monsoon season, whereas 474 mg/L and 15344 mg/L for post-monsoon season. High variation in TDS concentration was observed during both the seasons. Sample IDs 4, 7, 9, 11, 14, 15, 16, 17 and 10, 11, 14, 16, 17, 18 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 6.4 Seasonal variation of TDS

Total Suspended Solids (mg/L)

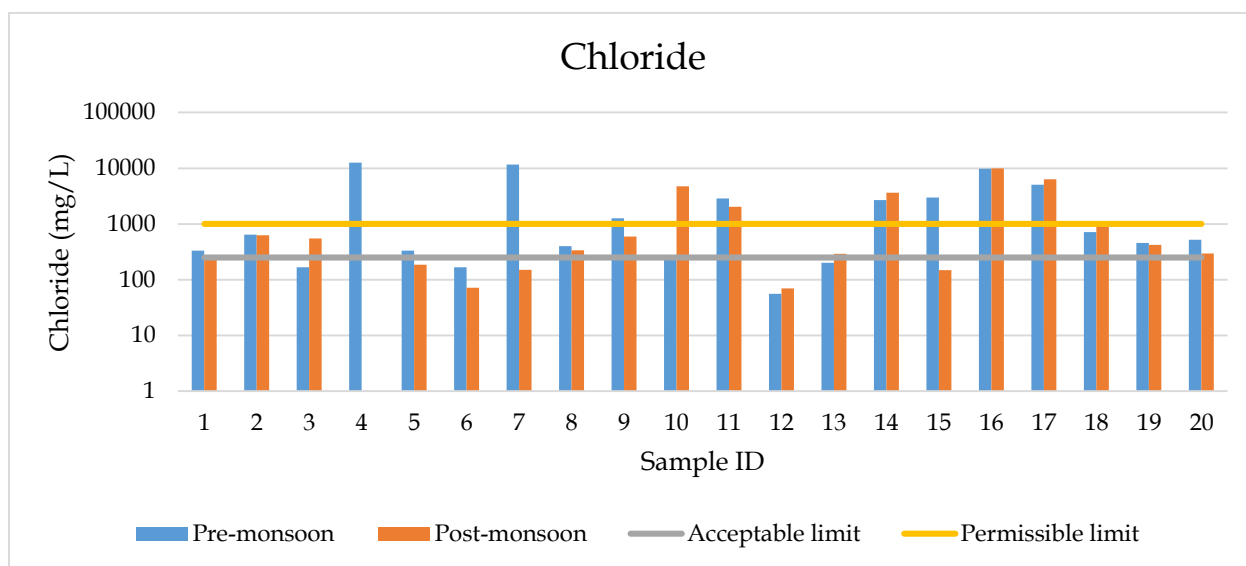
Total suspended solids (TSS) are particles that are larger than 2 microns found in the water column. Anything smaller than 2 microns (average filter size) is considered a dissolved solid. Most suspended solids are made up of inorganic materials, though bacteria and algae can also contribute to the total solids concentration. In the present study, TSS ranged between 0 and 780 mg/L for pre-monsoon season, whereas 0 and 4 mg/L for post-monsoon season. Sample ID 4 and 7 showed excess amount of TSS present in water for pre-monsoon season, remaining samples were resulted within 10 mg/L concentration.



Graph 6.5 Seasonal variation of TSS

Chloride (mg/L)

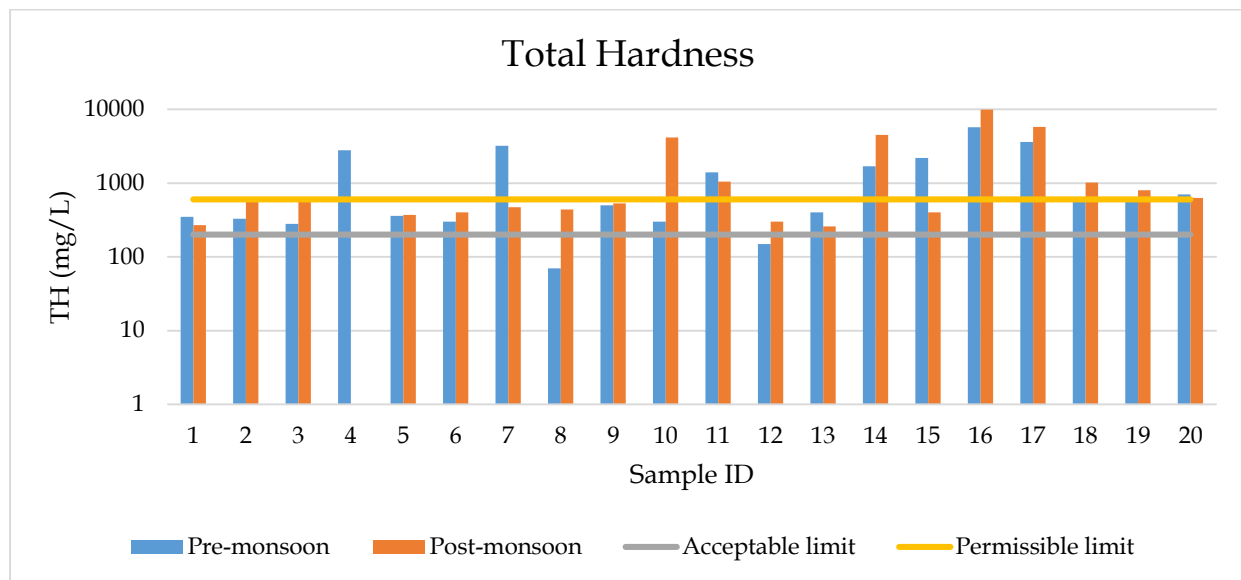
Chloride occurs in all natural water in widely varying concentration. The chloride content normally increases as the mineral content increases and it is also important in detecting the contamination of groundwater by sewage. ^[17] In the present study, Chloride ranged between 56 mg/L and 12617 mg/L for pre-monsoon season, whereas 70 mg/L and 9940 mg/L for post-monsoon season. Sample IDs 4, 7, 9, 11, 14, 15, 16, 17 and 10, 11, 14, 16, 17, 18 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 6.6 Seasonal variation of Chloride

Total Hardness (mg/L)

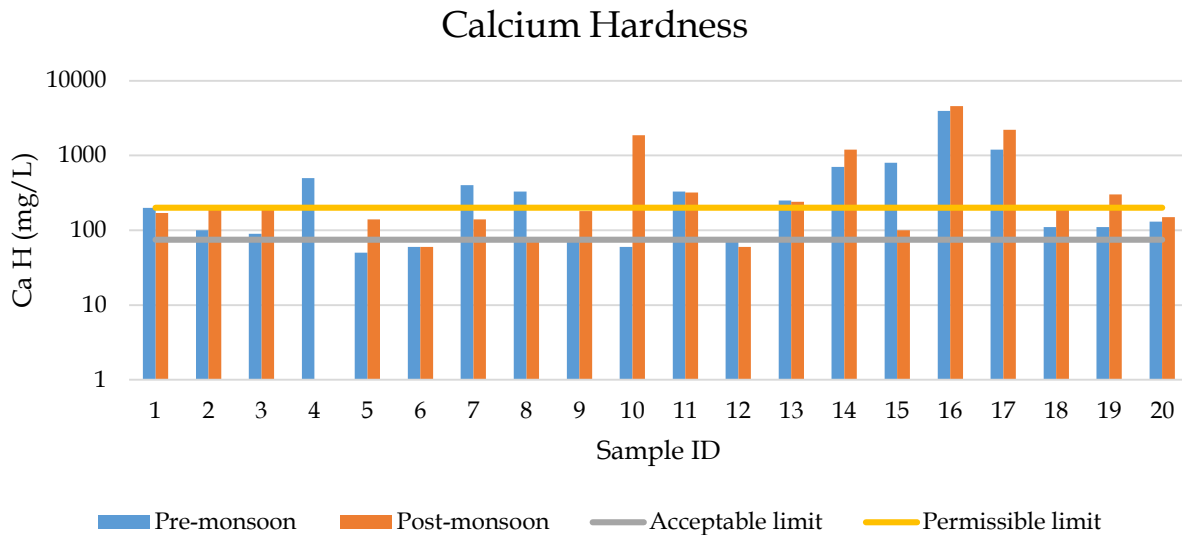
The optimum range of hardness in drinking water is from 80 mg/L to 100 mg/L. Water with hardness greater than 200 mg/L is considered poor in most regions of the province and water with hardness greater than 500 mg/L is normally considered unacceptable for domestic purposes. [21] In the present study, TH ranged between 70 mg/L and 5750 mg/L for pre-monsoon season, whereas 260 mg/L and 9880 mg/L for post-monsoon season. Sample IDs 4, 7, 11, 14, 15, 16, 17, 20 and 2, 10, 11, 14, 16 to 20 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 6.7 Seasonal variation of TH

Calcium Hardness (mg/L)

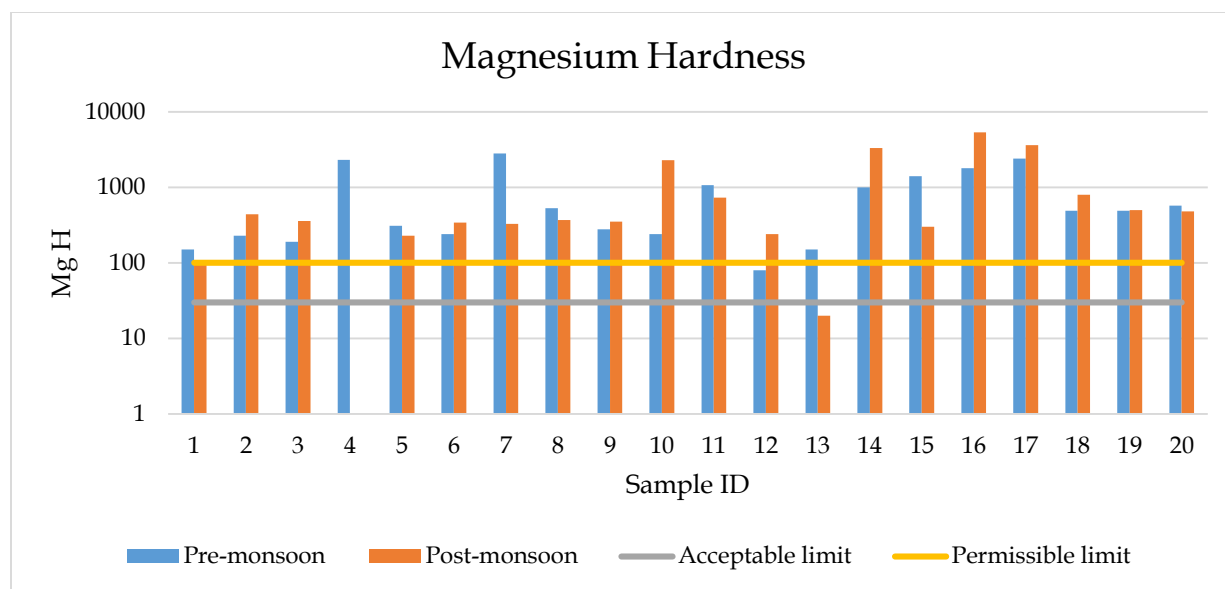
In the present study, Ca-H ranged between 50 mg/L and 3950 mg/L for pre-monsoon season, whereas 60 mg/L and 4560 mg/L for post-monsoon season. The lowest value found in Talangpor and Vaktana for pre and post monsoon season, however the highest value found in Pardi Kande for both pre and post monsoon season respectively. Sample IDs 4, 7, 8, 11, 13, 14, 15, 16, 17 and 10, 11, 13, 14, 16, 17, 19 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 6.8 Seasonal variation of Ca H

Magnesium Hardness (mg/L)

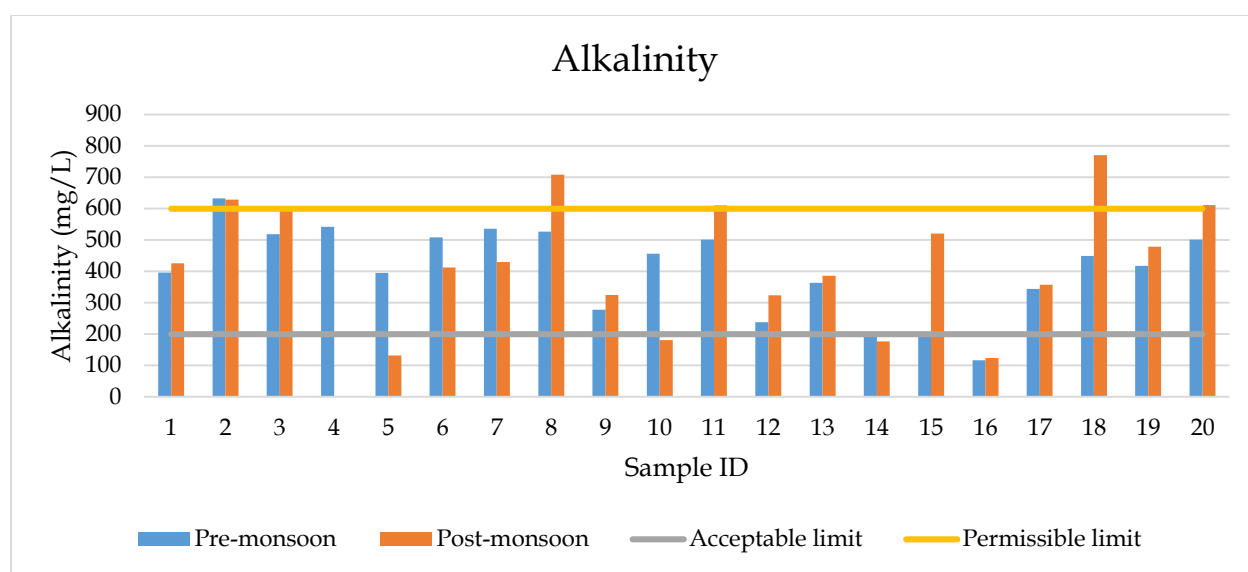
In the present study, Mg-H ranged between 80 mg/L and 2800 mg/L for pre-monsoon season, whereas 20 mg/L and 5320 mg/L for post-monsoon season. The lowest value found in Ganesh Nagar and Budiya for pre and post monsoon season, however the highest value found in Kharvasa and Pardi Kande for pre and post monsoon season respectively. Sample IDs 1 to 11, 13 to 20 and 2, 3, 5 to 12, 14 to 20 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 6.9 Seasonal variation of Mg H

Alkalinity (mg/L)

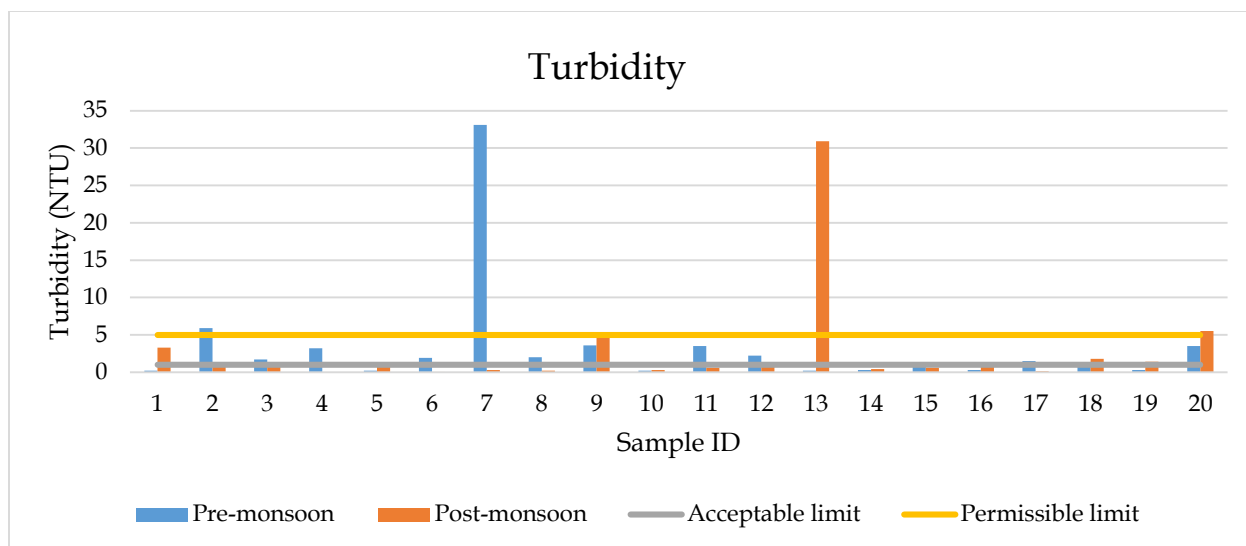
Alkalinity refers to the capacity of water to neutralize acid. In uncontaminated water, alkalinity is primarily a measure of dissolved bicarbonate and carbonate; concentrations of other acid-consuming solutes such as hydroxide, silicate, borate, and dissolved organic compounds are usually small compared to the amount of bicarbonate and carbonate. Alkalinity is reported as the equivalent amount of calcium carbonate in mg/L. [25] In the present study, Alkalinity ranged between 117 mg/L and 632 mg/L for pre-monsoon season, whereas 124 mg/L and 771 mg/L for post-monsoon season. Most of the analyzed samples resulted within permissible limit except for samples IDs 2 and 2, 3, 8, 11, 18, 20; which exceeded permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 6.10 Seasonal variation of Alkalinity

Turbidity (NTU)

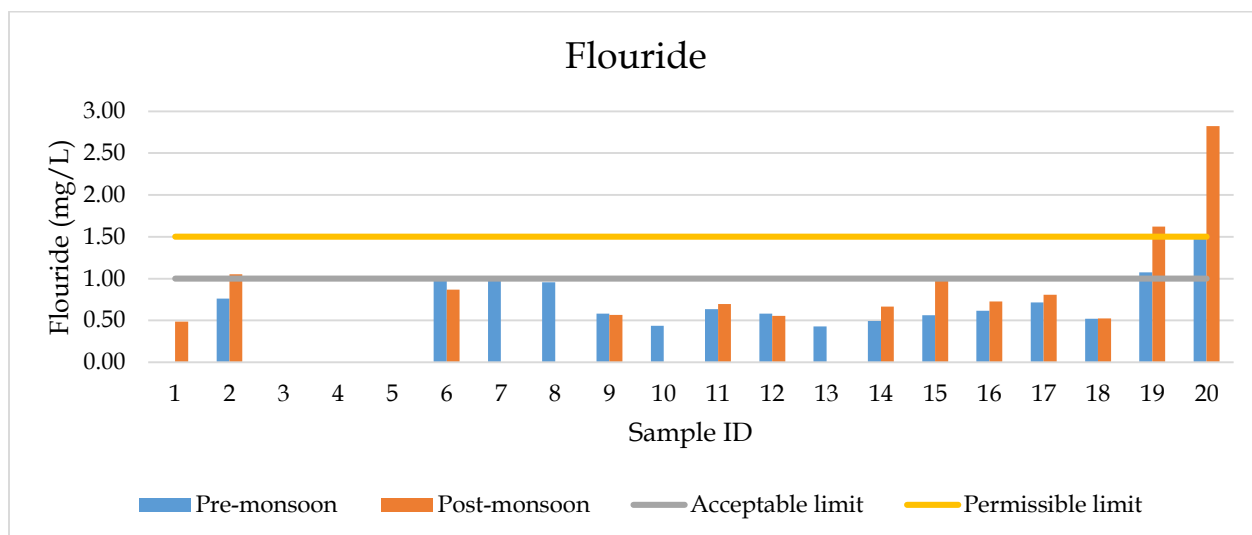
In the present study, Turbidity ranged between 0.2 NTU and 33.1 NTU for pre-monsoon season, whereas 0.01 NTU and 30.9 NTU for post-monsoon season. Turbidity level in all the analyzed samples were found within permissible limit of drinking water standards except for sample IDs 2, 7 and 13, 20; which were exceeding permissible limit for pre-monsoon and post-monsoon respectively.



Graph 6.11 Seasonal variation of Turbidity

Fluoride (mg/L)

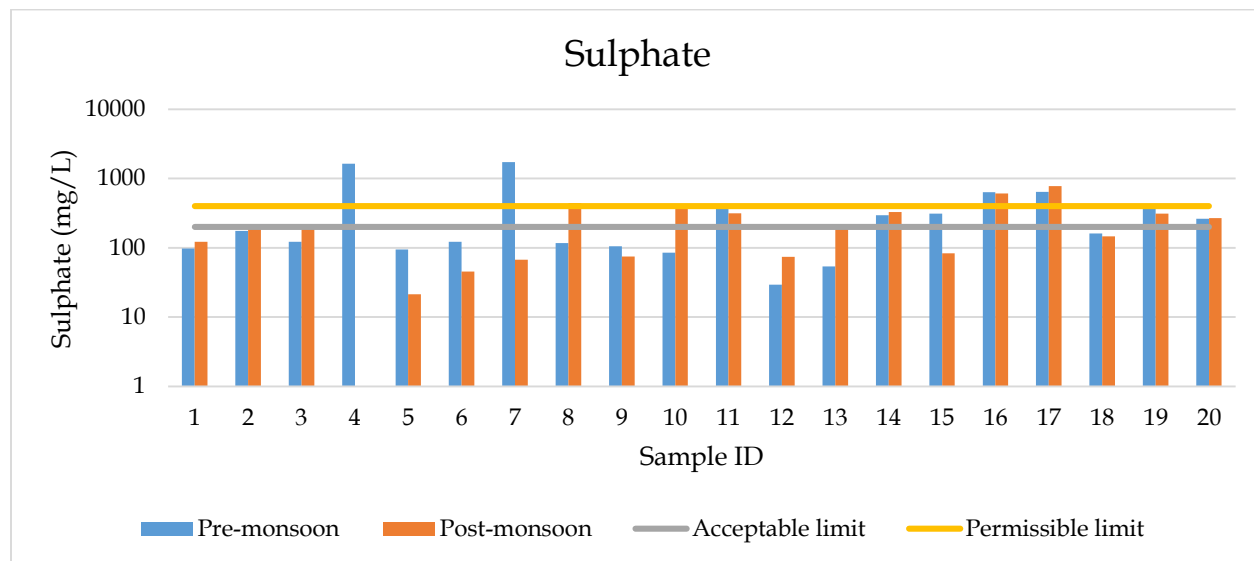
Fluorine is the lightest member of the halogen group and is one of the most reactive of all chemical elements. In groundwater, low or high concentrations of fluoride can occur, depending on the nature of the rocks and the occurrence of fluoride-bearing minerals. Fluoride is necessary in humans in minute quantities during tooth formation, to harden the tooth enamel. In the present study, Fluoride ranged between 0.43 mg/L and 1.51 mg/L for pre-monsoon season, whereas 0.49 mg/L and 2.82 mg/L for post-monsoon season. Sample IDs 20 and 19, 20 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 6.12 Seasonal variation of Flouride

Sulphate (mg/L)

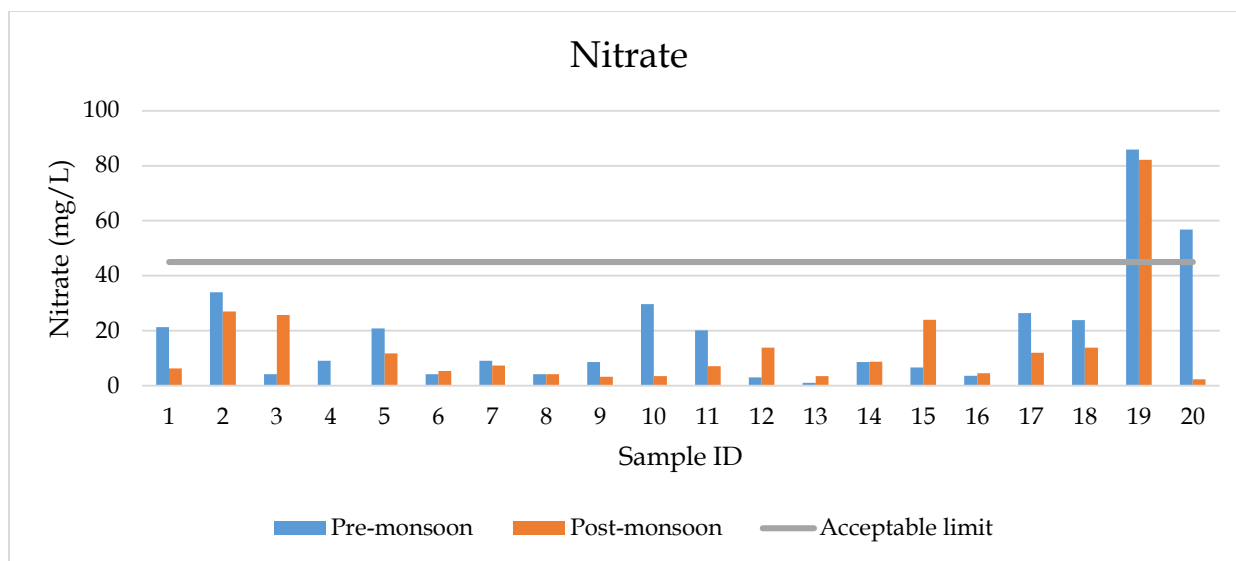
Sulphate concentration may be caused by the oxidation process of their ores, they are also present in industrial wastes. The Sulphate ion is one of the major anions occurring in natural water. In the present study, Sulphate ranged between 29 mg/L and 1737 mg/L for pre-monsoon season, whereas 21 mg/L and 780 mg/L for post-monsoon season. All the samples analyzed were found well within permissible limit except for sample IDs 4, 7, 16, 17 and 8, 16, 17; which were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 6.13 Seasonal variation of Sulphate

Nitrate (mg/L)

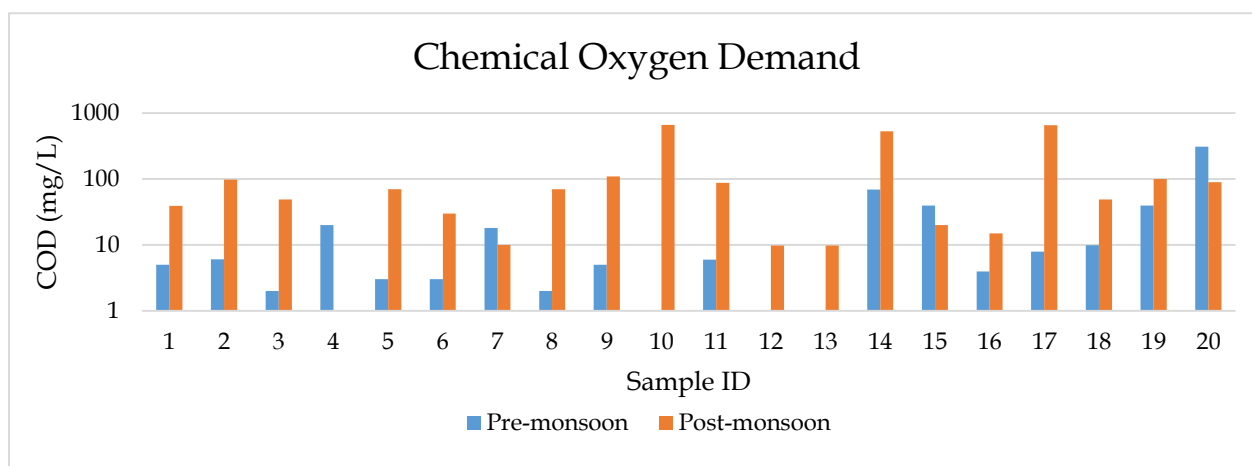
Relatively little of the nitrate found in natural waters is of mineral origin, most coming from organic and inorganic sources, the former including waste discharges and the latter comprising chiefly artificial fertilizers. However, bacterial oxidation and fixing of nitrogen by plants can both produce nitrate. Groundwater contamination can directly affect human health because excessive levels of nitrate in drinking water can produce negative health impacts on human well-being. In the present study, Nitrate ranged between 1 mg/L and 86 mg/L for pre-monsoon season, whereas 2 mg/L and 82 mg/L for post-monsoon season. Sample IDs 19, 20 and 19 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 6.14 Seasonal variation of Nitrate

Chemical Oxygen Demand (mg/L)

COD is defined as the amount of oxygen equivalents consumed in oxidizing the organic compounds of samples by strong oxidizing agents such as dichromate or permanganate. It is expressed in milligrams per liter (mg/L) that indicates the mass of oxygen consumed per liter of solution. The higher the chemical oxygen demand, the higher the amount of pollution in the water sample. ^[20] In the present study, COD ranged between 2 mg/L and 307 mg/L for pre-monsoon season, whereas 10 mg/L and 657 mg/L for post-monsoon season. Excess amount of COD was present in few samples which indicates presence pollution of groundwater.



Graph 6.15 Seasonal variation of COD

Ammonical Nitrogen

It is also commonly found in groundwater due to anthropogenic activities, primarily due to leaching from fertilizers, organic waste disposal or leaking sewage systems. ^[18] $\text{NH}_3\text{-N}$ in groundwater aquifer reacts through complex physical- chemical and biological processes to form other associated nitrogen bearing compounds such as NO_2 , NO_3 which makes the risk even more complicated. Their toxicity is attributed to the influence of pH, temperature, and dissolved oxygen concentration. ^[19] In the present study, Ammonical Nitrogen was analyzed for all the collected samples for both pre-monsoon and post-monsoon season. However, none of the sample exceeded concentration of 1 mg/L for both pre-monsoon and post-monsoon season.

Heavy Metals

Heavy Metals Zn, Cd, As, Pb, Cr, Fe and Cr(VI) were analyzed in a few samples collected during sampling for both pre and post monsoon seasons. The parameter wise inference drawn is covered in table 6.6 along with the details of sample IDs exceeding the permissible limit for respective seasons. Heavy metals in general were analyzed for 7 samples and 10 samples for pre and post monsoon season respectively.

Table 6.6 Details of Heavy Metals analyzed

Sample IDs analyzed for Heavy Metals	
Pre-monsoon	3, 7, 12, 14, 16, 19, 20
Post-monsoon	7, 9, 11, 12, 14, 15, 16, 17, 19, 20
Parameters	Inferences drawn
Zinc	Each sample analyzed showed concentration below 0.5 mg/L and hence no trace of zinc pollution is evident.
Cadmium	Cadmium with acceptable limit of 0.003 mg/L was not detected in any of samples collected for both pre and post monsoon seasons and hence monitored sources are free from arsenic pollution.
Arsenic	Arsenic was not found in any of samples analyzed for both pre and post monsoon seasons except for sample ID 19 which showed 0.0059 mg/L of arsenic present in the water, however it was not exceeding the permissible limit of 0.05 mg/L.
Chromium	Concentration of Chromium was within acceptable limit of 0.05 mg/L for every sample analyzed for both pre-monsoon and post-monsoon seasons.
Lead	Sample IDs 8 and 11 with 0.016 mg/L and 0.177 mg/L were recorded and also exceeded acceptable limit of 0.01 mg/L for post-monsoon sampling.
Iron	Two samples i.e. 11 and 12 showed 2.12 mg/L and 0.36 mg/L iron concentration exceeding acceptable limit of 0.3 mg/L during post-monsoon sampling.

Hexavalent Chromium	Sample IDs 19 and 20 were analyzed for Hexavalent Chromium for post-monsoon season only, however they were found below 0.01 mg/L.
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6.4 Conclusion

Groundwater sampling was carried out around the industrial cluster of Sachin wherein; 20 samples were collected from 20 selected locations for both pre and post monsoon seasons. From the analytical results and monitoring carried out following conclusions are drawn:

- The overall groundwater quality around industrial cluster is found to be good for most of the samples collected expect a few with sample IDs and location as 4 (Bore well of Gabheni), 7 (Bore well of Kharvasa), 10 (Bore well of Pali gam), 11 (Bore well of Bamroli), 14 (Bore well of raj nagar), 15 (Bore well of Vanz), 16 (Bore well of Pardi Kande) and 17 (Bore well of Pandesara) whose quality is found to be inferior with respect to all physico-chemical parameters analyzed for the samples including heavy metal contamination in terms of Iron and Lead contamination. Chemical Oxygen Demand analyzed in the samples reflects presence of pollution in groundwater ranging from smaller to greater extent.

- In respect to information collected about sources during monitoring it is known that the objectionable sources are used for mainly domestic purposes expect two sources which are used for drinking (around 120 population uses that source) and irrigation purposes.

- Thus the overall groundwater quality of the monitored cluster is found to be fairly good expect for few samples.

Chapter: 7 Nandesari GIDC

7.1 Overview of Vadodara (Nandesari) Industrial Cluster

The industrial clusters in Vadodara especially in Nandesari include Chemicals & fertilizers, Pharmaceuticals, Biotechnology, Cotton Textiles, Machine Tools, Glass, Engineering, Tobacco, Fisheries and Dairy. There are over 18,000 small scale industrial units in Vadodara, in which repairs & services units are maximum in numbers accounting nearly 5,713 units, 1,923 textiles, 1,615 metal works, 1,357 chemicals, 1,316 equipments\machinery, 1,145 rubber & its products, 1,047 food products and 3,840 are misc. units. Other key small scale industries include textiles, metal works, chemicals, equipments, rubber products and food products etc. 885 glass, ceramics & cement, 829 equipments related to electricity, 753 papers & its products, 601 non-ferrous metals, 543 leather, 173 tobaccos and 56 are misc. units. ^[13]

Table 7.1: Vadodara district at a glance

S. No.	Particulars	Information
1.	Geographical information	21°49'19" & 22°48'37" N and 72°51'05" & 74°16'55" E
2.	Geographical area (sq. km)	7548.50
3.	Average annual rainfall	922.70 mm
4.	Average temperature	34.79 °C (Summer), 20.58 °C (Winter)
5.	Population	4,223,568
6.	Literacy	78.92 %
7.	Stage of Groundwater development	-

(Source: District Ground Water Brochure Vadodara ^[12])

Products Manufactured in Nandesari GIDC

Chemical industries in Vadodara manufacture diverse range of products like dyes and dye intermediates, pigments, alkalis, pesticides/insecticides, petrochemicals, pharmaceuticals, and so on. Some of the major categories of chemicals is shown in table:

Table 7.2: Chemical products manufactured in Nandesari GIDC

Category	Chemicals produced
Pharmaceuticals	Pharmaceuticals
Pesticides	Pesticides
Fertilisers	Caprolactum, Ammonia, Melamine
Petrochemicals	Fluconazole bulk, Azithromycin, Polybutadiene Rubber, LPG, Petroleum Products
Organic chemicals	Propylonyl Chloride

Inorganic chemicals	Sodium Hydrosuphide
Alkali	Hydrogen Peroxide, Caustic Soda

(Source: Brief Industrial Profile of Vadodara District ^[13])

7.2 Sampling and monitoring

For the purpose of study sampling sites were selected within 5 km buffer region across the GIDC area. The details of the sampling sites selected is described in table 7.3 and figure 7.1.

Table 7.3 Sampling locations around Nandesari GIDC

Villages/locations selected for Sampling & Monitoring		
Within 1 km (3 samples)	Within 3 km (9 samples)	Within 5 km (9 samples)
Nandesari, Rupapura, Radhiyapur	Angadh(2), Dhanora, Padamla(2), Rayka, Sakariyapura, Rampura Petro Chemical INA	Vasad, Sakarda, Ranoli (2), Dodka, Dasarath, Karachiya, Babarpur (Kevalnagar), Mandvapura



Figure 7.1 Sampling locations around Nandesari GIDC

Samples were collected from the identified location for both pre and post monsoon seasons. The information about sampling location, source type, color appearance at field, depth of source, population dependent on source, water usage and name of person contacted during the time of sampling is summarized in table 7.4.

Table 7.4 Details collected during sampling around Nandesari GIDC

S. ID	Village/ Site	Location	Latitute Longitude	Source type	Color Appearance @ Field	Depth of well	Popula tion depend ent on source	Water Usage	Persons Contacted	Remarks
1.	Ranoli	Near Post Office	22.397733 73.127608	Bore well	Clear	210	350	Drinking	Mohan bhai Harijan	Open sewage line is nearby the source
2.	Vasad	Near Vasad Toll tax	22.450633 72.069706	Bore well	Clear	160	-	Domestic	Gautam bhai	-
3.	Radhiyapur	Near Radhiyapur-Damapura Road	22.417482 73.10082	Bore well	Orange	180	10 farms	Irrigation	Jashvant bhai Gohil	Deepak Aeromatic, Deepak Nitrate and other multi-chemical companies and their outlets are nearby.
4.	Rupapura	Madhavsinh Bhimsinh Bore	22.422127 73.088498	Bore well	Clear	200	-	Irrigation	Madhavsinh Gohil	-
5.	Sakariyapura	Near Nandesari Chokdi, Sakariyapura village road	22.426927 73.082936	Bore well	Clear	260	35	Drinking, Irrigation	Laxamansinh Ramansinh	Source is surrounded by agricultural land; however, pesticide is not used as said by informer.
6.	Angadh	Near Chamunda mata Temple	22.398741 73.086447	Bore well	Yellowish	180	40	Irrigation, Domestic	Takhatsinh Shankarsinh	-
7.	Nandesari	Near Tebi Vistar	22.414105 73.085351	Bore well	Clear	130	200	Irrigation, Domestic	Amarsinh Gohil	-
8.	Rampura	Near Anganwadi	22.383079 73.094994	Bore well	Yellowish	130	10-15 farmers	Irrigation	Natvarsinh Padiyar	Post-monsoon sampling was not carried out due to power cut

9.	Dhanora	Near Shiv Temple	22.380986 73.107534	Bore well	Clear	60	-	Domestic	Kanti bhai Makwana	-
10.	Padamala	Somakaka's bore well	22.414061 73.129619	Bore well	Clear	330	20	Drinking, Irrigation	Soma bhai Parmar	-
11.	Babarpur	Near Primary School	22.438937 73.1127	Tube well	Clear	225	60	Drinking	Kesarsinh & Dineshbhai	-
12.	Sakarda	Near Meldimata Temple	22.429347 73.115565	Bore well	Clear	150	2-5 House, 3-5 Farms	Drinking, Irrigation	Natu bhai Narsinh Patel	-
13.	Rayka	Near Rayka village gate	22.438286 73.096935	Bore well	Clear	170	15	Irrigation	Thakor bhai Shah	-
14.	Dodka	Near Dodka Post Office	22.45354 73.105543	Bore well	Clear	225	600	Domestic	Pravinsinh Rathod	-
*	Dodka	Near village gate	22.4534177 3.108189	Bore well	Clear	120	600	-	Pravinsinh Rathod	Due to lack of water, villagers dug new bore (not in use yet), hence sample taken from new source.
15.	Padamala	Near Highway	22.417267 73.119851	Bore well	Green	200	15-18 farms	Irrigation, Tent wash	Rajiv bhai Patel	-
16.	Ranoli	Near Gangeshwar Mahadev Temple	22.405171 73.115537	Bore well	Clear	210	400	Drinking	Bharat bhai Parmar	Near to industrial area, Dmark Company
17.	Dasarath	VRL Logistic, Yogiraj Green	22.387436 73.148178	Bore well	Clear	150	600	Drinking, Domestic	Hemant bhai	Salty taste of water, hence used for drinking after filtration. There are 3 sewage sump in the society.
18.	Karachia	Near Water Tank	22.379507 73.131806	Open well	Clear	60	160	Domestic	Mohan bhai Parmar	-

19.	Petro Chemicals INA	GACL (Ranoli) Mota Mahadev Temple	22.397227 3.113306	Bore well	Clear	190	240	Domestic	Mahesh bhai	-
20.	Mandvapura	Near household area	22.425687 73.060888	Bore well	Clear	220	-	Irrigation	-	-
*	Mandvapura	Near agriculture area	22.4428217 3.061949	Bore well	Clear	180	-	Drinking, Irrigation	-	Sample taken from another source
21.	Angadh (Wadiya Vago)	Near Meldimata Temple	22.3975177 3.089152	Bore well	Reddish	360	-	Irrigation	-	-

Note:

1. * different sample source has taken for post-monsoon season as previously monitored source was found sealed (not in use condition) due to contamination.
2. Color code given to sample ID



Color Code is given to samples collected within 1 km buffer



Color Code is given to samples collected within 3 km buffer



Color Code is given to samples collected within 5 km buffer



7.3 Result and Discussion

The physico-chemical parameters of groundwater samples with their acceptable and permissible limits as recommended by Indian Standards Drinking Water - Specification (Second Revision) (IS 10500: 2012) are shown in below table:

Table 7.5 Result range of pre-monsoon and post-monsoon sampling

Sr. No.	Parameters	Measured Units	Requirement (Acceptable Limit)	Permissible limit in the absence of alternate source	Range of present study (Pre-monsoon)	Range of present study (Post-monsoon)
1	pH	-	6.5 – 8.5	No Relaxation	6.6 – 7.84	6.66 – 7.77
2	Colour	Hazen	5	15	5 – 500	5 – 500
3	Electrical Conductivity	µS/cm	-	-	1300 – 5770	1034 – 5570
4	TDS	mg/L	500	2000	678 – 2936	602 – 3156
5	TSS	mg/L	-	-	2 – 6	0 - 392
6	COD	mg/L	-	-	7 – 170	10 – 186
7	Chloride as Cl ⁻	mg/L	250	1000	88 – 1500	61 – 1452
8	Total Hardness	mg/L	200	600	380 – 1500	350 – 1900
9	Ca ⁺⁺ Hardness	mg/L	75	200	100 – 640	100 – 1180
10	Mg ⁺⁺ Hardness	mg/L	30	100	180 – 860	200 – 1400
11	Alkalinity	mg/L	200	600	238 – 582	368 – 623
12	Turbidity	NTU	1	5	0.1 – 6.4	0.01 – 75.5
13	Fluoride	mg/L	1.0	1.5	0.42 – 2.71	0.45 – 1.21
14	Sulphate	mg/L	200	400	21 – 991	18 – 787
15	Nitrate	mg/L	45	No Relaxation	12 – 487	10 – 338
16	Ammonical Nitrogen	mg/L	-	-	0 – 1.8	0 – 2.95
17	Lead	mg/L	0.01	No Relaxation	0 – 0.003	0 – 0.029
18	Cadmium	mg/L	0.003	No Relaxation	0	0
19	Iron	mg/L	0.3	No Relaxation	0 – 0.0002	0 – 1.402

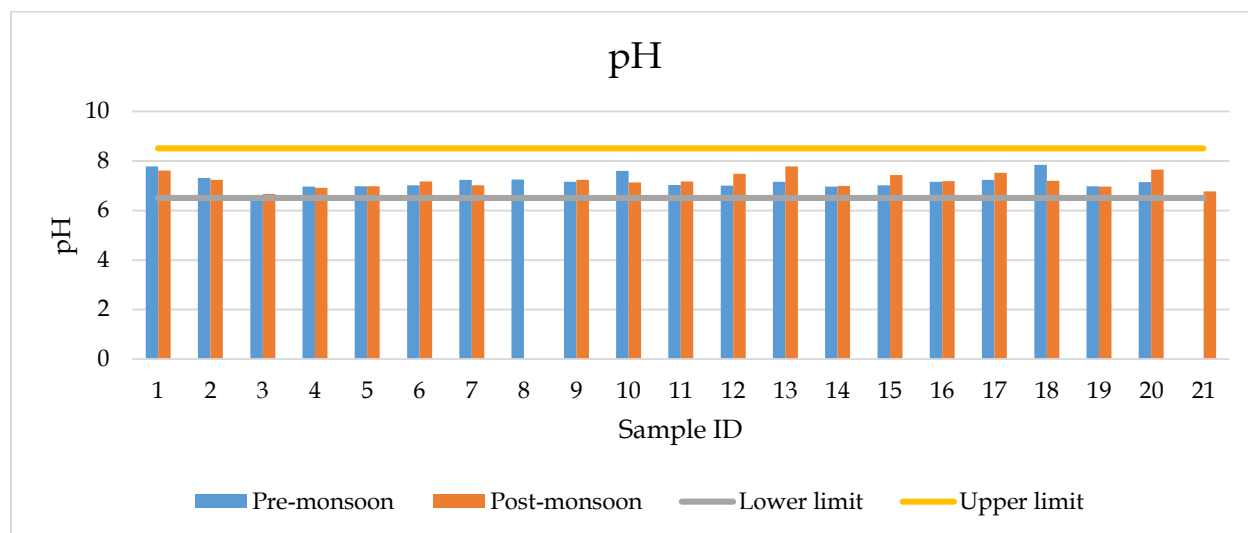
20	Chromium	mg/L	0.05	No Relaxation	0 – 0.009	0.0003 – 0.026
21	Zinc	mg/L	5	15	0	0
22	Arsenic	mg/L	0.01	0.05	0	0
23	Hexavalent Chromium	mg/L	-	-	-	0

Note: The limits are as per the **Indian Standards Drinking Water - Specification (Second Revision) (IS 10500: 2012)** ^[5]

Discussion

pH

In common, water with a pH < 7 is considered acidic and with a pH > 7 is considered basic. The normal range for pH in surface water systems is 6.5 to 8.5 and for groundwater systems 6 to 8.5. The pH of pure water is 7 at 25° C. ^[23] In the present study, the pH ranged between 6.6 and 7.84 for pre-monsoon season whereas 6.66 and 7.77 for post-monsoon season. All samples resulted within permissible limit. Each sample analyzed resulted below permissible limit of drinking water standards.

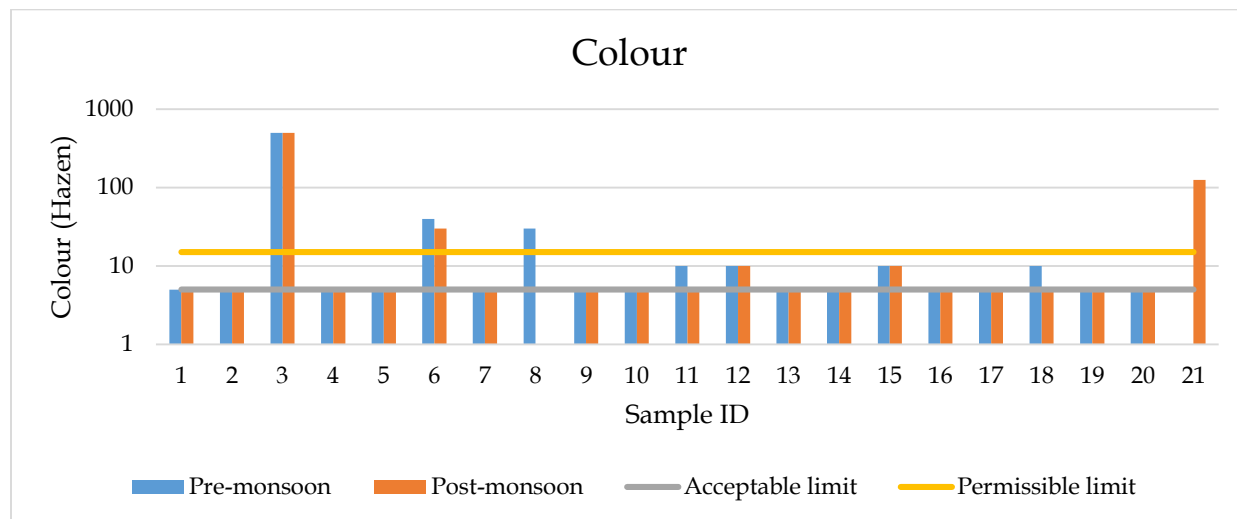


Graph 7.1 Seasonal variation of pH

Colour (Hazen)

Suspended and dissolved particles in water influence color. Suspended material in water bodies may be a result of natural causes and/or human activity. Transparent water with a low accumulation of dissolved materials appears blue and indicates low productivity. ^[22] In the present study, color ranged between 5 Hazen and 500 Hazen for both pre-monsoon and post-monsoon seasons. High variations observed in the result; however most of the samples were resulted within permissible limit except for few samples which

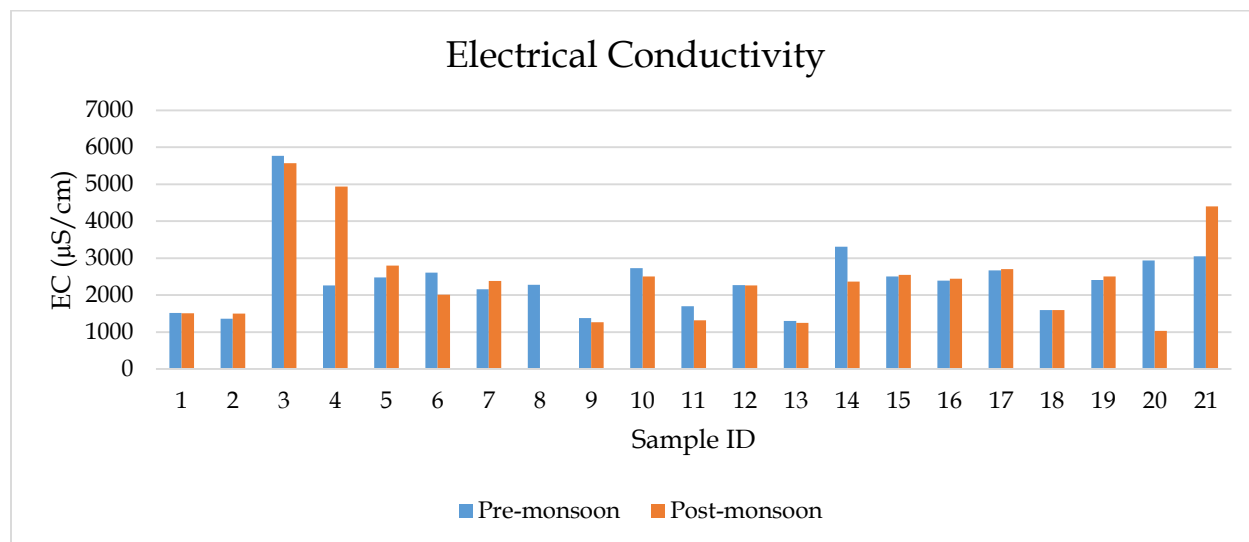
showed excess amount of color in water. Sample IDs 3, 6, 8 and 3, 6, 21 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 7.2 Seasonal variation of Colour

Electrical Conductivity ($\mu\text{S}/\text{cm}$)

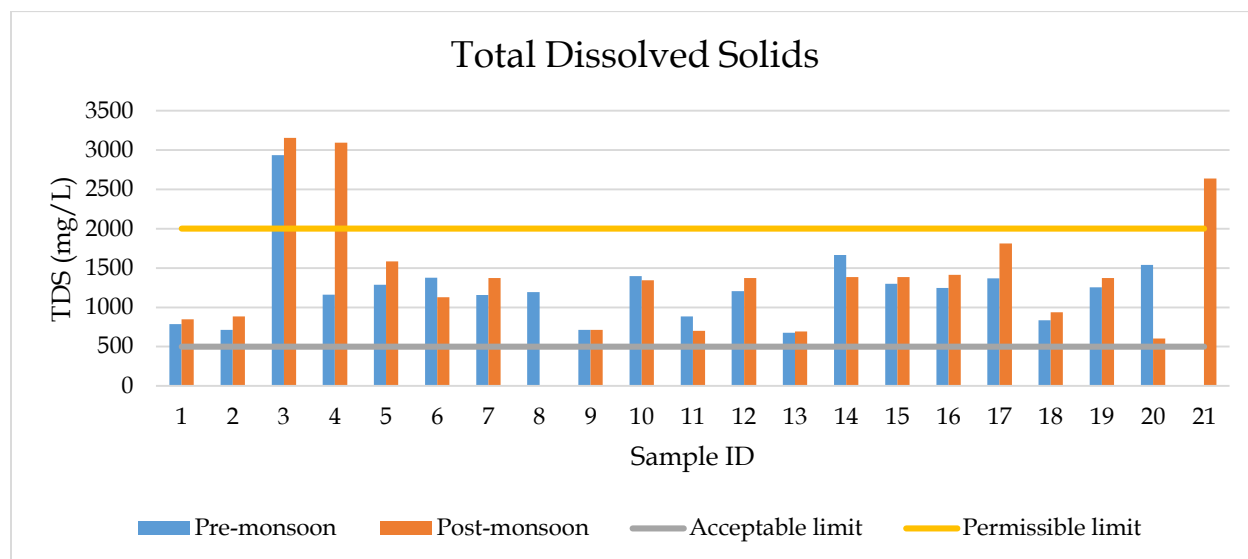
Electrical conductivity is an indirect measurement of salinity, and it is temperature dependent and good indicator of the total salinity, but it does not provide any information about the ionic composition within the water sample. ^[16] In the present study, EC ranged between 1300 $\mu\text{S}/\text{cm}$ and 5770 $\mu\text{S}/\text{cm}$ for pre-monsoon season, whereas 1034 $\mu\text{S}/\text{cm}$ and 5570 $\mu\text{S}/\text{cm}$ for post-monsoon season. High variations observed in the result; however, it does not indicate pollution of water.



Graph 7.3 Seasonal variation of EC

Total Dissolved Solids (mg/L)

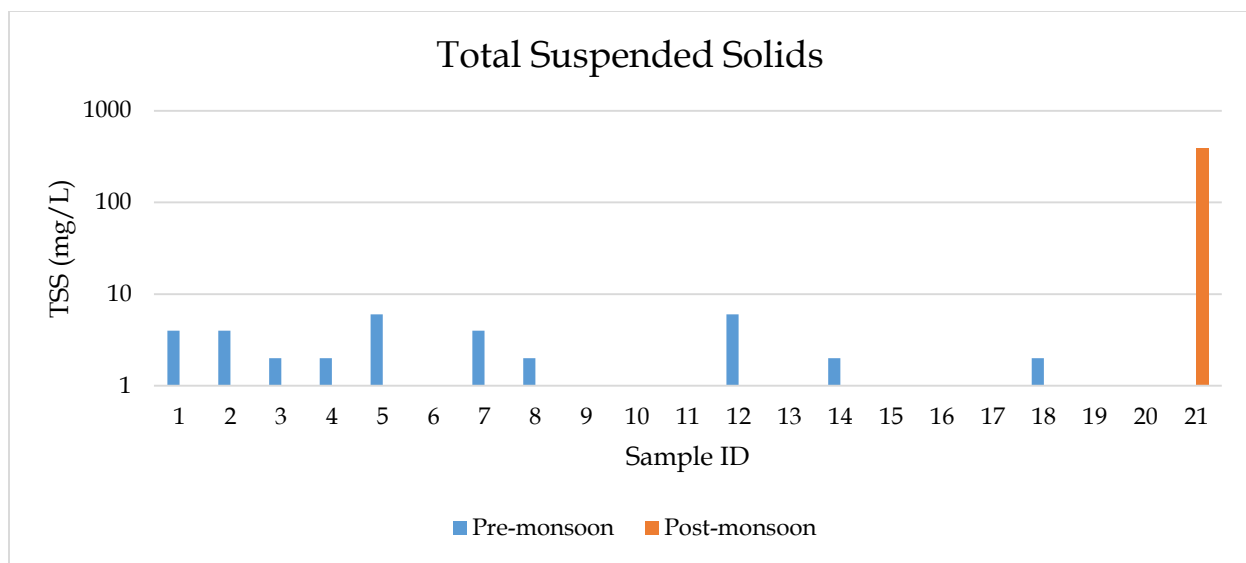
In general, the total dissolved solids concentration is the sum of the cations (positively charged) and anions (negatively charged) ions in the water. Therefore, the total dissolved solids test provides a qualitative measure of the amount of dissolved ions but does not tell us the nature or ion relationships. [24] In the present study, TDS ranged between 678 mg/L and 2936 mg/L for pre-monsoon season, whereas 602 mg/L and 3156 mg/L for post-monsoon season. Most of the analyzed samples showed concentration of TDS within permissible limit of drinking water standards except for sample IDs 3 and 3, 4, 21; which were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 7.4 Seasonal variation of TDS

Total Suspended Solids (mg/L)

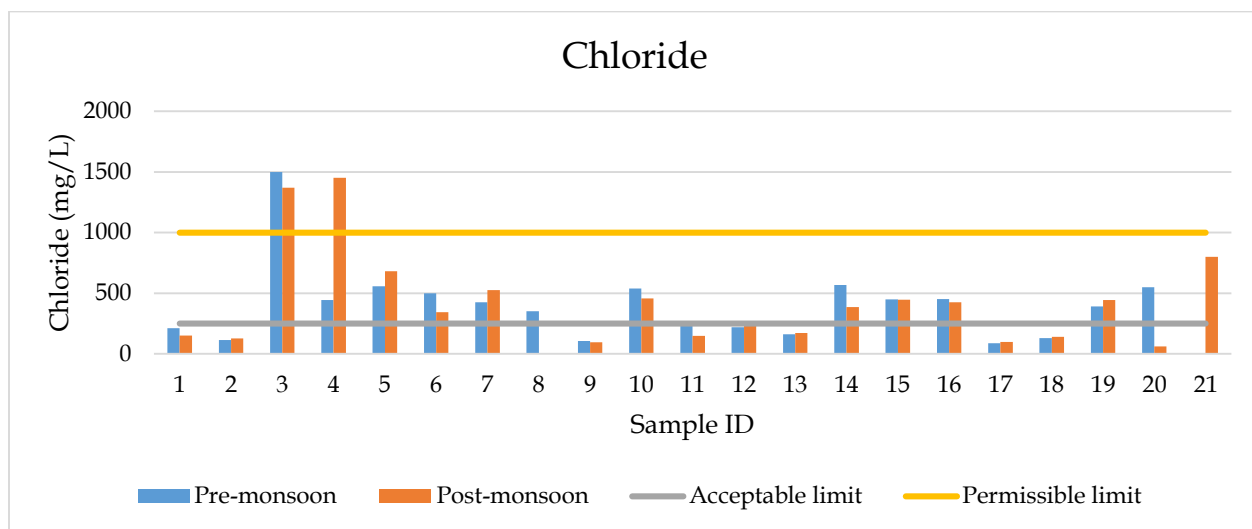
Total suspended solids (TSS) are particles that are larger than 2 microns found in the water column. Anything smaller than 2 microns (average filter size) is considered a dissolved solid. Most suspended solids are made up of inorganic materials, though bacteria and algae can also contribute to the total solids concentration. In the present study, TSS ranged between 0 and 6 mg/L for pre-monsoon season, whereas 0 and 392 mg/L for post-monsoon season. Concentration of TSS for pre-monsoon season was found well within 10 mg/L; however, for post-monsoon season, sample ID 21 showed excess amount of TSS present in water, while remaining samples showed less concentration of TSS present in water.



Graph 7.5 Seasonal variation of TSS

Chloride (mg/L)

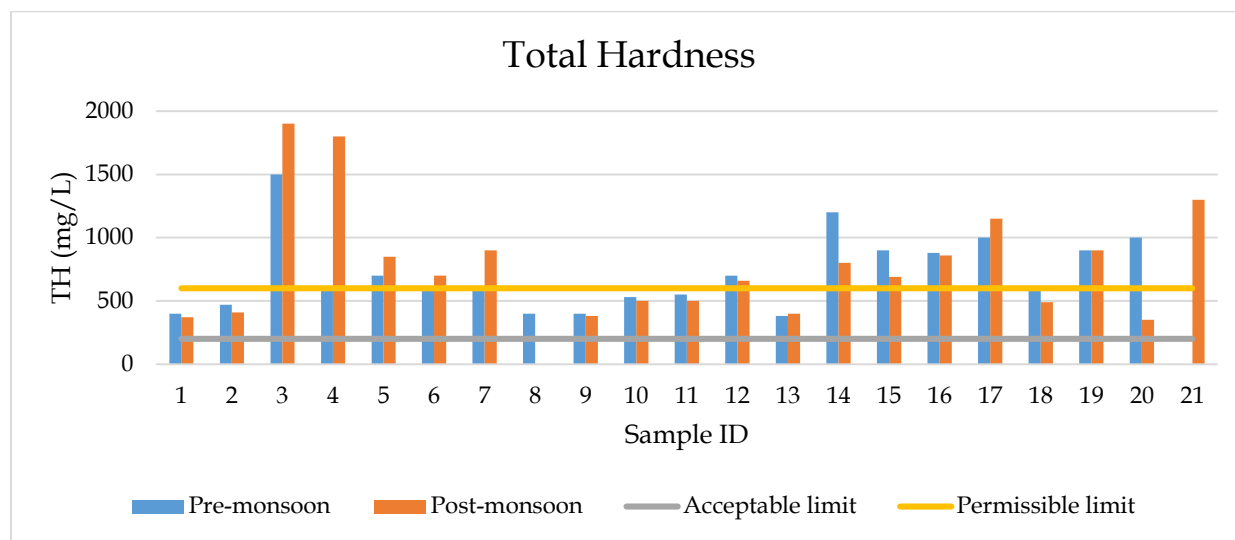
Chloride occurs in all natural water in widely varying concentration. The chloride content normally increases as the mineral content increases and it is also important in detecting the contamination of groundwater by sewage. ^[17] In the present study, Chloride ranged between 88 mg/L and 1500 mg/L for pre-monsoon season, whereas 61 mg/L and 1452 mg/L for post-monsoon season. Almost all the analyzed samples were found well within permissible limit except for sample IDs 3 and 3, 4; which were exceeding permissible limit for pre-monsoon and post-monsoon season respectively. Thus, most of the analyzed sources are free from Chloride and ultimately less TDS concentration in water.



Graph 7.6 Seasonal variation of Chloride

Total Hardness (mg/L)

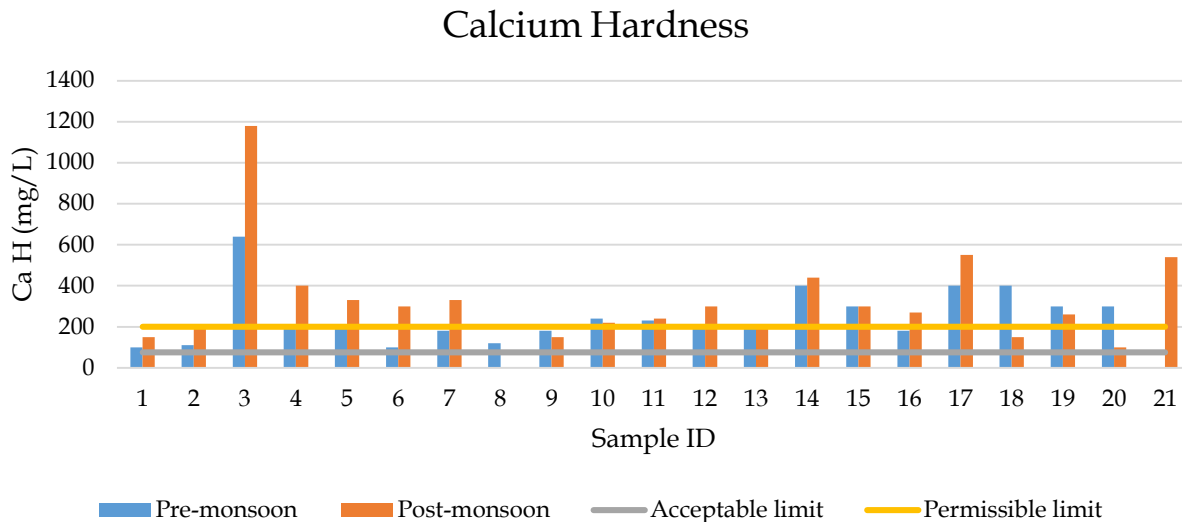
The optimum range of hardness in drinking water is from 80 mg/L to 100 mg/L. Water with hardness greater than 200 mg/L is considered poor in most regions of the province and water with hardness greater than 500 mg/L is normally considered unacceptable for domestic purposes. [21] In the present study, TH ranged between 380 mg/L and 1500 mg/L for pre-monsoon season, whereas 350 mg/L and 1900 mg/L for post-monsoon season. High variations were observed in result and sample IDs 3, 5, 6, 12, 14, 15, 16, 17, 19, 20 and 3 to 7, 12, 14 to 17, 19, 21 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 7.7 Seasonal variation of TH

Calcium Hardness (mg/L)

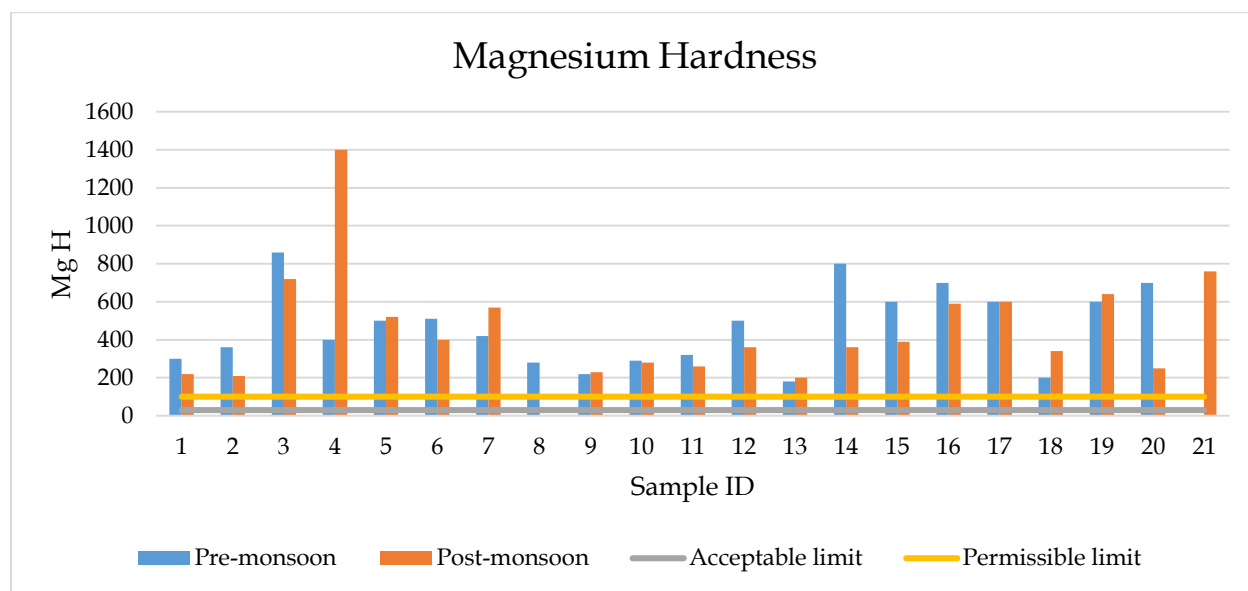
In the present study, Ca-H ranged between 100 mg/L and 640 mg/L for pre-monsoon season, whereas 100 mg/L and 1180 mg/L for post-monsoon season. The lowest value found in Angadh and Mandvapura for pre and post monsoon season, however the highest value found in Radhiyapur for both pre and post monsoon season respectively. Sample IDs 3, 10, 11, 14, 15, 17, 18, 19, 20 and 3 to 7, 10, 11, 12, 14 to 17, 19, 21 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 7.8 Seasonal variation of Ca H

Magnesium Hardness (mg/L)

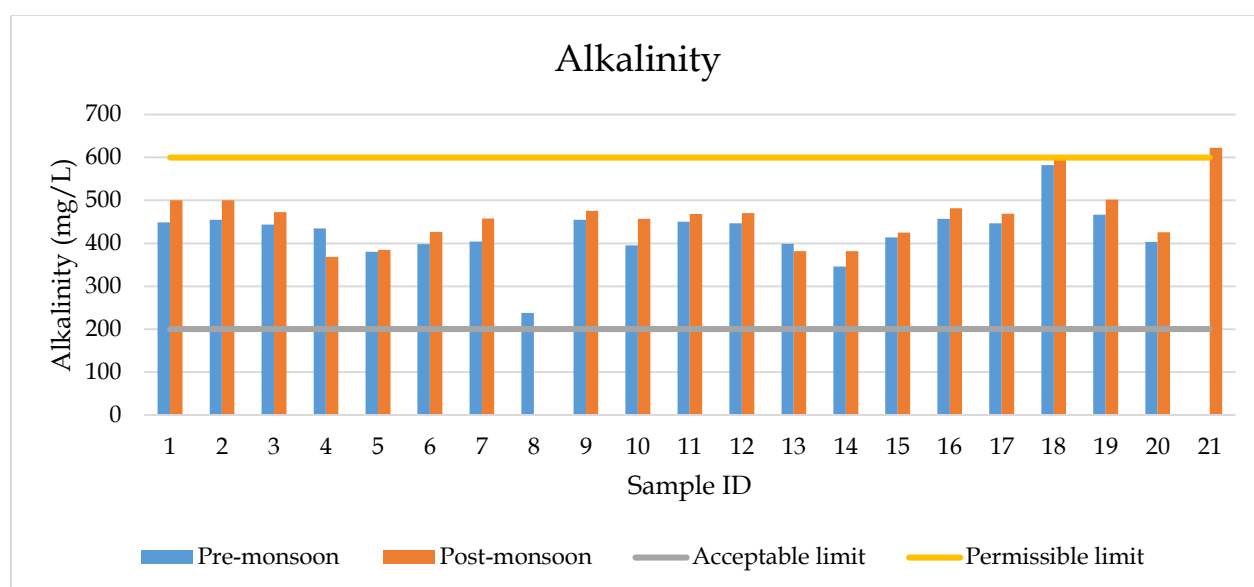
In the present study, Mg-H ranged between 180 mg/L and 860 mg/L for pre-monsoon season, whereas 200 mg/L and 1400 mg/L for post-monsoon season. The lowest value found in Rayka for both pre and post monsoon season, however the highest value found in Radhiyapur and Rupapura for pre and post monsoon season respectively. Sample IDs 1 to 20 and 1 to 7, 9 to 21 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 7.9 Seasonal variation of Mg H

Alkalinity (mg/L)

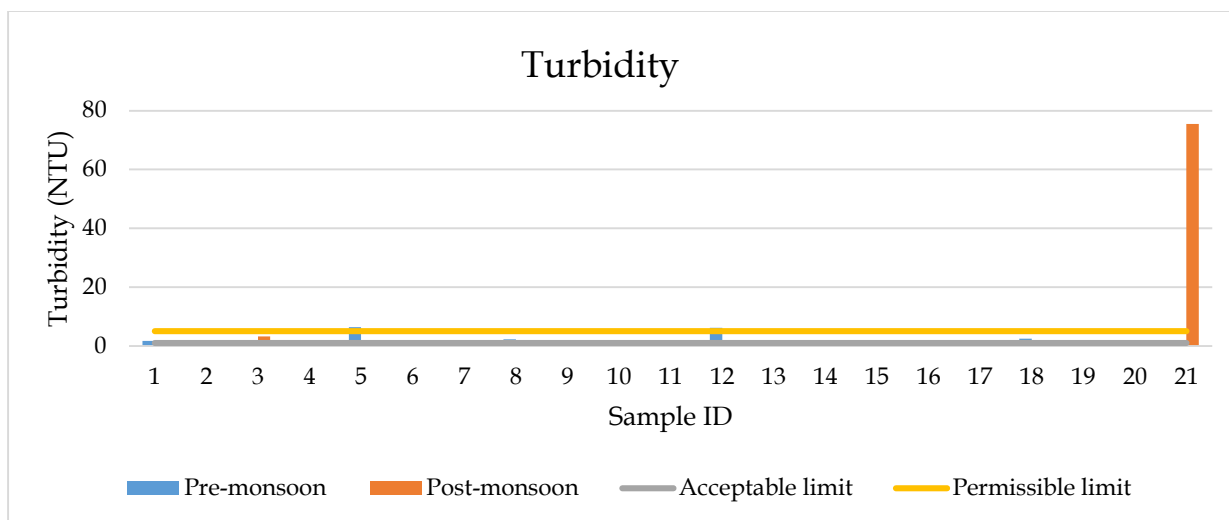
Alkalinity refers to the capacity of water to neutralize acid. In uncontaminated water, alkalinity is primarily a measure of dissolved bicarbonate and carbonate; concentrations of other acid- consuming solutes such as hydroxide, silicate, borate, and dissolved organic compounds are usually small compared to the amount of bicarbonate and carbonate. Alkalinity is reported as the equivalent amount of calcium carbonate in mg/L. [25] In the present study, Alkalinity ranged between 238 mg/L and 582 mg/L for pre-monsoon season, whereas 368 mg/L and 623 mg/L for post-monsoon season. Sample IDs 18, 21 were exceeding permissible limit for post-monsoon season, whereas no sample reported beyond permissible limit for pre-monsoon season. Hence, it can be said that almost all the analyzed samples are free from traces of Alkalinity.



Graph 7.10 Seasonal variation of Alkalinity

Turbidity (NTU)

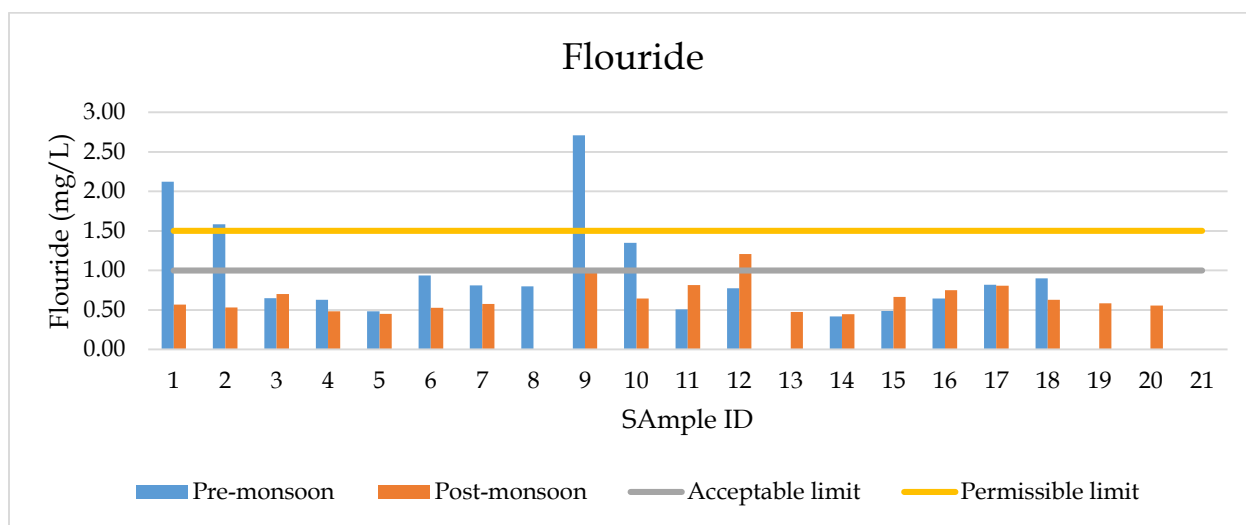
In the present study, Turbidity ranged between 0.1 NTU and 6.4 NTU for pre-monsoon season, whereas 0.01 NTU and 75.5 NTU for post-monsoon season. A few samples have been seen turbid during field sampling, whereas most of the sources have clear water. Sample IDs 5, 12 and 21 were exceeding permissible limit for pre-monsoon and post-monsoon respectively.



Graph 7.11 Seasonal variation of Turbidity

Fluoride (mg/L)

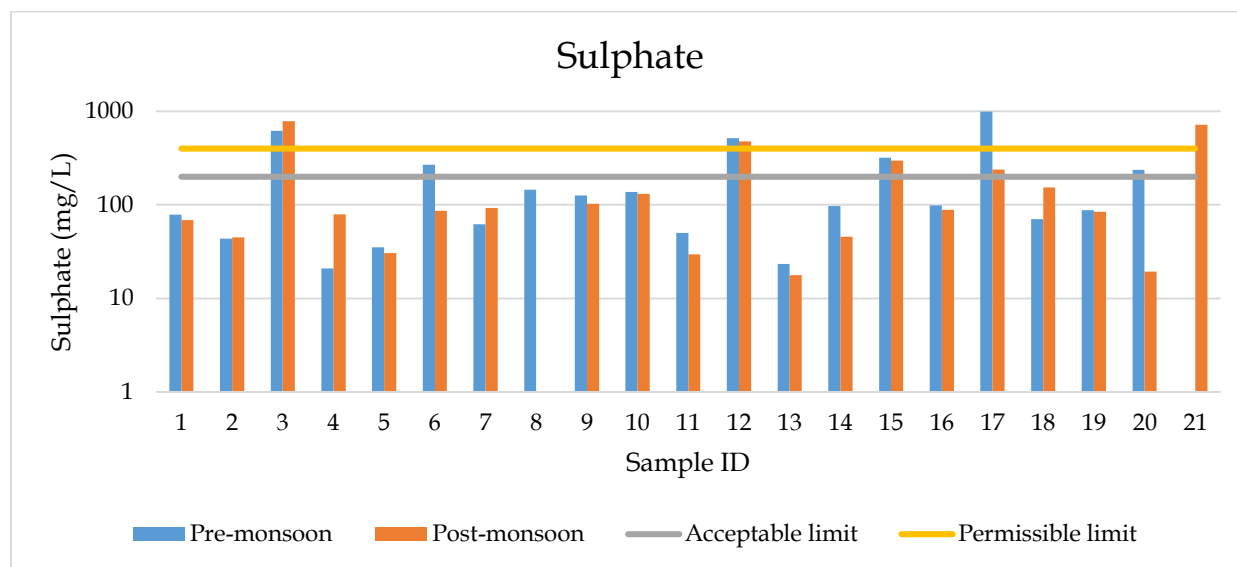
Fluorine is the lightest member of the halogen group and is one of the most reactive of all chemical elements. In groundwater, low or high concentrations of fluoride can occur, depending on the nature of the rocks and the occurrence of fluoride-bearing minerals. Fluoride is necessary in humans in minute quantities during tooth formation, to harden the tooth enamel. In the present study, Fluoride ranged between 0.42 mg/L and 2.71 mg/L for pre-monsoon season, whereas 0.45 mg/L and 1.21 mg/L for post-monsoon season. Most of the analyzed samples were found within permissible limit except for sample IDs 1, 2, 9; which were exceeding permissible limit for pre-monsoon season, however no sample resulted above permissible limit for post-monsoon season. Hence, almost all the samples are free from Fluoride pollution.



Graph 7.12 Seasonal variation of Flouride

Sulphate (mg/L)

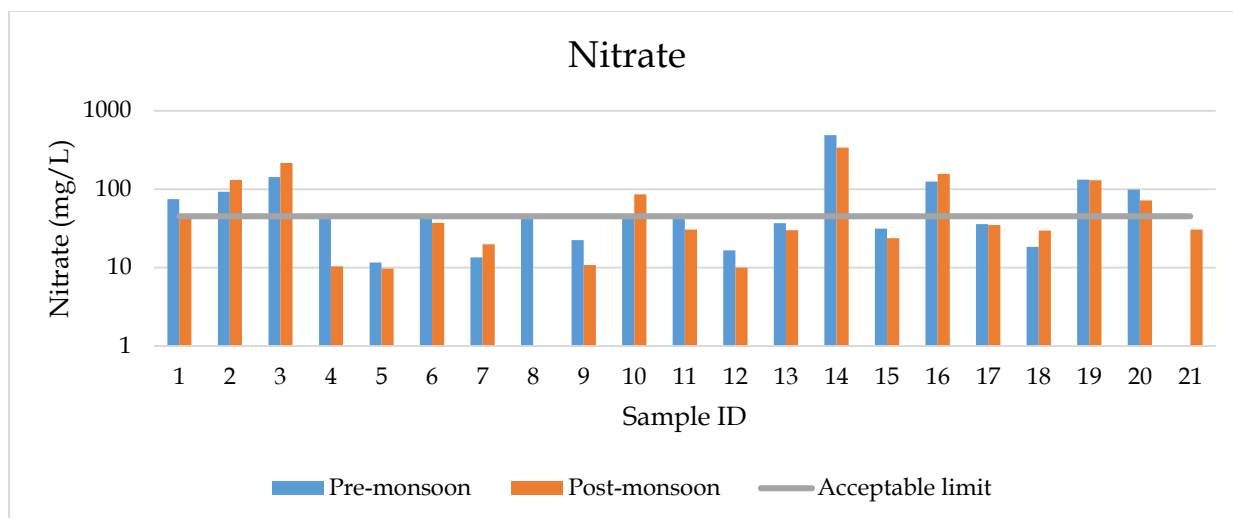
Sulphate concentration may be caused by the oxidation process of their ores, they are also present in industrial wastes. The sulphate ion is one of the major anions occurring in natural water. In the present study, sulphate ranged between 21 mg/L and 991 mg/L for pre-monsoon season, whereas 18 mg/L and 787 mg/L for post-monsoon season. Almost all the analyzed samples were resulted within permissible limit except for a few. Sample IDs 3, 12, 17 and 3, 12, 21 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 7.13 Seasonal variation of Sulphate

Nitrate (mg/L)

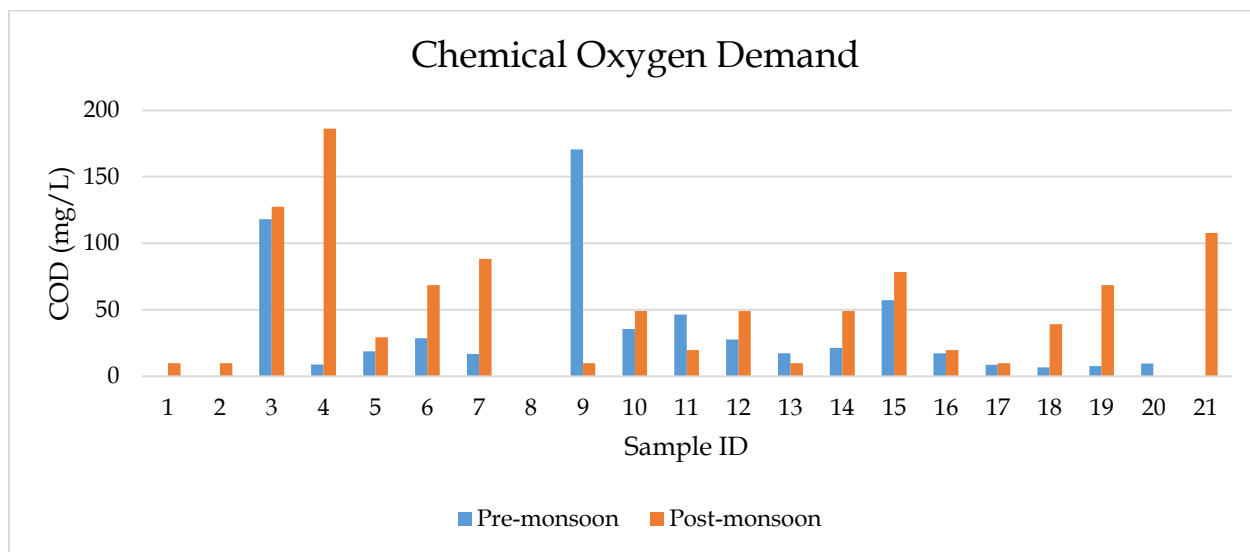
Relatively little of the nitrate found in natural waters is of mineral origin, most coming from organic and inorganic sources, the former including waste discharges and the latter comprising chiefly artificial fertilisers. However, bacterial oxidation and fixing of nitrogen by plants can both produce nitrate. Groundwater contamination can directly affect human health because excessive levels of nitrate in drinking water can produce negative health impacts on human well-being. In the present study, Nitrate ranged between 12 mg/L and 487 mg/L for pre-monsoon season, whereas 10 mg/L and 338 mg/L for post-monsoon season. Sample IDs 1, 2, 3, 6, 10, 11, 14, 16, 19, 20 and 1, 2, 3, 10, 14, 16, 19, 20 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively. Several analyzed samples were exceeding acceptable limit and hence it can be said that sampled sources are polluted with Nitrate concentration.



Graph 7.14 Seasonal variation of Nitrate

Chemical Oxygen Demand (mg/L)

COD is defined as the amount of oxygen equivalents consumed in oxidizing the organic compounds of samples by strong oxidizing agents such as dichromate or permanganate. It is expressed in milligrams per liter (mg/L) that indicates the mass of oxygen consumed per liter of solution. The higher the chemical oxygen demand, the higher the amount of pollution in the water sample. ^[20] In the present study, COD ranged between 7 mg/L and 170 mg/L for pre-monsoon season, whereas 10 mg/L and 186 mg/L for post-monsoon season. COD levels reported for most of the sampled sources reflects contamination of groundwater from anthropogenic/industrial activities across the industrial cluster.



Graph 7.15 Seasonal variation of COD

Ammonical Nitrogen

It is also commonly found in groundwater due to anthropogenic activities, primarily due to leaching from fertilizers, organic waste disposal or leaking sewage systems. ^[18] $\text{NH}_3\text{-N}$ in groundwater aquifer reacts through complex physical- chemical and biological processes to form other associated nitrogen bearing compounds such as NO_2 , NO_3 which makes the risk even more complicated. Their toxicity is attributed to the influence of pH, temperature, and dissolved oxygen concentration. ^[19] In the present study, Ammonical Nitrogen was analyzed for all the collected samples for both pre-monsoon and post-monsoon season. However, none of the sample exceeded concentration of 1 mg/L for both pre-monsoon and post-monsoon season except for sample ID 17 which showed concentration of 1.8 mg/L and 2.95 mg/L Ammonical Nitrogen is present in the water for pre-monsoon and post-monsoon season respectively.

Heavy Metals

Heavy Metals Zn, Cd, As, Pb, Cr, Fe and Cr(VI) were analyzed in a few samples collected during sampling for both pre and post monsoon seasons. The parameter wise inference drawn is covered in table 7.6 along with the details of sample IDs exceeding the permissible limit for respective seasons. Heavy metals in general were analyzed for 5 samples and 14 samples for pre and post monsoon season respectively.

Table 7.6 Details of Heavy Metals analyzed

Sample IDs analyzed for Heavy Metals	
Pre-monsoon	3, 6, 8, 15, 19
Post-monsoon	1, 2, 3, 6, 9, 10, 11, 14, 15, 16, 17, 19, 20, 21
Parameters	Inferences drawn
Zinc	All of the samples analyzed were resulted below 0.5 mg/L for both pre-monsoon and post-monsoon seasons.
Cadmium	Concentration of Cadmium was not exceeding 0.002 mg/L for any of the sample for both pre-monsoon and post-monsoon seasons, hence each sampled source is free from cadmium pollution.
Arsenic	Each analyzed source is free from arsenic pollution as all the samples showed arsenic concentration below 0.005 mg/L.
Chromium	Acceptable limit for chromium concentration in drinking water is 0.05 mg/L and all the samples analyzed resulted within acceptable limit.
Lead	Two sample IDs i.e. 3 and 21 showed lead concentration of 0.013 mg/L and 0.029 mg/L which exceeds acceptable limit of 0.01 mg/L. Remaining samples resulted below 0.002 mg/L.
Iron	Iron concentration of 1.40 mg/L was reported for sample ID 21 during post-monsoon sampling, which is exceeding acceptable limit of 0.3 mg/L. Remaining samples found within the limit.

Hexavalent Chromium	Sample ID 6 was analyzed for this parameter for post-monsoon season only. However, it resulted in concentration below 0.01 mg/L.
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7.4 Conclusion

Groundwater sampling was carried out around the industrial cluster of Nandesari wherein; 20 samples were collected from 20 selected locations for both pre and post monsoon seasons. From the analytical results and monitoring carried out following conclusions are drawn:

- The overall groundwater quality around industrial cluster is found to be good but varied with respect to all physico-chemical parameters except for hardness wherein a few sample exceeded limits for different parameters. Sample IDs and location as 3 (Bore well of Radhiyapur), 4 (Bore well of Rupapura) and 21 (additional sample collected during post-monsoon near Angadh near Meldi Mata temple) were found inferior with respect to all physico-chemical parameters including heavy metal contamination in terms of Iron, and Lead contamination raising a concern on application which was mainly reported as irrigation for contaminated sources with IDs 3, 4 and 21.
- The color as well as chemical oxygen demand analyzed in the samples reflects presence of pollution in groundwater ranging from smaller to greater extent. Many samples have reported elevated concentration of Nitrate in groundwater raising a concern on its respective application.
- Thus the overall groundwater quality of the monitored cluster is found to be good except for a few samples and is also found to be moderately hard.

Chapter: 8 Vatva GIDC

8.1 Overview of Ahmedabad (Vatva) Industrial Cluster

Vatva is located in one of the industrially developed district Ahmedabad having Large, Micro, Small and Medium scale industries of various types. Ahmedabad is an Industrial hub for textiles and is popularly known as the “Manchester of India”. A well-developed infrastructure, prudent industrial policy of the state government and a peaceful industrial atmosphere have been the vital factors which have contributed to the industrial growth in the district. There are 12 main Industrial Estates, 12 Special Economic Zones, and 10 Industrial Parks/Developers in Ahmedabad district. ^[15]

Table 8.1: Ahmedabad district at a glance

S. No.	Particulars	Information
1.	Geographical information	21.58 to 23.3 N and 71.37 to 72.50 E
2.	Geographical area (sq. km)	8087.59
3.	Average annual rainfall	612
4.	Average temperature	34.4 °C (Summer), 20.5 °C (Winter)
5.	Population	72,08,200
6.	Literacy	88.29%
7.	Stage of Groundwater Development (Dascroi tehsil)	103.73% - Over Exploited

(Source: Ground Water Brochure Ahmedabad District ^[14])

Types of industries

Various industries in Ahmedabad manufacture diverse range of products like dyes and dye intermediates, pigments, pharmaceuticals, paints, and so on. Some of the major categories of industries is shown in table:

Table 8.2: Chemical products manufactured in Vatva GIDC

Types of industries	Textiles, chemicals, machinery, metal products, pharmaceutical, engineering, plastics, electrical appliances, electronics, passenger cars etc.
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(Source: Brief Industrial Profile of Ahmedabad District ^[15])

8.2 Sampling and monitoring

For the purpose of study sampling sites were selected within 5 km buffer region across the GIDC area. The details of the sampling sites selected is described in table 8.3 and figure 8.1.

Table 8.3 Sampling locations around Vatva GIDC

Villages/locations selected for Sampling & Monitoring		
Within 1 km (7 samples)	Within 3 km (4 samples)	Within 5 km (9 samples)
Vinzol, Parsi Dungaravadi, Ayojan Nagar, Jashoda Nagar, Ghodasar, Ramol, New Maninagar	Vatva, Paras Nagar, Bibipura, Memadpura	Geratpur, Vanch, Ropda, Piplaj, Gulab Nagar, Al-Mubarak Raw House (A-Block), Vrindavan Nagar, Ganesh Nagar, Sana Residency

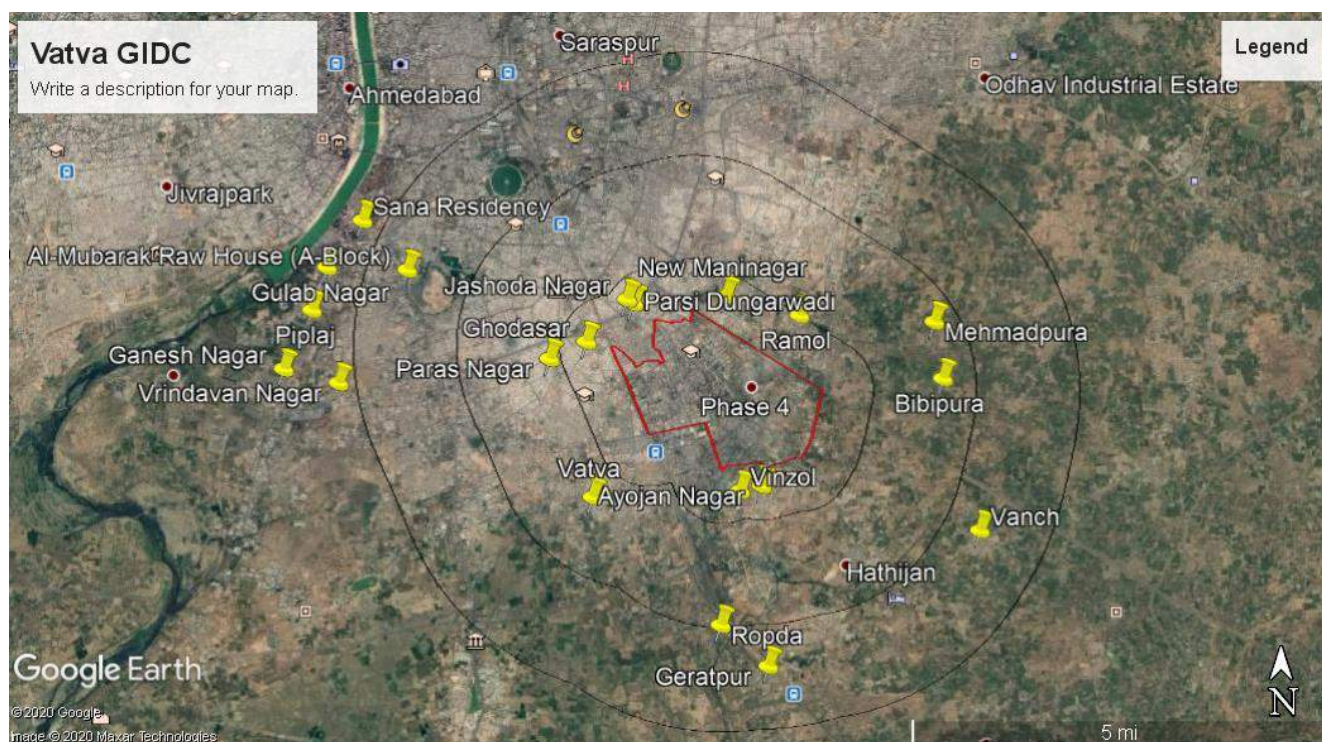


Figure 8.1 Sampling locations around Vatva GIDC

Samples were collected from the identified location for both pre and post monsoon seasons. The information about sampling location, source type, color appearance at field, depth of source, population dependent on source, water usage and name of person contacted during the time of sampling is summarized in table 8.4.

Table 8.4 Details collected during sampling around Vatva GIDC

S. ID	Village/ Site	Location	Latitude Longitude	Source type	Color Appearance@ Field	Depth of well	Population dependent on source	Water Usage	Persons Contacted	Remarks
1.	Vinzol	Near lake	22.951808 72.643002	Bore well	Clear	500	Industries in GIDC	Drinking, Industrial	Shambhu bhai Rathod	-
2.	Vatva	Roopvila bungalows	22.948307 72.616685	Bore well	Clear	220	15 veegha	Irrigation	Bhadresh bhai Patel	Irrigation activity nearby
3.	Paras Nagar	Near Pushpak society	22.971882 72.608553	Bore well	Clear	360	500	Drinking, Domestic	Shavji bhai Bharwad	Source is located near Canal. Water is used for drinking after filtration.
4.	Ghodasar	Nr. Ghanshyam Nagar society	22.983227 72.610539	Bore well	Clear	60	Household	Household uses	Shyam bhai	-
5.	Jashoda Nagar	Nr.Jashodanaga r Police Chowki	22.983069 72.625257	Bore well	Clear	300	Household	Drinking, Construct ion work	Jagdish bhai	-
6.	Parsi Dungarwadi	Near Jashodanagar Police Chowki	22.981987 72.624377	Bore well	Clear	900	110	Domestic	Parag bhai	-
7.	Vanch	Nr. Brhamani Mata Temple	22.943619 72.690366	Bore well	Clear	350	-	Drinking	Fataji Thakor	-
8.	Bibipura	Near Lord Shiva Temple	22.965889 72.67909	Bore well	Clear	270	-	Drinking, Domestic	Mahesh bhai Zala	-
*	Bibipura	Near Lord Shiva Temple	22.9658897 2.67909	Bore well	Clear	270	-	Drinking	Sarpanch of Bibipura	Previous source is not in working condition, hence sample taken from newly dug bore well
9.	Mehmadpura	Near Joganimata Tample	22.978403 72.681208	Bore well	Clear	450	-	Drinking	Ambalal Thakor	-
10.	Ropda	Near Angamwadi	22.92927 72.639468	Bore well	Turbid	200	-	Irrigatio, Domestic	Pintu bhai	Near Agriculture area
11.	Geratpur	Near Bariyadev Temple	22.921559 72.646643	Bore well	Turbid	300	-	Irrigation	Bharat bhai Patel	Near Agriculture area

12.	Ayojan Nagar	Near Shiv Temple	22.950365 72.648629	Bore well	Clear	500	-	Drinking, Domestic	Kanaksinh Zala	-
13.	New Maninagar	Trimurti Sanidhya	22.483152 72.638998	Bore well	Clear	300	170	Drinkin, Domestic	Laxmanbhai Bhavsar	Water is used for drinking after filtration.
14.	Ramol	Near lake	22.991536 72.660108	Bore well	Clear	700	-	Drinking, Industrial uses	Nawaz Saiyad	-
15.	Piplaj	Near S. S. Motors, Narol-Piplaj cross road	22.980803 72.563603	Bore well	Slight Turbid	350	-	Construct ion activities	-	Salty taste
16.	Gulab Nagar	Near Masjid	22.988292 72.565974	Bore well	Clear	620	-	Drinking, Domestic	M. K. Patel	-
17.	Al-Mubarak Raw House (A-Block)	Near Chandola lake	22.98792 72.58154	Bore well	Turbid	260	-	Drinking, Domestic	Nasir bhai	Change in color during summer season
18.	Vrindavan Nagar, Narol	Near Shubham Industrial Park	22.967985 72.56845	Bore well	Clear	200	-	Drinking, Domestic	-	Slight salty taste
19.	Ganesh Nagar, Piplaj	Near Anganwadi	22.970602 72.558284	Bore well	Clear	500	-	Drinking, Domestic	Ambaram bhai	Turbid in summer season
20.	Sana Residency	Near Shell Petrol Pump, Piplaj-Pirana Road	22.996359 72.572982	Bore well	Turbid	300	-	Domestic	-	Water seems turbid for half the year

Note:

- * different sample source has taken for post-monsoon season as previously monitored source was found sealed (not in use condition) due to contamination.
- Color code given to sample ID



Color Code is given to samples collected within 1 km buffer



Color Code is given to samples collected within 3 km buffer



Color Code is given to samples collected within 5 km buffer



8.3 Result and Discussion

The physico-chemical parameters of groundwater samples with their acceptable and permissible limits as recommended by Indian Standards Drinking Water - Specification (Second Revision) (IS 10500: 2012) are shown in below table:

Table 8.5 Result range of pre-monsoon and post-monsoon sampling

Sr. No.	Parameters	Measured Units	Requirement (Acceptable Limit)	Permissible limit in the absence of alternate source	Range of present study (Pre-monsoon)	Range of present study (Post-monsoon)
1	pH	-	6.5 – 8.5	No Relaxation	7.05 – 8.28	7.19 – 7.84
2	Colour	Hazen	5	15	5 – 40	5 – 50
3	Electrical Conductivity	µS/cm	-	-	407 – 4850	1027 – 4350
4	TDS	mg/L	500	2000	208 – 2512	584 – 2436
5	TSS	mg/L	-	-	2 – 20	0 – 2
6	COD	mg/L	-	-	8 – 48	10 – 60
7	Chloride as Cl ⁻	mg/L	250	1000	42 – 941	70 – 814
8	Total Hardness	mg/L	200	600	100 – 1240	130 -540
9	Ca ⁺⁺ Hardness	mg/L	75	200	50 – 300	60 – 200
10	Mg ⁺⁺ Hardness	mg/L	30	100	50 – 940	60 – 340
11	Alkalinity	mg/L	200	600	141 – 730	183 – 711
12	Turbidity	NTU	1	5	0.1 – 19.1	0.2 – 30.2
13	Fluoride	mg/L	1.0	1.5	0.71 – 1.66	0.60 – 3.78
14	Sulphate	mg/L	200	400	16 – 411	28 – 350
15	Nitrate	mg/L	45	No Relaxation	11 – 76	5 – 170
16	Ammonical Nitrogen	mg/L	-	-	0	0
17	Lead	mg/L	0.01	No Relaxation	0	0 – 0.002
18	Cadmium	mg/L	0.003	No Relaxation	0	0

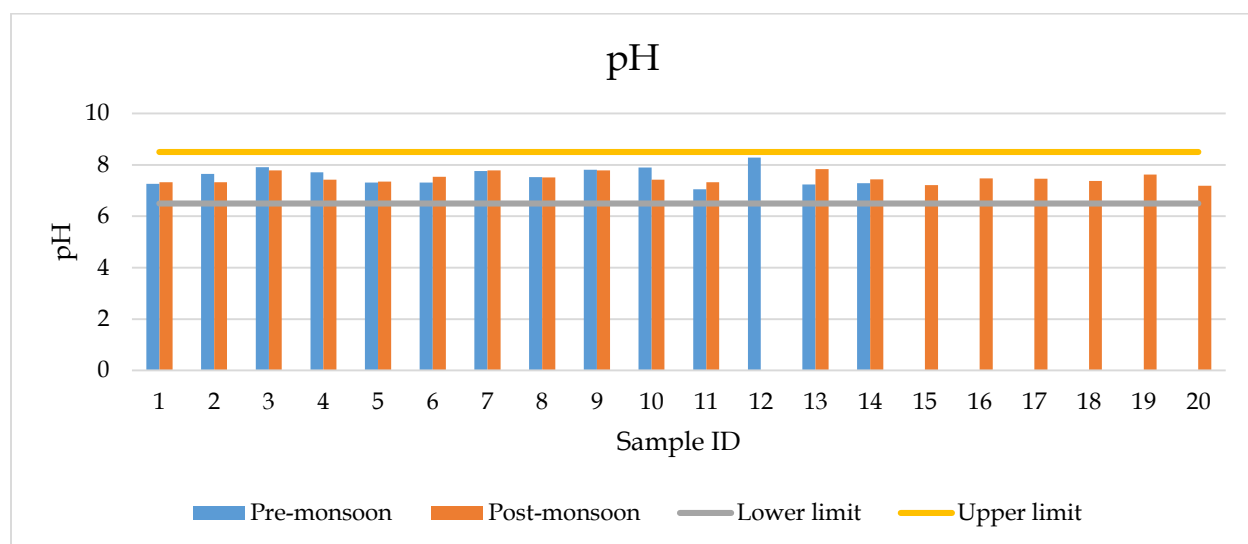
19	Iron	mg/L	0.3	No Relaxation	0	0 – 0.719
20	Chromium	mg/L	0.05	No Relaxation	0	0 – 0.006
21	Zinc	mg/L	5	15	0	0
22	Arsenic	mg/L	0.01	0.05	0	0

Note: The limits are as per the **Indian Standards Drinking Water - Specification (Second Revision) (IS 10500: 2012)** ^[5]

Discussion

pH

In common, water with a pH < 7 is considered acidic and with a pH > 7 is considered basic. The normal range for pH in surface water systems is 6.5 to 8.5 and for groundwater systems 6 to 8.5. The pH of pure water is 7 at 25° C. ^[23] In the present study, the pH ranged between 7.05 and 8.28 for pre-monsoon season whereas 7.19 and 7.84 for post-monsoon season. All the analyzed samples were resulted within permissible limit.

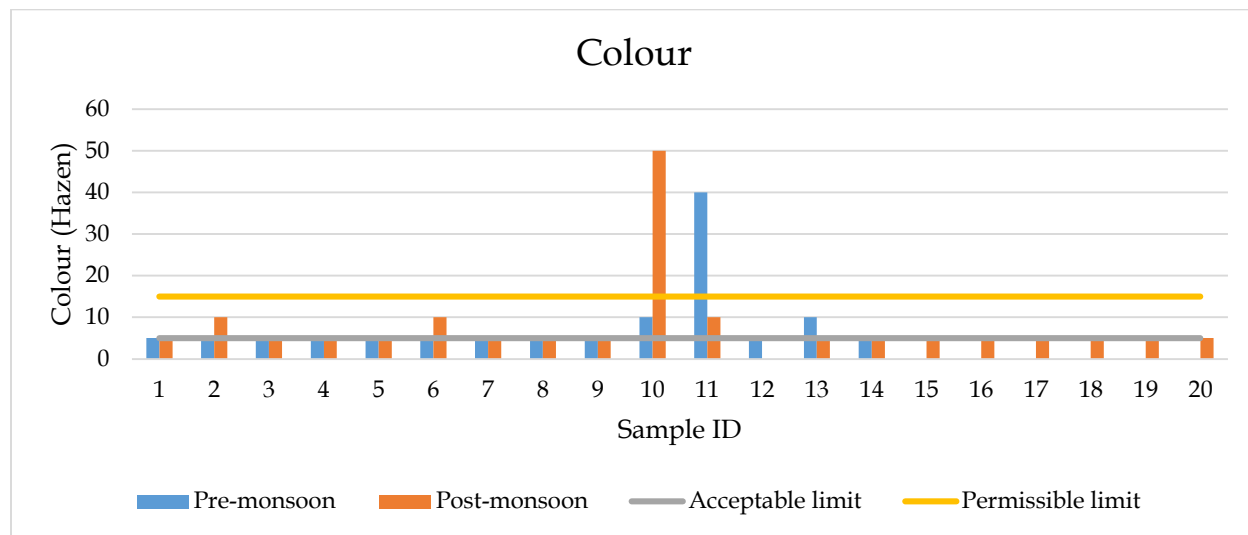


Graph 8.1 Seasonal variation of pH

Color (Hazen)

Suspended and dissolved particles in water influence color. Suspended material in water bodies may be a result of natural causes and/or human activity. Transparent water with a low accumulation of dissolved materials appears blue and indicates low productivity. ^[22] In the present study, color ranged between 5 Hazen and 40 Hazen for pre-monsoon season whereas 5 Hazen and 50 Hazen for post-monsoon season. Almost all the analyzed samples were found well within permissible limit except for a few. Sample IDs 11 and 10

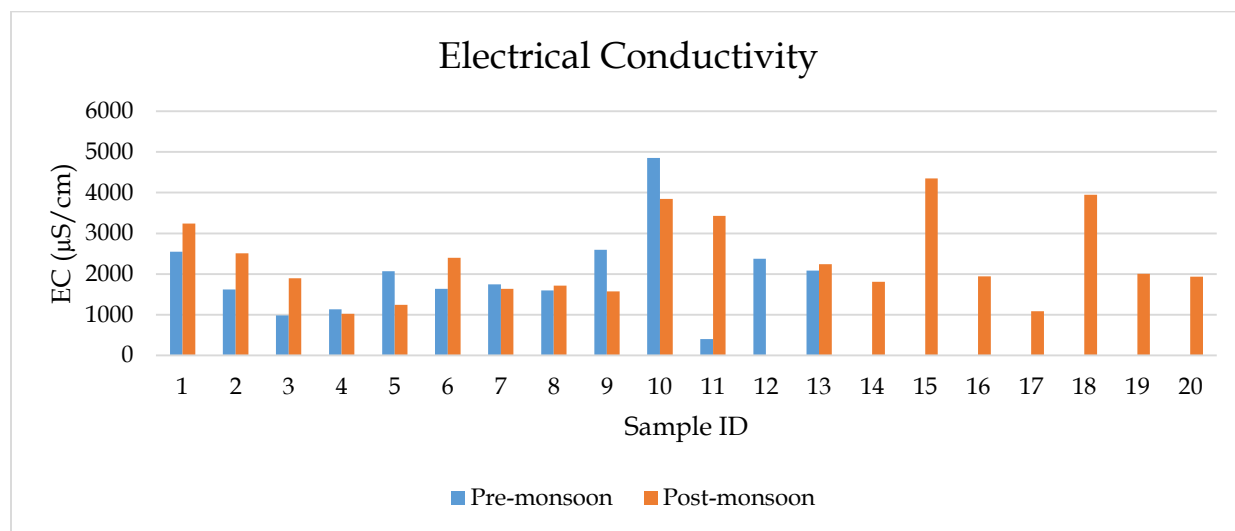
were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 8.2 Seasonal variation of Colour

Electrical Conductivity ($\mu\text{S}/\text{cm}$)

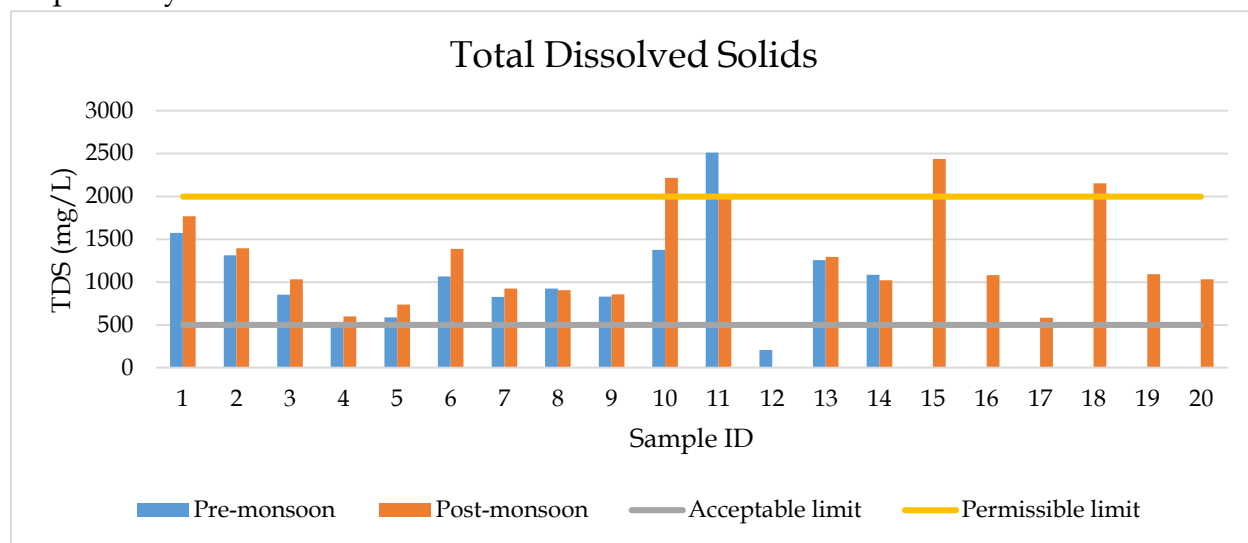
Electrical conductivity is an indirect measurement of salinity, and it is temperature dependent and good indicator of the total salinity, but it does not provide any information about the ionic composition within the water sample. ^[16] In the present study, EC ranged between 407 $\mu\text{S}/\text{cm}$ and 4850 $\mu\text{S}/\text{cm}$ for pre-monsoon season, whereas 1027 $\mu\text{S}/\text{cm}$ and 4350 $\mu\text{S}/\text{cm}$ for post-monsoon season. High variations were observed in the results; however, it does not indicate quality of groundwater.



Graph 8.3 Seasonal variation of EC

Total Dissolved Solids (mg/L)

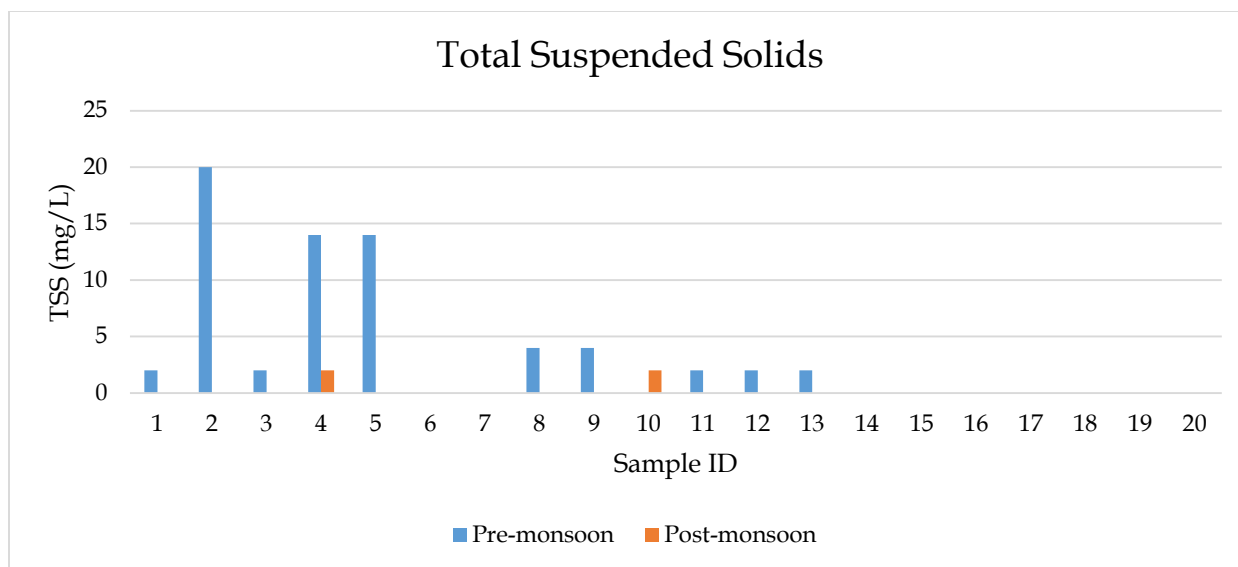
In general, the total dissolved solids concentration is the sum of the cations (positively charged) and anions (negatively charged) ions in the water. Therefore, the total dissolved solids test provides a qualitative measure of the amount of dissolved ions but does not tell us the nature or ion relationships. [24] In the present study, TDS ranged between 208 mg/L and 2512 mg/L for pre-monsoon season, whereas 584 mg/L and 2436 mg/L for post-monsoon season. Most of the analyzed samples showed TDS concentration below permissible limit of drinking water standards except for sample IDs 11 and 10, 15, 18; which were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 8.4 Seasonal variation of TDS

Total Suspended Solids (mg/L)

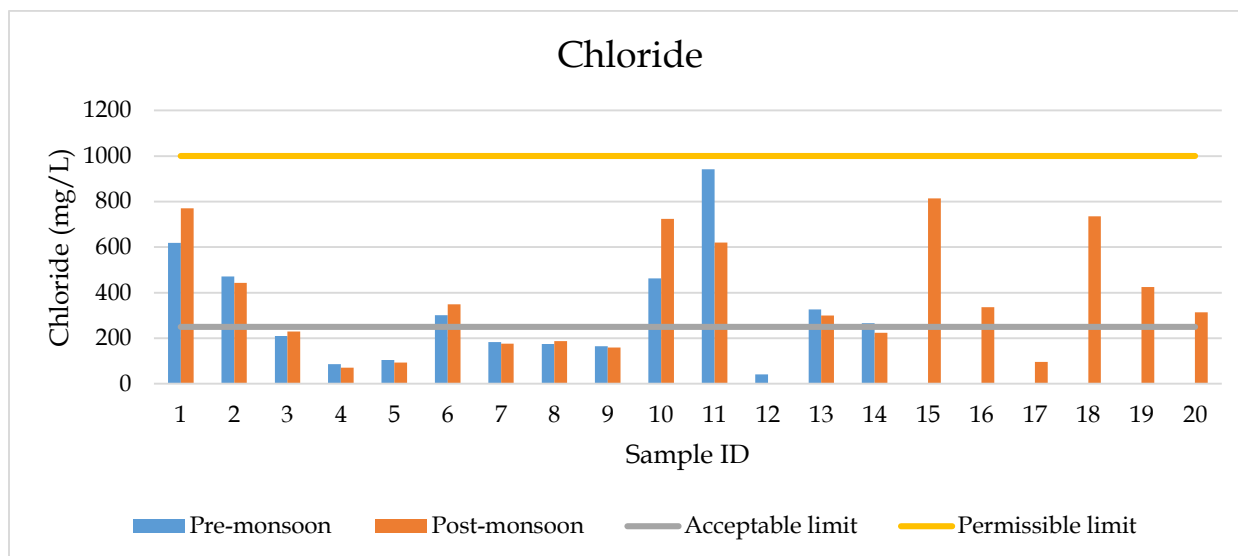
Total suspended solids (TSS) are particles that are larger than 2 microns found in the water column. Anything smaller than 2 microns (average filter size) is considered a dissolved solid. Most suspended solids are made up of inorganic materials, though bacteria and algae can also contribute to the total solids concentration. In the present study, TSS ranged between 0 and 20 mg/L for pre-monsoon season, whereas 0 and 2 mg/L for post-monsoon season. Variations were shown in result during pre and post monsoon sampling.



Graph 8.5 Seasonal variation of TSS

Chloride (mg/L)

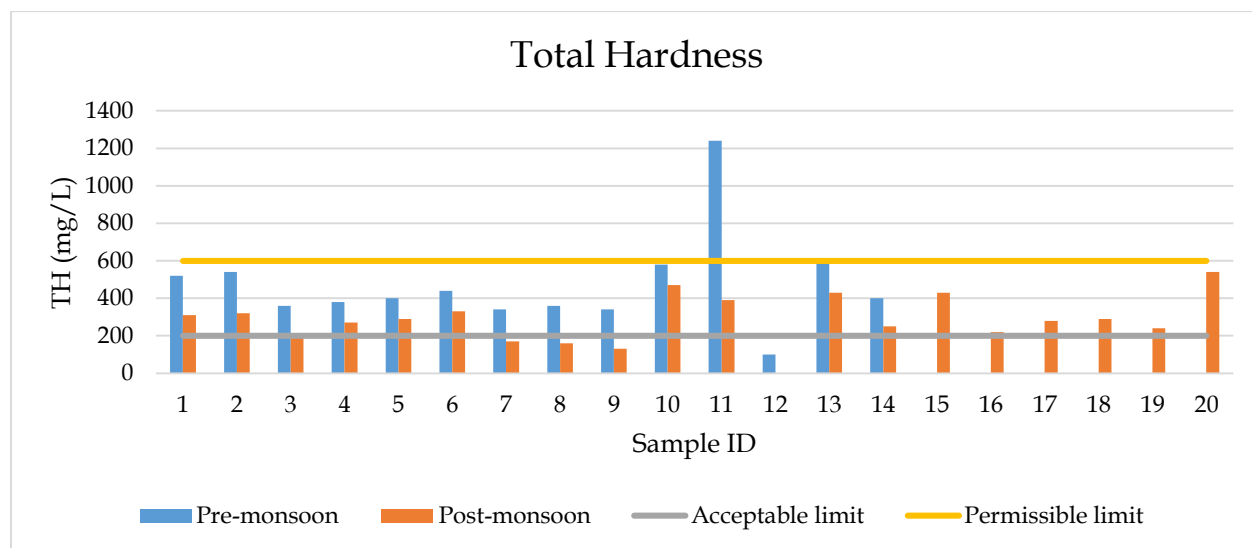
Chloride occurs in all natural water in widely varying concentration. The chloride content normally increases as the mineral content increases and it is also important in detecting the contamination of groundwater by sewage. ^[17] In the present study, Chloride ranged between 42 mg/L and 941 mg/L for pre-monsoon season, whereas 70 mg/L and 814 mg/L for post-monsoon season. All the analyzed samples were found well within permissible limit of drinking water standards for both pre-monsoon and post-monsoon seasons and hence all the sampled sources are free from Chloride traces.



Graph 8.6 Seasonal variation of Chloride

Total Hardness (mg/L)

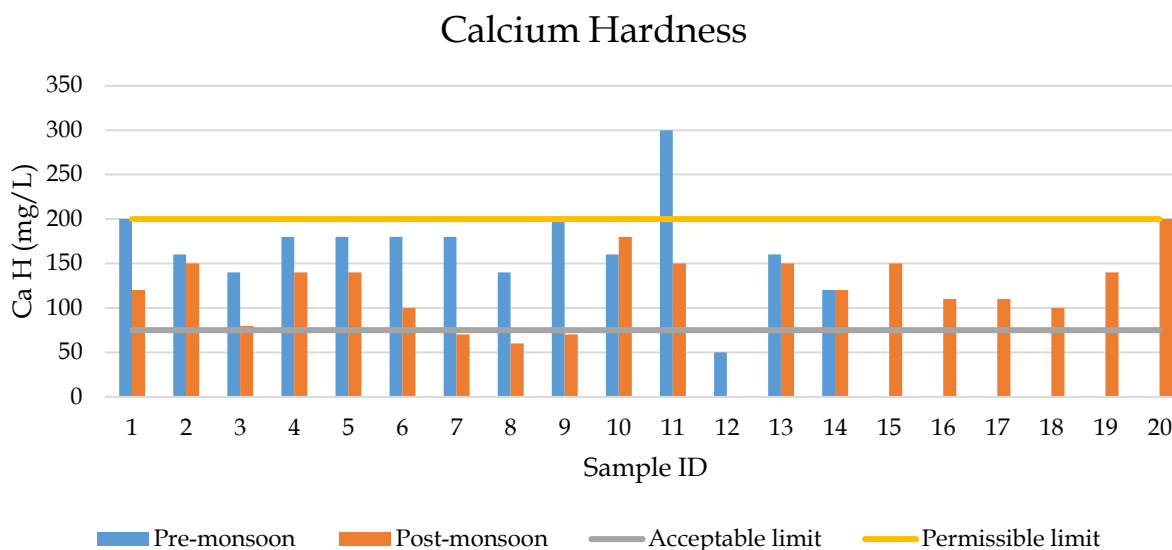
The optimum range of hardness in drinking water is from 80 mg/L to 100 mg/L. Water with hardness greater than 200 mg/L is considered poor in most regions of the province and water with hardness greater than 500 mg/L is normally considered unacceptable for domestic purposes. [21] In the present study, TH ranged between 100 mg/L and 1240 mg/L for pre-monsoon season, whereas 130 mg/L and 540 mg/L for post-monsoon season. All the analyzed samples resulted within permissible limit except for one. Sample ID 11 was exceeding permissible limit for pre-monsoon season, however no sample resulted beyond permissible limit for post-monsoon season. Hence, water seems to be soft water.



Graph 8.7 Seasonal variation of TH

Calcium Hardness (mg/L)

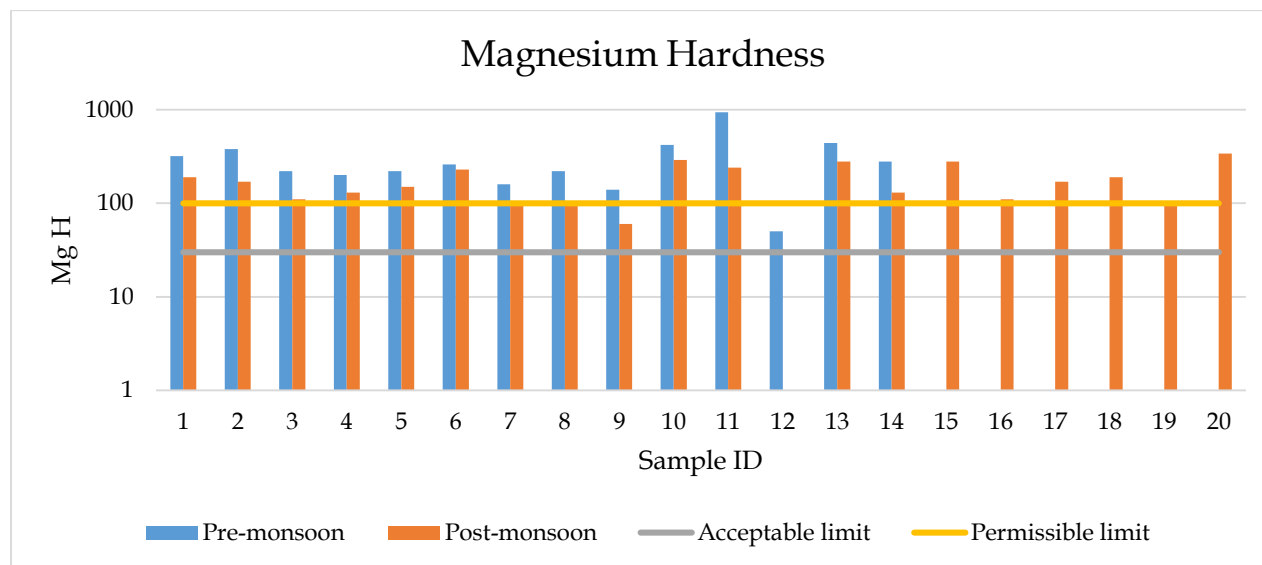
In the present study, Ca-H ranged between 50 mg/L and 300 mg/L for pre-monsoon season, whereas 60 mg/L and 200 mg/L for post-monsoon season. The lowest value found in Ayojan Nagar and Bibipura for pre and post monsoon season, however the highest value found in Geratpur and Sana Residency for pre and post monsoon season respectively. Sample ID 11 was exceeding permissible limit for pre-monsoon season, however no sample resulted beyond permissible limit for post-monsoon season.



Graph 8.8 Seasonal variation of Ca H

Magnesium Hardness (mg/L)

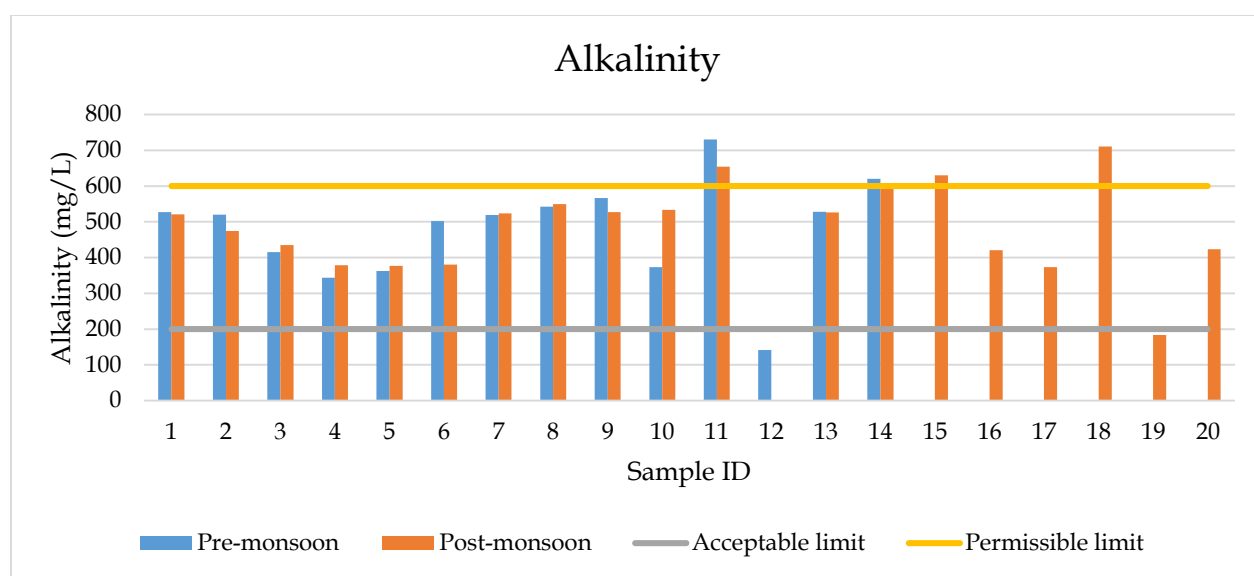
In the present study, Mg-H ranged between 50 mg/L and 940 mg/L for pre-monsoon season, whereas 60 mg/L and 340 mg/L for post-monsoon season. The lowest value found in Ayojan Nagar and Mehmadpura for pre and post monsoon season, however the highest value found in Geratpur and Sana Residency for pre and post monsoon season respectively. Sample IDs 1 to 11, 13, 14 and 1, 2, 10, 11, 13 to 18, 20 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 8.9 Seasonal variation of Mg H

Alkalinity (mg/L)

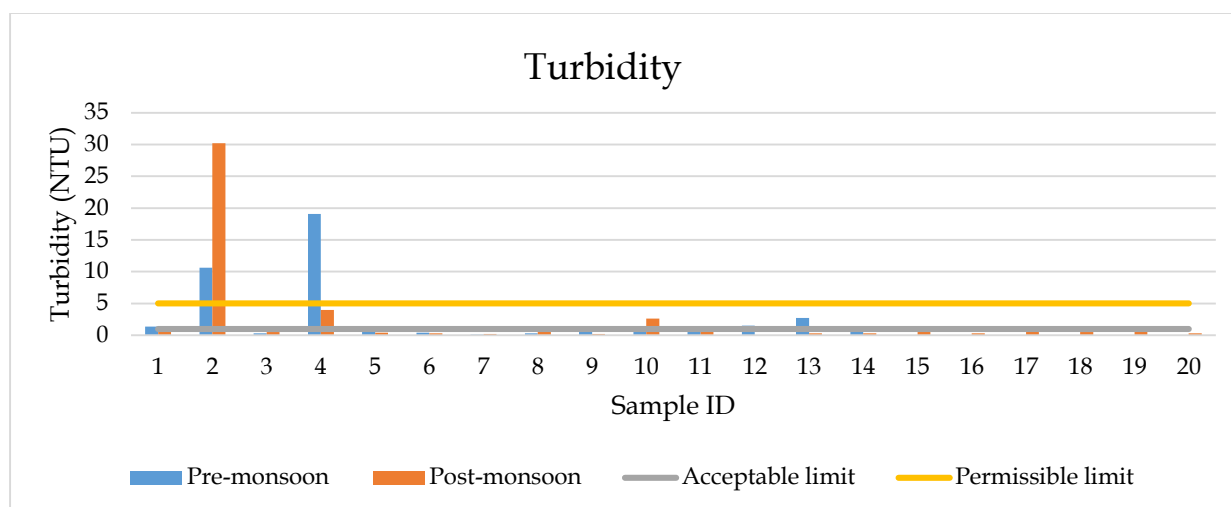
Alkalinity refers to the capacity of water to neutralize acid. In uncontaminated water, alkalinity is primarily a measure of dissolved bicarbonate and carbonate; concentrations of other acid- consuming solutes such as hydroxide, silicate, borate, and dissolved organic compounds are usually small compared to the amount of bicarbonate and carbonate. Alkalinity is reported as the equivalent amount of calcium carbonate in mg/L. [25] In the present study, Alkalinity ranged between 141 mg/L and 730 mg/L for pre-monsoon season, whereas 183 mg/L and 711 mg/L for post-monsoon season. Most of the analyzed samples resulted within permissible limit of drinking water standards except for sample IDs 11, 14 and 11, 14, 15, 18; which were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 8.10 Seasonal variation of Alkalinity

Turbidity (NTU)

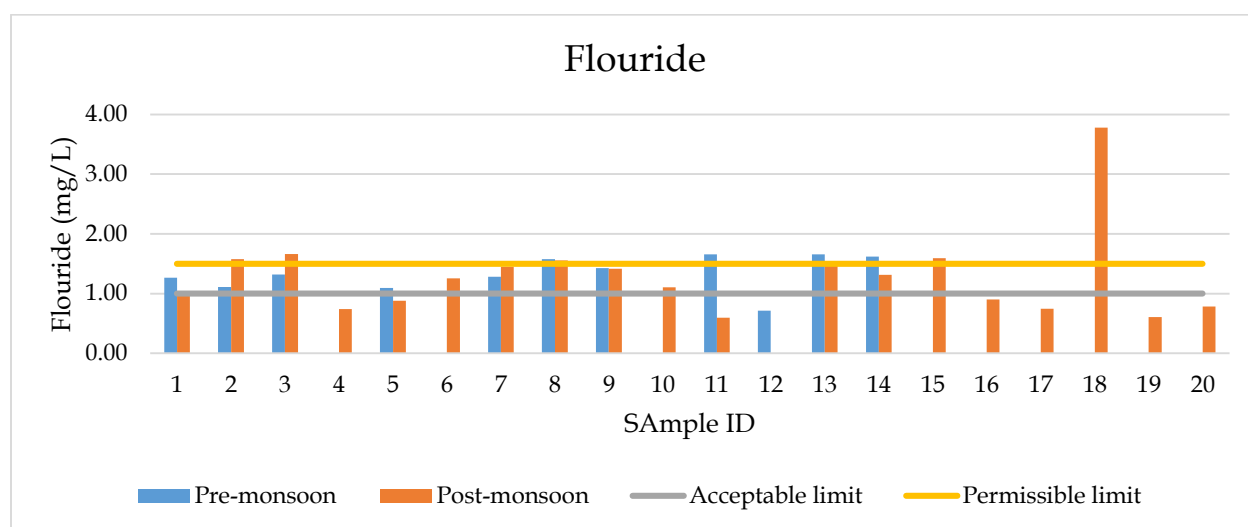
In the present study, Turbidity ranged between 0.1 NTU and 19.1 NTU for pre-monsoon season, whereas 0.2 NTU and 30.2 NTU for post-monsoon season. Almost all the analyzed samples were found well within permissible limit of drinking water standards except for sample IDs 2, 4 and 2; which were exceeding permissible limit for pre-monsoon and post-monsoon season respectively.



Graph 8.11 Seasonal variation of Turbidity

Fluoride (mg/L)

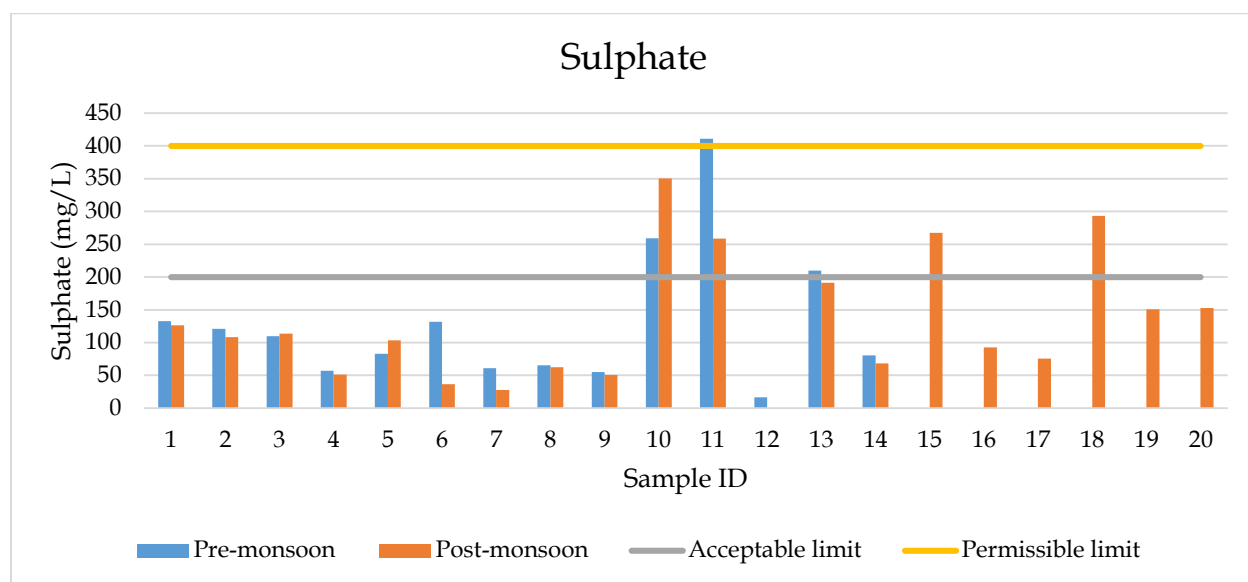
Fluorine is the lightest member of the halogen group and is one of the most reactive of all chemical elements. In groundwater, low or high concentrations of fluoride can occur, depending on the nature of the rocks and the occurrence of fluoride-bearing minerals. Fluoride is necessary in humans in minute quantities during tooth formation, to harden the tooth enamel. In the present study, Fluoride ranged between 0.71 mg/L and 1.66 mg/L for pre-monsoon season, whereas 0.60 mg/L and 3.78 mg/L for post-monsoon season. Sample IDs 8, 11, 13, 14 and 2, 13, 15, 18 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively. Thus, few analyzed sources are not free from Fluoride contamination.



Graph 8.12 Seasonal variation of Fluoride

Sulphate (mg/L)

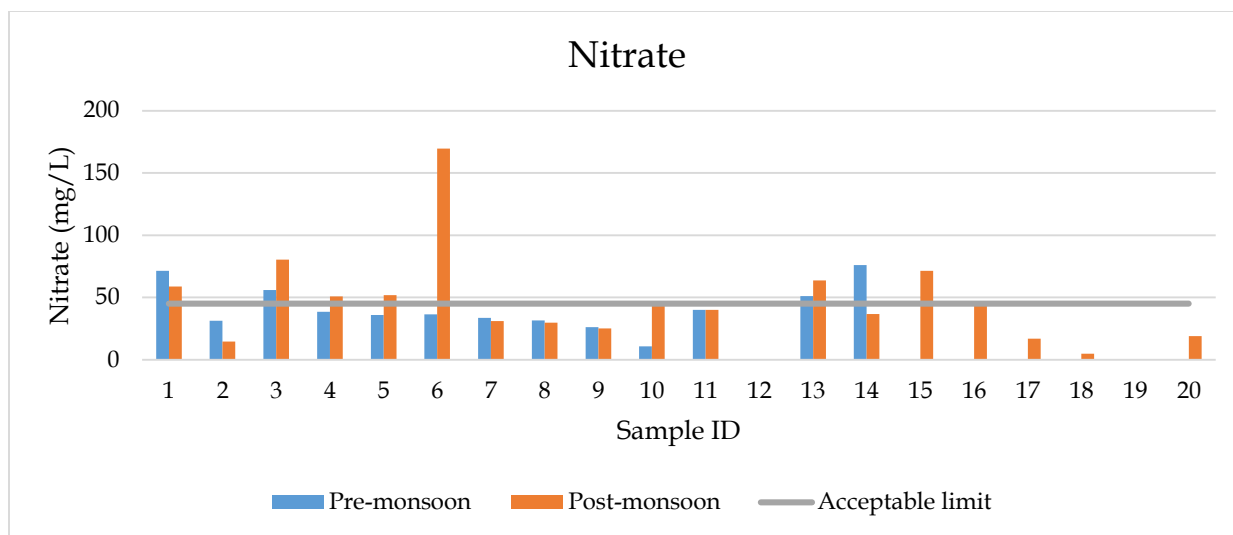
Sulphate concentration may be caused by the oxidation process of their ores, they are also present in industrial wastes. The sulphate ion is one of the major anions occurring in natural water. In the present study, Sulphate ranged between 16 mg/L and 411 mg/L for pre-monsoon season, whereas 28 mg/L and 350 mg/L for post-monsoon season. Each analyzed sources resulted within permissible limit except for sample ID 11; which was exceeding permissible limit for pre-monsoon season, however all samples were within permissible limit for post-monsoon season. Hence, most of the sampled sources are free from Sulphate contamination.



Graph 8.13 Seasonal variation of Sulphate

Nitrate (mg/L)

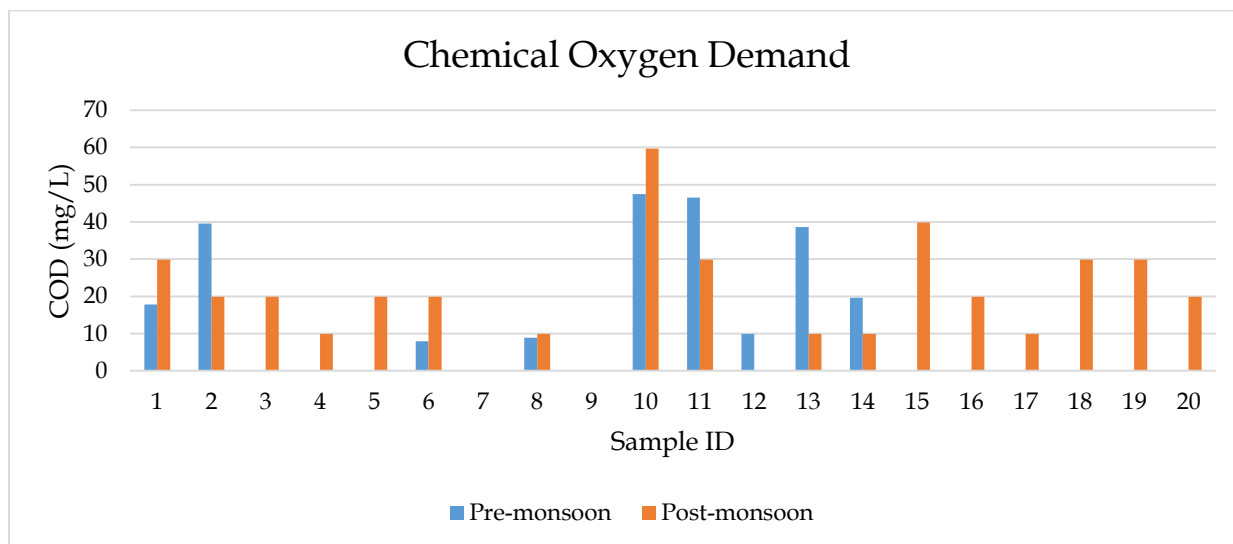
Relatively little of the nitrate found in natural waters is of mineral origin, most coming from organic and inorganic sources, the former including waste discharges and the latter comprising chiefly artificial fertilizers. However, bacterial oxidation and fixing of nitrogen by plants can both produce nitrate. Groundwater contamination can directly affect human health because excessive levels of nitrate in drinking water can produce negative health impacts on human well-being. In the present study, Nitrate ranged between 11 mg/L and 76 mg/L for pre-monsoon season, whereas 5 mg/L and 170 mg/L for post-monsoon season. Most of the analyzed sources resulted within permissible limit except for few. Sample IDs 1, 3, 13, 14 and 1, 3 to 6, 13, 15 were exceeding permissible limit for pre-monsoon and post-monsoon season respectively and hence traces of Nitrate pollution are present in few sources.



Graph 8.14 Seasonal variation of Nitrate

Chemical Oxygen Demand (mg/L)

COD is defined as the amount of oxygen equivalents consumed in oxidizing the organic compounds of samples by strong oxidizing agents such as dichromate or permanganate. It is expressed in milligrams per liter (mg/L) that indicates the mass of oxygen consumed per liter of solution. The higher the chemical oxygen demand, the higher the amount of pollution in the water sample. ^[20] In the present study, COD ranged between 8 mg/L and 48 mg/L for pre-monsoon season, whereas 10 mg/L and 60 mg/L for post-monsoon season. High variations were observed in results; however, presence of COD in water indicates pollution of groundwater.



Graph 8.15 Seasonal variation of COD

Ammonical Nitrogen

It is also commonly found in groundwater due to anthropogenic activities, primarily due to leaching from fertilizers, organic waste disposal or leaking sewage systems. [18] $\text{NH}_3\text{-N}$ in groundwater aquifer reacts through complex physical- chemical and biological processes to form other associated nitrogen bearing compounds such as NO_2 , NO_3 which makes the risk even more complicated. Their toxicity is attributed to the influence of pH, temperature, and dissolved oxygen concentration. [19] In the present study, Ammonical Nitrogen was analyzed for all the collected samples for both pre-monsoon and post-monsoon season. However, none of the sample exceeded concentration of 1 mg/L for both pre-monsoon and post-monsoon season.

Heavy Metals

Heavy Metals Zn, Cd, As, Pb, Cr and Fe were analyzed in a few samples collected during sampling for both pre and post monsoon seasons. The parameter wise inference drawn is covered in table 8.6 along with the details of sample IDs exceeding the permissible limit for respective seasons. Heavy metals in general were analyzed for 2 samples and 9 samples for pre and post monsoon season respectively.

Table 8.6 Details of Heavy Metals analyzed

Sample IDs analyzed for Heavy Metals	
Pre-monsoon	10, 11
Post-monsoon	1, 2, 10, 11, 13, 14, 15, 17, 20
Parameters	Inferences drawn
Zinc	All of the samples analyzed resulted below 0.5 mg/L for both pre-monsoon and post-monsoon seasons, hence no trace of zinc contamination is evident.
Cadmium	Concentration of Cadmium was not exceeding 0.002 mg/L for any of the sample for both pre-monsoon and post-monsoon seasons and hence the sources monitored are free from cadmium contamination.
Arsenic	Source monitored were analyzed for arsenic and were found free from arsenic pollution.
Chromium	Acceptable limit for chromium concentration in drinking water is 0.05 mg/L and all the samples analyzed resulted within acceptable limit.
Lead	Each monitored source found within acceptable limit of 0.01 mg/L and hence monitored sources are free from lead contamination.
Iron	Iron concentration of 0.71 mg/L was reported for sample ID 2 during post-monsoon sampling, which is exceeding acceptable limit of 0.3 mg/L. Remaining all samples are found within the limit.

8.4 Conclusion

Groundwater sampling was carried out around the industrial cluster of Vatva wherein; 20 samples were collected from 20 selected locations for both pre and post monsoon seasons. From the analytical results and monitoring carried out following conclusions are drawn:

- The groundwater quality around industrial cluster is found to be good in respect to all monitored physico-chemical and heavy metal parameters for almost all the samples collected during both pre and post monsoon seasons. Hence, no traces of major or minor contamination is observed during a period of study. In contradiction, the analytical results of chemical oxygen demand indicate presence of pollution in most of the samples.
- However, the overall groundwater quality of the monitored cluster is found to be fairly good in respect to all physico-chemical parameters except chemical oxygen demand.

Chapter: 9 References

1. Kartikey Tiwari et al., A Groundwater Quality Analysis of GIDC Naroda, International Journal of Engineering Science Invention Research & Development; Vol. III, Issue IX, March 2017, e-ISSN: 2349-6185.
2. Gursimran Singh et al., Effect of Polluted Surface Water On Groundwater: A Case Study of Budha Nullah, IOSR Journal of Mechanical and Civil Engineering (IOSR-JMCE) Volume 5, Issue 5 (Mar. - Apr. 2013), PP 01-08, e-ISSN: 2278-1684.
3. M. Vasanthavigar et al., Application of water quality index for groundwater quality assessment: Thirumanimuttar sub-basin, Tamilnadu, India, Environ Monit Assess (21/01/2010) 171:595–609.
4. GEMI's Sampling Protocol for Water and Wastewater, 2014, Gandhinagar, Gujarat: Gujarat Environment Management Institute.
5. Indian Standard Specifications for Drinking Water 10500:2012 (Second Revision), 2012.
6. Groundwater Brochure Bharuch District, Government of India Ministry of Water Resources Central Ground Water Board West Central Region Ahmedabad, March 2014.
7. Cluster Profile Report – Ankleshwar Chemical Cluster, New Delhi: The Energy and Resources Institute 31pp. [Project Report No. 2012IE03], TERI 2012.
8. Groundwater Brochure Valsad District, Government of India Ministry of Water Resources Central Ground Water Board West Central Region Ahmedabad, March 2013.
9. Brief Industrial Profile of Valsad District, MSME-Development Institute Government of India.
10. Ashok Kumar, District Ground Water Brochure Surat District, Gujarat State, Government of India Ministry of Water Resources Central Ground Water Board, December – 2013.
11. Brief Industrial Profile of Surat District, MSME-Development Institute Government of India.
12. District Ground Water Brochure: Vadodara.
13. Brief Industrial Profile of Vadodara District, MSME-Development Institute Government of India.
14. R. K. Verma, Ground Water Brochure Ahmedabad District Gujarat, Government of India Central Ground Water Board West Central Region Ahmedabad, March 2014.
15. Brief Industrial Profile of Ahmedabad District, MSME-Development Institute Govt. of India.

16. S. A. M. Al Dahaan et al., Influence of Groundwater Hypothetical Salts on Electrical Conductivity Total Dissolved Solids, Scientific Research Publishing, Engineering, 2016, 8, 823-830 ISSN: 1947-3931.
17. Sameer V et al., Assessment of chloride concentration in groundwater: A case study for Belgaum City, International Journal of Environmental Sciences Volume 2, No 1, 2011, ISSN 0976 – 4402.
18. Mohd Zul Asyraf Mohd Kamal et al., The Effect of the Ammonium Concentration in The Groundwater, People: International Journal of Social Sciences, Special Issue, 2015, pp. 313-319, ISSN 2454-5899.
19. Mustapha Atta et al., Ammoniacal Nitrogen Contaminated Groundwater at Taman Beringin Ex-Landfill site: Implication to Health and the Environment, Advances in Environmental Biology, 8(15) Special 2014, Pages: 136-142.
20. Tasnim Alam, Estimation of Chemical Oxygen Demand in Wastewater using UV-VIS Spectroscopy, Summer 2015.
21. Hardness in Groundwater, Water Stewardship Information Series, February 2007.
22. Color of Water Fact Sheet, State Water Resources Control Board Clean Water Team Fact Sheet 3.1.5.9.
23. Website: Water Research Centre. The pH of Water. <https://water-research.net/index.php/ph> (accessed March, 2020).
24. Website: Water Research Centre. Water Testing Total Dissolved Solids Drinking Water Quality. <https://water-research.net/index.php/water-treatment/tools/total-dissolved-solids> (accessed March, 2020).
25. Website: Groundwater-Quality Analyte Descriptions. <https://kgs.uky.edu/kgsweb/datasearching/water/analyteDescr.asp> (accessed March, 2020).